

Removing Payment Friction: Evidence-Based Churn Prevention for PT Asuransi Nusantara Sejahtera

Final Stage 1 — Data Analytics Competition

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Executive Summary

PT Asuransi Nusantara Sejahtera loses 21.1% of its customers to lapse, surrender or non-renewal, which puts about IDR 10B of annual premium (20.8% of the premium base) at risk in the analysed sample of 8,000 customers, roughly IDR 1,494.8B across the full portfolio of 1,200,000. Using these records, we combined statistical driver analysis, survival analysis, interpretable regression, a cross-validated machine-learning ensemble and doubly robust causal inference to answer the four questions posed by management: who leaves, when they leave, how accurately this can be predicted and which actions are worth funding.

Churn is most strongly associated with premium change at last renewal, payment method and auto-renewal; customers whose premium change at last renewal is between 13.2% and 30.6% churn at 33.5%, 1.6 times the average. Risk is concentrated in time: the churn hazard peaks between 18 and 24 months of tenure (0.99% per month, about 1.4 times the average). The final model reaches an out-of-fold ROC-AUC 0.760 (95% CI 0.748–0.772); the top 20% of scored customers contain 46% of all churners (2.3× random targeting). A retention campaign aimed at the top 50% of customers by expected loss is estimated to return IDR 810.8M net (ROI 1.4×), whereas random targeting of the same size yields IDR 514.8M. Causal analysis indicates that keeping the premium change at last renewal (%) at or below its median (4.1) would lower churn by about 11.7 percentage points, a result that survives placebo and sensitivity tests.

We recommend three actions: (1) smarter renewal pricing; (2) shift manual payers to automatic payment; (3) make renewal effortless (default auto-renewal). Each action should start as a randomised pilot; detecting a reduction of 11.7 percentage points requires about 277 customers per arm.

For ANS, whose renewal premiums are 58% of premium income and whose lost policies each leave about Rp1.1 million of acquisition cost unrecovered, the programme protects revenue and recovers acquisition spend at the same time.

1. Introduction

PT Asuransi Nusantara Sejahtera (ANS) is a multi-line insurer that sells life, health, motor and property policies through agents, bancassurance partners, brokers and an online channel. Policy non-renewal and lapse rose from 15.2% in 2022 to 20.6% in 2024, and the Board has asked for an explanation of who is leaving, when and why, together with a retention programme for 2026 within a budget of Rp1.5 billion.

ANS manages around 1.2 million active policies and booked gross written premium of Rp4.8 trillion in 2024; renewal premiums represent 58% of its premium income.

Management estimates that every lost policyholder leaves about Rp1.1 million of acquisition cost unrecovered. Renewal premiums accounted for Rp77.07 trillion, about 42% of Indonesian life insurance premiums, in 2024 ("AAJI catat premi," 2025). Surrender claims paid by Indonesian life insurers still amounted to Rp62.72 trillion in 2025, 19% lower than in 2024 ("AAJI catat nilai klaim surrender," 2026).

For an individual insurer, churn (lapse, surrender or non-renewal) affects several parts of the business at once. It erodes the renewal premium base, leaves acquisition costs unrecovered when policies exit early, distorts the persistency assumptions used in pricing (Eling & Kochanski, 2013) and is often a signal of unmet needs or poor service (Brockett et al., 2008). The economics of retention are well documented: Reichheld and Sasser (1990) reported that reducing defections by 5% raised profits by 25% to 85% across several service businesses. Not every exit is equally avoidable, however: Gottlieb and Smetters (2021) found that forgetting to pay explained 37.8% of recent life-insurance lapses, so part of churn can be addressed through process design rather than price.

Management's objective is to maximise the profit of retention spending. To support this objective, the study answers four research questions (RQ):

- RQ1. What are the main drivers of churn, and how large is each effect?
- RQ2. When during the customer lifecycle is churn risk highest?
- RQ3. How accurately can churn be predicted for the customers in the test set?
- RQ4. Which retention programme within the Rp1.5 billion budget maximises the financial return?

2. Data and Methodology

2.1 Data and preparation

The training data contain 8,000 customers (plus 2,000 customers to be scored) and 27 original variables covering January 2022 – June 2025 (snapshot 30 June 2025). The target is 'churn', defined as a policy that was not renewed or lapsed within the observation window (January 2022 – June 2025) (churn = Yes); the churn rate is 21.1%, so models are evaluated with ROC-AUC, precision–recall and profit rather than accuracy. One related table (claims) was aggregated to one row per customer (counts, sums and recency).

Preparation was automated and audited: 6 type corrections and harmonisations were applied (for example numbers or ranges stored as text, date parsing and inconsistent category labels), 15 duplicate records were removed and impossible values were set to missing. Leakage was the most serious data problem: cancellation reason is recorded

only around or after the churn event; such post-event variables leak the outcome, so they and all derived features were removed before any analysis (single-feature and missingness tests flagged them automatically). We engineered 18 insurance-specific features, such as tenure (years), tenure band, new customer (<12 months), age band and premium-to-income ratio.

2.2 Analytical framework

The analysis follows five layers (Figure B.1). (1) Univariate tests (χ^2 and Mann–Whitney U) with effect sizes, Information Value and Benjamini–Hochberg correction screen candidate drivers (Benjamini & Hochberg, 1995). (2) Because churn is a time-to-event outcome with censored observations, Kaplan–Meier curves, log-rank tests, a Cox model, an accelerated failure time model and restricted mean survival time describe when customers leave (Kaplan & Meier, 1958; Cox, 1972). (3) A logistic regression gives adjusted odds ratios that managers can read directly. (4) Nine algorithms, CatBoost, Explainable boosting machine (EBM), Extra trees, Hist. gradient boosting, LightGBM, Logistic regression, Neural network (MLP), Random forest and XGBoost, were trained with stratified 5-fold cross-validation, tuned with Bayesian optimisation and combined by ensemble selection; tree ensembles remain the strongest family on tabular data (Grinsztajn et al., 2022), and SHAP values explain each prediction (Lundberg & Lee, 2017). (5) Doubly robust estimators (AIPW and the DR-learner) estimate the average and heterogeneous effects of controllable levers (Chernozhukov et al., 2018; Kennedy, 2023). Results are translated into money with customer lifetime value, the expected maximum profit measure (Verbraken et al., 2013) and a campaign simulation.

2.3 Validation

All reported performance figures are out-of-fold estimates; target-dependent encodings and hyperparameter search were fitted inside training folds only. Models are compared with DeLong tests for correlated AUCs (DeLong et al., 1988) with Holm correction, uncertainty is reported with bootstrap and DeLong confidence intervals, and calibration is checked with the Brier score and expected calibration error. Sanity checks support the validity of the pipeline: a model trained on shuffled labels scores an AUC of 0.519, and an adversarial classifier separates training from scoring data with AUC 0.50. Appendix B documents the full validation design.

3. Findings

3.1 Who churns? (RQ1)

The strongest correlates of churn are premium change at last renewal (IV 0.16), where customers whose premium change at last renewal is between 13.2% and 30.6% churn at

33.5%, 1.6× the average; payment method (IV 0.10), where customers with payment method 'Bank Transfer' churn at 25.5%, 1.2× the average; auto-renewal (IV 0.08), where customers without auto-renewal churn at 26.4%, 1.3× the average (Table 1). Several variables (age, claims status Approved count, sales channel and policy start date month) are statistically detectable but practically negligible (IV < 0.02), a reminder that with 8,000 records significance alone is not evidence of importance.

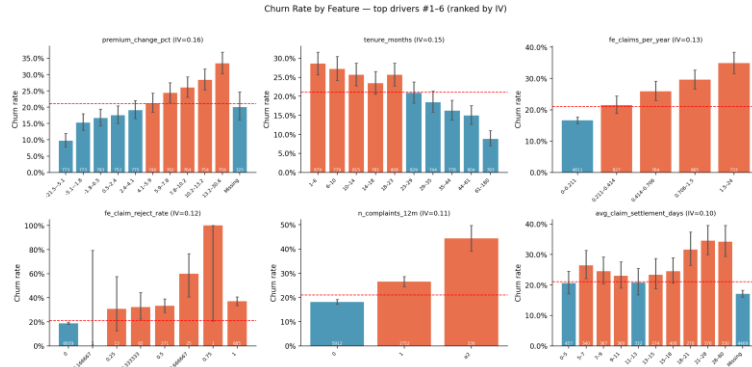


Figure 1. Churn rate by the strongest drivers (bars: churn rate with 95% Wilson CI; dashed line: overall churn rate).

Table 1. Evidence for churn drivers across methods (overall churn rate 21.1%; 'methods agreeing' counts significant univariate test, logistic regression, Cox model and top-10 SHAP).

Driver	Riskiest group	Churn rate	Lift	IV	Odds ratio	Hazard ratio	SHAP rank	Methods agreeing
premium change at last renewal (%)	13.2–30.6	33.5%	1.59×	0.16	1.64	1.39	1	4/4
payment method	Bank Transfer	25.5%	1.21×	0.10	2.33	1.83	2	4/4
auto-renewal	No	26.4%	1.25×	0.08	0.51	0.61	3	4/4
complaints in the last 12 months	≥2	44.3%	2.10×	0.11	1.33	1.23	8	4/4
average claim settlement time (days)	21–28	34.6%	1.64×	0.10	1.19	1.11	9	4/4
tenure (months)	1–6	28.6%	1.35×	0.15	0.63	—	4	3/4
number of ANS products	1	23.8%	1.13×	0.05	0.76	—	7	3/4

3.2 When do customers churn? (RQ2)

Survival analysis treats active customers as censored observations. According to the Kaplan–Meier estimates, 96.8% after 6 months, 92.2% after 12 months, 82.8% after 24

months of customers are still active; in the industry's persistency terms, the 13-month ratio is 91.5% and 25-month ratio is 82.0%. The churn hazard peaks between 18 and 24 months of tenure at 0.99% per month, against an average of 0.72% (Figure 2).

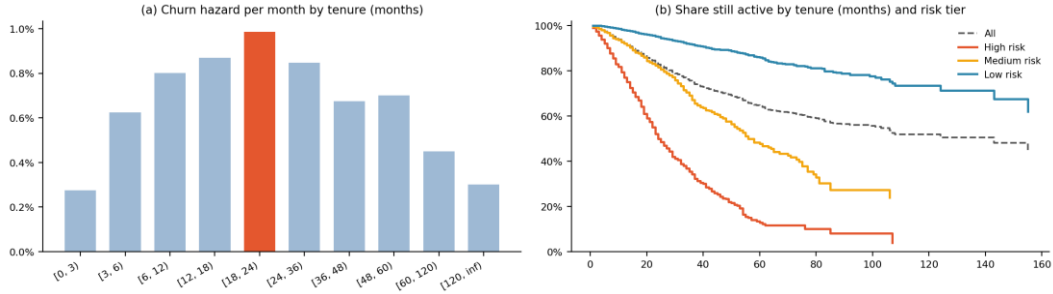


Figure 2. (a) Churn hazard per month by tenure interval; (b) Kaplan–Meier survival by predicted risk tier (out-of-sample validation of the score over time).

3.3 Who is most at risk? Interpretable segments

A shallow decision tree turns the drivers into segments operations teams can act on (Table F.1). The riskiest segment, claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method $\neq$ Auto-debit AND claim rejection rate > 0.12 , has 266 customers (3.3% of the base) churning at 55.3% (2.6 $\times$) and contains 8.7% of all churners, whereas the most loyal segment churns at only 6.1%.

3.4 How well can churn be predicted? (RQ3)

Table 2. Out-of-fold performance (5-fold stratified CV; random baseline ROC-AUC 0.500, PR-AUC 0.211). EMPC = expected maximum profit per customer (Verbraken et al., 2013); p-values Holm-corrected.

Model	ROC-AUC (95% CI)	PR-AUC	Brier	EMPC (IDR)	DeLong p vs final
Final ensemble: hill climb	0.760 (0.747–0.773)	0.474	0.140	114.99K	—
xgb_fs	0.758 (0.745–0.771)	0.470	0.141	114.61K	0.022
Logistic regression	0.757 (0.744–0.769)	0.470	0.141	114.50K	0.016
xgb_multiseed	0.756 (0.743–0.769)	0.467	0.142	114.24K	<0.001
xgb_tuned	0.755 (0.743–0.768)	0.466	0.142	114.16K	<0.001

The final model reaches an out-of-fold ROC-AUC 0.760 (95% CI 0.748–0.772) and a PR-AUC of 0.474, 2.2 times the random baseline (Table 2). The interpretable logistic regression reaches 0.757; the ensemble is significantly better (Δ AUC +0.003, DeLong p 0.016), so the added complexity buys real accuracy. A glass-box Explainable Boosting Machine reaches 0.748. Ranking by expected maximum profit agrees with ranking by

AUC. Concentration is strong: the highest-risk decile churns at 55.9% and the top 20% contain 46% of churners.

3.5 Why is a customer at risk? Explaining the model

SHAP values attribute each prediction to the features (Lundberg & Lee, 2017). The model relies mostly on premium change at last renewal (%) (11%), payment method (10%) and auto-renewal (10%) of total attribution (Figure G.4), consistent with the statistical evidence, so the model behaves in a way managers can audit.

4. Business Impact (RQ4)

4.1 Value at risk and prioritisation

Weighting each customer's churn probability by annual premium, about IDR 10B of annual premium (20.8% of the premium base) is expected to be lost in the analysed sample, or roughly IDR 1,494.8B when scaled to the portfolio of 1,200,000 customers. Weighted by premium, churn is 20.8% against 21.1% by headcount, so leavers hold about average policies. Crossing risk with value (Figure 3b) isolates a 'Priority Save' group (9.8% of customers with high risk and above-median premium) that holds 29% of the value at risk and warrants personal contact; high-risk, low-value customers can be served through automated digital touchpoints.

4.2 Retention campaign economics

Each customer's value is the discounted margin earned while the policy stays in force, and the campaign contacts the set $S(k)$ of k customers with the highest expected loss:

$$CLV_i = m \cdot P_i \cdot (1 - \rho_i^H) / (1 - \rho_i), \quad \rho_i = (1 - p_i) / (1 + d) \quad (1)$$

$$\Pi(k) = \sum (\gamma \cdot y_i \cdot V_i - c), \quad \text{summed over } i \text{ in } S(k) \quad (2)$$

We simulated a retention campaign on the out-of-fold predictions, using the parameters given in the case: a contact-and-incentive cost of IDR 150K per customer, that 30% of contacted would-be churners are retained and a 20% margin over a 5-year horizon (discount rate 10%). Contacting the top 50% by expected loss maximises net benefit at IDR 810.8M (ROI 1.4×; Figure 3a). The conclusion holds under different assumptions: the campaign stays profitable in 9 of 9 combinations of success rate ($\pm 50\%$) and cost ($\times 0.5$ – $\times 2$) in the sensitivity analysis. Scaled to the full portfolio of 1,200,000 customers, the same rule would contact about 600,000 customers at a cost of IDR 90B for a net benefit of about IDR 121.6B. The budget of IDR 1.5B is the binding constraint: it pays for about 10,000 contacts (0.8% of the portfolio). Spending it on the customers with the highest expected loss (among whom 65.7% would otherwise churn) is estimated to return IDR 11.5B net (ROI 7.6×), so the budget should be concentrated on the top of the ranking rather than spread thinly.

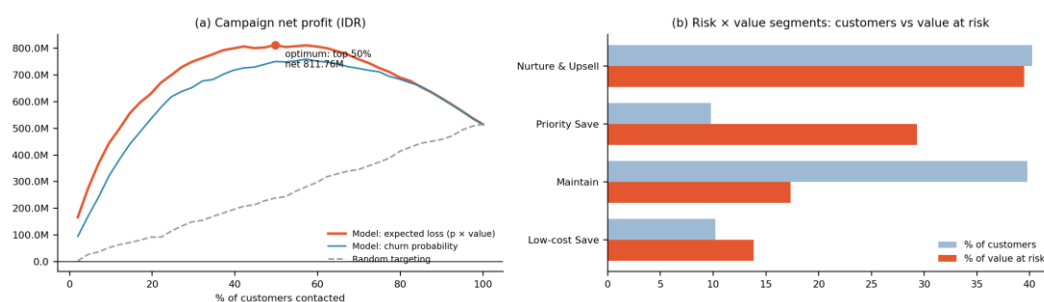


Figure 3. (a) Net profit of a retention campaign by share of customers contacted under three targeting rules; (b) share of customers vs share of value at risk by risk $\times$ value segment.

4.3 What works? Causal evidence and uplift

For each controllable lever we estimated the average effect of the favourable condition on churn with a cross-fitted doubly robust (AIPW) estimator that adjusts for observed confounders (Chernozhukov et al., 2018). The largest effects are 'premium change at last renewal (%) $\leq$ median (4.1)' (-11.7 pp, 95% CI -13.6 to -9.9; naive difference -11.1 pp), 'auto-renewal = Yes' (-10.0 pp, 95% CI -11.8 to -8.1; naive difference -9.7 pp) and 'payment method = Auto-debit' (-9.9 pp, 95% CI -11.9 to -8.0; naive difference -10.4 pp). 5 of 5 effects pass all robustness checks (placebo treatment $\approx$ 0, adequate overlap and E-values above 1.25; VanderWeele & Ding, 2017). We also tested whether some customers respond more than others (DR-learner; Kennedy, 2023). For 'premium change at last renewal (%) $\leq$ median (4.1)' the effect is broadly homogeneous (most vs least responsive fifth: 12.0 vs 11.7 pp, p 0.940), and targeting the top 20% by churn risk avoids 66.5 churners per 1,000 eligible customers versus 30.6 with uplift-based targeting, so risk-based targeting is appropriate for this lever; the pilots should still re-test responsiveness (Ascarza, 2018).

5. Recommendations and Implementation

Table 3. Recommended retention actions, ranked by evidence and expected value.

#	Action	Key evidence	Expected impact	KPI	When
1	Smarter renewal pricing	premium change at last renewal (%) = 13.2–30.6: churn 33.5% (1.6 $\times$ avg); IV 0.16	-2.98 pp churn at full adoption	Churn rate among customers with increases	Medium term
2	Shift manual payers to automatic payment	payment method = Bank Transfer: churn 25.5% (1.2 $\times$ avg); IV 0.10	-6.19 pp churn at full adoption	% of policies on auto-pay	Quick win
3	Make renewal effortless (default auto-renewal)	auto-renewal = No: churn 26.4% (1.3 $\times$ avg); IV 0.08	-3.56 pp churn at full adoption	Auto-renew adoption	Quick win
4	Complaint recovery	complaints in the last 12 months =	-1.11 pp churn at full adoption	Complaint resolution time	Quick win

#	Action	Key evidence	Expected impact	KPI	When
	programme	≥ 2 : churn 44.3% (2.1 $\times$ avg); IV 0.11			
5	Improve the claims experience	average claim settlement time (days) = 21–28: churn 34.6% (1.6 $\times$ avg); IV 0.10	-0.76 pp churn at full adoption	Claim settlement days	Medium term

1. Smarter renewal pricing. Cap or phase in premium increases for at-risk loyal customers, explain the value behind increases, and offer coverage/deductible adjustments instead of lapse.

2. Shift manual payers to automatic payment. Offer a small incentive (e.g. premium discount/cashback) and one-click enrolment for auto-debit/recurring card payment; send pre-due reminders to remaining manual payers.

Implementation follows a pilot $\rightarrow$ measure $\rightarrow$ scale logic (Figure J.4). Every action keeps a randomised control group: with a baseline churn of 48.2% in the High-risk tier, detecting an 11.7 pp reduction requires 277 customers per arm ($\alpha = 0.05$, power 80%). Pilot results then train uplift models so that future campaigns target customers by expected response $\times$ value rather than by risk alone (Ascarza, 2018). The model is deployed as a monthly early-warning score with drift and calibration monitoring and quarterly retraining (Appendix J).

Two company initiatives make these actions cheaper to deliver: the self-service app planned for Q1 2026 can carry onboarding and payment reminders, and the auto-debit partnership under negotiation removes a major source of payment friction. We therefore recommend running the first-quarter pilots through these two channels, with agents reserved for Priority Save customers.

6. Discussion and Conclusion

The four research questions can now be answered. RQ1: Churn is driven mainly by premium change at last renewal, payment method and auto-renewal. RQ2: Risk peaks between 18 and 24 months of tenure. RQ3: Churn can be predicted with ROC-AUC 0.760 (95% CI 0.748–0.772), concentrating 46% of churners in the top 20%. RQ4: A value-targeted campaign returns about IDR 810.8M net in the analysed sample (about IDR 121.6B at portfolio scale); within the IDR 1.5B budget, contacting the 10,000 highest-loss customers returns about IDR 11.5B net. The most promising lever is keeping the premium change at last renewal (%) at or below its median (4.1). These results are consistent with prior evidence. In a survey of recent lapsers, forgetting to pay explained 37.8% of life-insurance lapses (Gottlieb & Smetters, 2021). In the U.S.

property and casualty market, around half of customers satisfied with their claims experience intend to renew, compared with about a third of dissatisfied customers (Fallis et al., 2022). In an Indonesian insurer's portfolio, product type, premium payment frequency and policy age mattered more for lapse than macroeconomic factors (Hein et al., 2020).

Limitations. The data are a single snapshot from one insurer, so results may not generalise and may shift with economic conditions. Drivers identified by statistical tests, regression and SHAP are associations; the causal estimates rely on the assumption of no unmeasured confounding, which we probe with E-values but cannot rule out. The campaign economics use the parameters given in the case, which we stress-test in a sensitivity analysis. Variables such as competitor prices, service interactions and household income shocks are not observed. A randomised pilot of the recommended actions is the next step to measure their true effect.

References

- Ascarza, E. (2018). Retention futility: Targeting high-risk customers might be ineffective. *Journal of Marketing Research*, 55(1), 80–98.
<https://doi.org/10.1509/jmr.16.0163>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300.
<https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Brockett, P. L., Golden, L. L., Guillén, M., Nielsen, J. P., Parner, J., & Pérez-Marín, A. M. (2008). Survival analysis of a household portfolio of insurance policies: How much time do you have to stop total customer defection? *Journal of Risk and Insurance*, 75(3), 713–737. <https://doi.org/10.1111/j.1539-6975.2008.00281.x>
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1–C68.
<https://doi.org/10.1111/ectj.12097>
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2), 187–202.
<https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>
- DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3), 837–845.
<https://doi.org/10.2307/2531595>

- Eling, M., & Kochanski, M. (2013). Research on lapse in life insurance: What has been done and what needs to be done? *The Journal of Risk Finance*, 14(4), 392–413. <https://doi.org/10.1108/JRF-12-2012-0088>
- Fallis, J., Barro, J., Leibovitz, A., & Lafortune, M. (2022, December 20). How to improve customer retention in property and casualty insurance. Bain & Company. <https://www.bain.com/insights/how-to-improve-customer-retention-in-property-and-casualty-insurance-snap-chart/>
- Gottlieb, D., & Smetters, K. (2021). Lapse-based insurance. *American Economic Review*, 111(8), 2377–2416. <https://doi.org/10.1257/aer.20160868>
- Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems 35* (pp. 507–520). <https://doi.org/10.52202/068431-0037>
- Hein, K. Z. P. W., Keovilay, J., Adinegoro, W., & Ko, B. (2020). Analisis empiris perilaku lapse pada nasabah asuransi jiwa di Indonesia. *Jurnal Statistika dan Aplikasinya*, 4(2), 73–81. <https://doi.org/10.21009/JSA.04202>
- Kaplan, E. L., & Meier, P. (1958). Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282), 457–481. <https://doi.org/10.1080/01621459.1958.10501452>
- Kennedy, E. H. (2023). Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2), 3008–3049. <https://doi.org/10.1214/23-EJS2157>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*. https://papers.nips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html
- Reichheld, F. F., & Sasser, W. E., Jr. (1990). Zero defections: Quality comes to services. *Harvard Business Review*, 68(5), 105–111.
- VanderWeele, T. J., & Ding, P. (2017). Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4), 268–274. <https://doi.org/10.7326/M16-2607>
- Verbraken, T., Verbeke, W., & Baesens, B. (2013). A novel profit maximizing metric for measuring classification performance of customer churn prediction models. *IEEE Transactions on Knowledge and Data Engineering*, 25(5), 961–973. <https://doi.org/10.1109/TKDE.2012.50>

Appendices

The appendices are not counted in the page limit. They document every analysis step so that each result in the main text can be traced and reproduced; all figures and tables were generated by the accompanying notebook.

Appendix A. Data Overview and Quality

This appendix documents the raw data, data quality checks and the transformations applied before analysis.

Key observations. Of 39 profiled columns, 15 contain missing values; the most incomplete is claims claim amount std (87.9% missing). Missingness was kept as information through indicator features rather than imputed away.

The leakage screen removed 1 variable(s) (cancellation reason); each one alone predicts churn almost perfectly because it is recorded after the event.

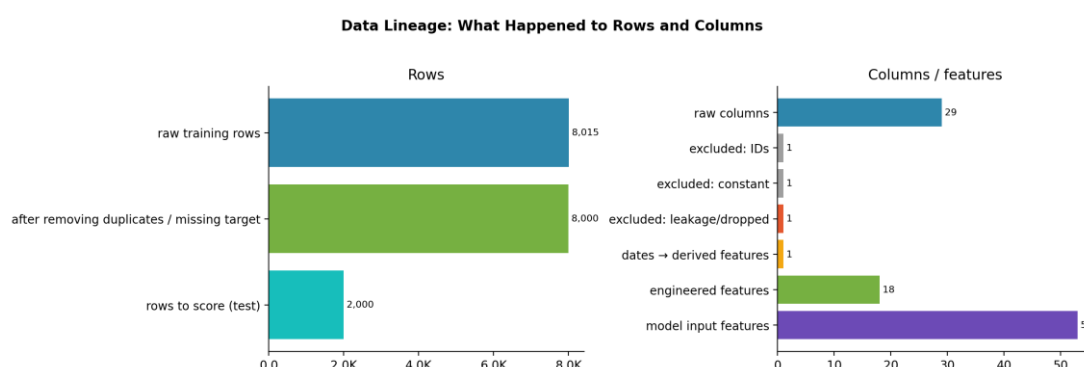


Figure A.1. Data Lineage: What Happened to Rows and Columns

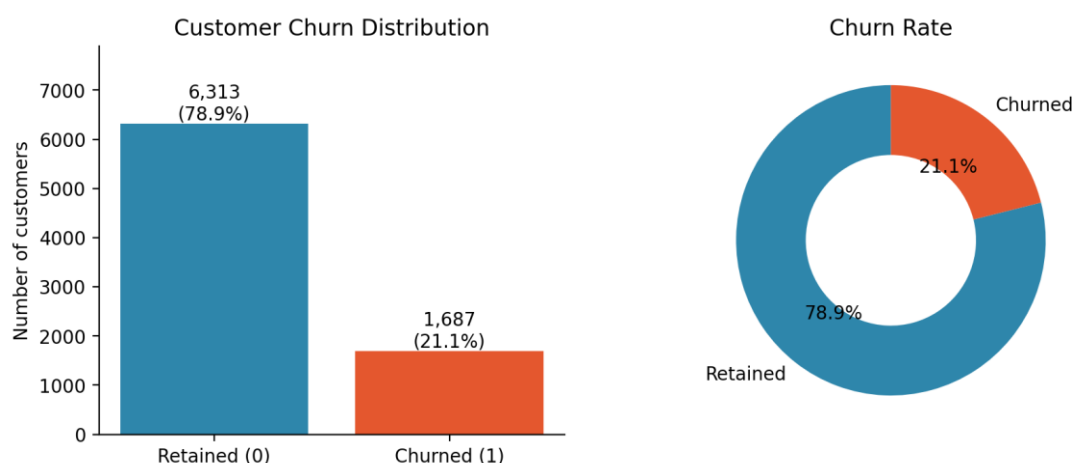


Figure A.2. Customer Churn Distribution

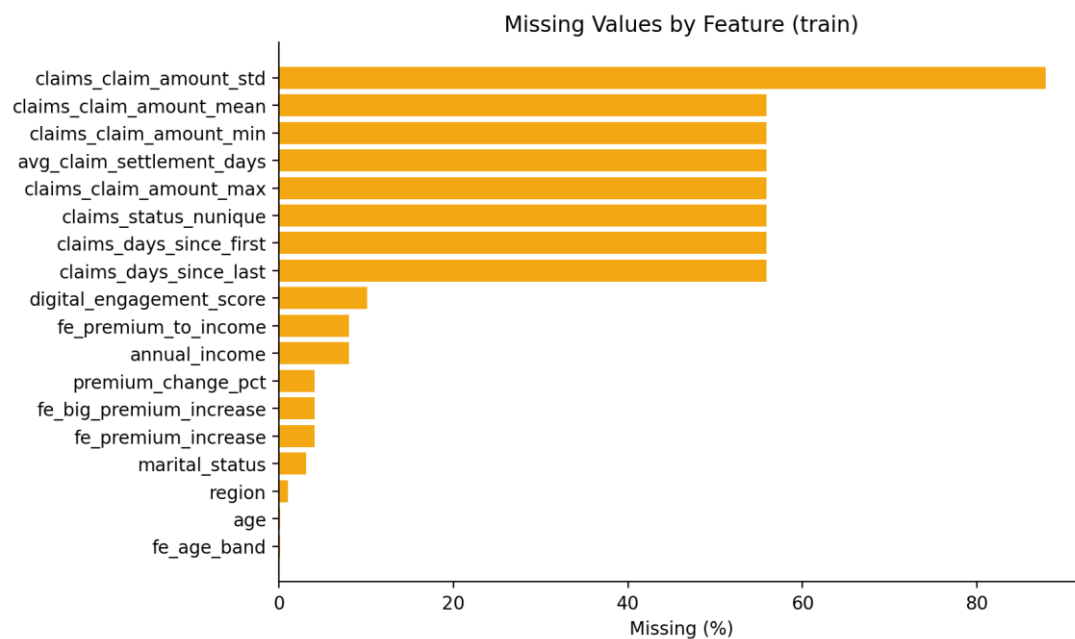


Figure A.3. Missing Values by Feature (train)

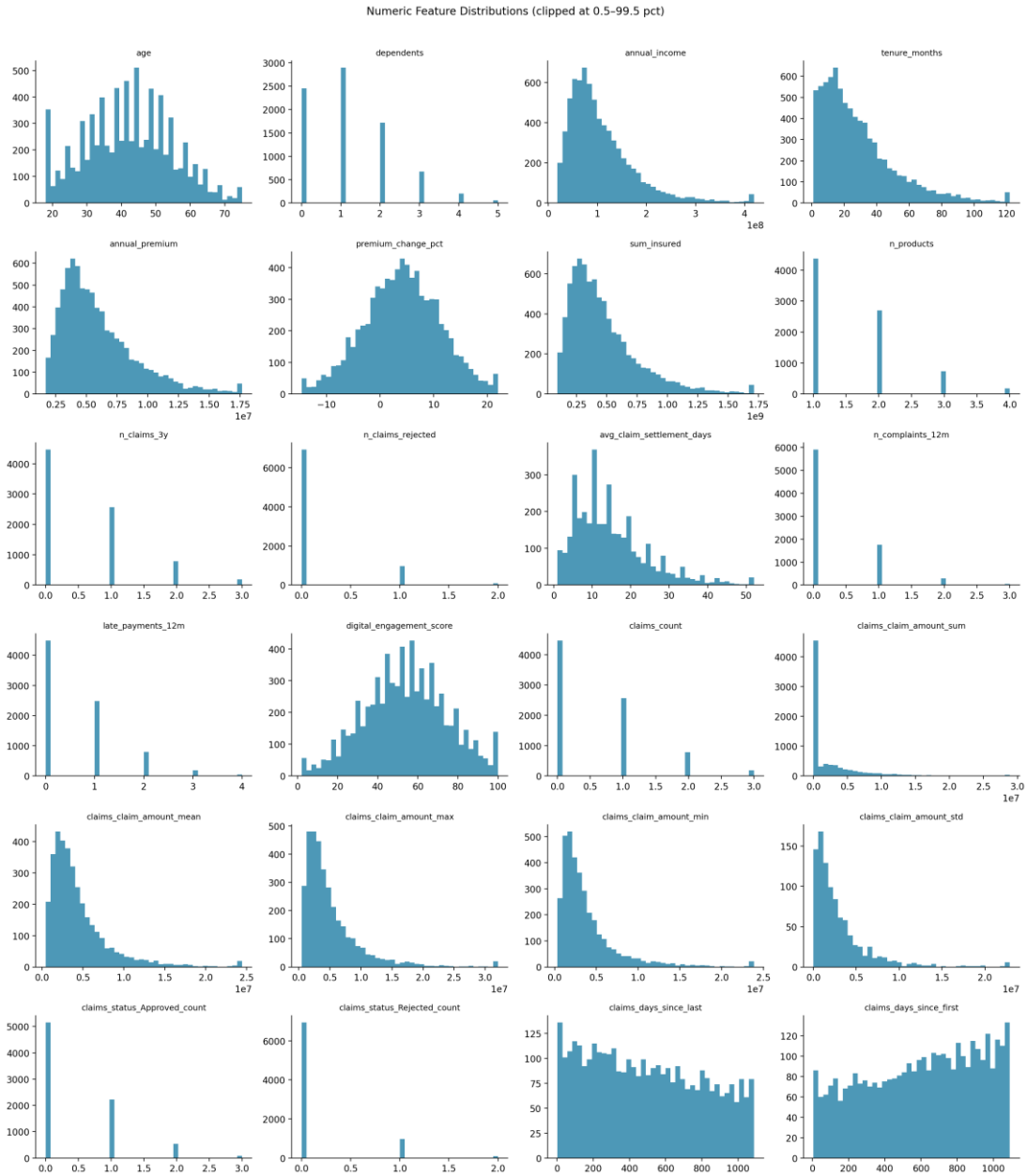


Figure A.4. Numeric Feature Distributions (clipped at 0.5–99.5 pct)

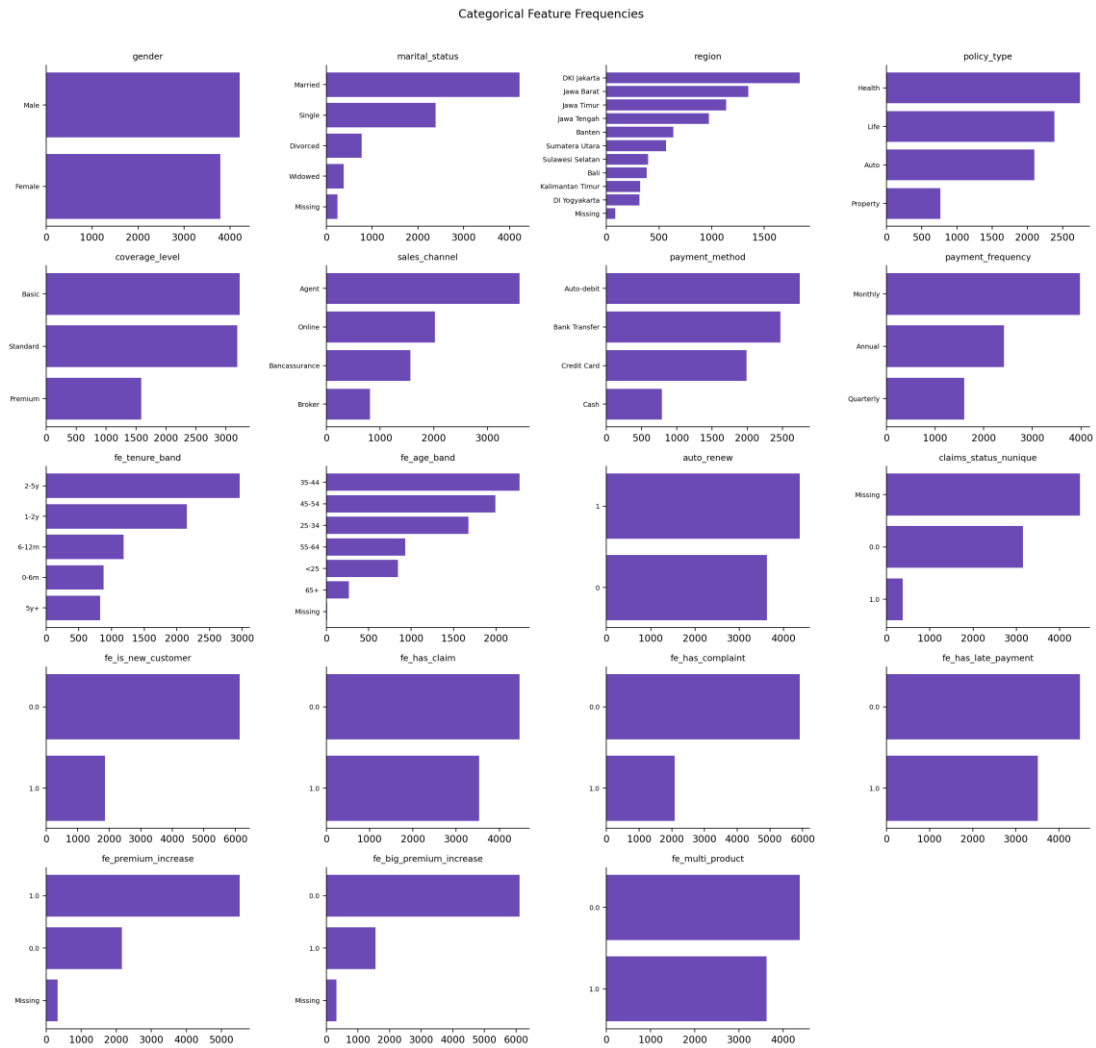


Figure A.5. Categorical Feature Frequencies

Table A.1. Data dictionary and quality report (inferred type, missing values, cardinality).

Column	Inferred type	Role	Missing %	N unique	Top value	Top freq %
customer id	id		0	8,000	CUST000001	0.010
age	numeric	age	0.100	68	18.0	3.670
gender	binary_cat	gender	0	2	Male	52.6
marital status	categorical		3.090	4	Married	54.4
dependents	numeric	dependents	0	8	1	36.2
region	categorical	region	1.040	10	DKI Jakarta	23.2
annual income	numeric	income	8.040	2,210	71.40M	0.200
policy type	categorical	policy type	0	4	Health	34.3
coverage level	categorical		0	3	Basic	40.3
sales channel	categorical	channel	0	4	Agent	45.0
agent id	high_card	agent	0	400	NONE	55.0
payment method	categorical	payment method	0	4	Auto-debit	34.4
payment frequency	categorical	payment frequency	0	3	Monthly	49.8
policy start date	date		0	146	2024-04-30	2.740

Column	Inferred type	Role	Missing %	N unique	Top value	Top freq %
					00:00:00	
tenure (months)	numeric	tenure	0	146	14.0	2.740
annual premium	numeric	premium	0	5,321	5.79M	0.090
premium change at last renewal (%)	numeric	premium_change	4.060	416	3.900	0.790
sum insured	numeric	sum insured	0	5,084	287.50M	0.090
number of ANS products	numeric	products	0	7	1	54.7
claims in the last 3 years	numeric	claims	0	7	0	55.9
rejected claims	numeric		0	4	0	86.7
average claim settlement time (days)	numeric		55.9	65	10.0	5.690
complaints in the last 12 months	numeric	complaints	0	7	0	73.9
late payments in the last 12 months	numeric	late_payment	0	7	0	56.1
digital engagement score	numeric	engagement	10.1	101	48.0	2.320
auto-renewal	binary	auto-renewal	0	2	1	54.6
currency	constant		0	1	IDR	100.0
cancellation reason	categorical		78.9	5	Moved	20.7
claims count	numeric		0	7	0	55.9
claims claim amount sum	numeric		0	3,033	0	55.9
claims claim amount mean	numeric		55.9	3,028	3.45M	0.110
claims claim amount max	numeric		55.9	2,948	2.12M	0.110
claims claim amount min	numeric		55.9	2,753	2.61M	0.140
claims claim amount std	numeric		87.9	927	1.40M	0.310
claims status Approved count	numeric		0	6	0	64.4
claims status Rejected count	numeric		0	4	0	86.7
claims status nunique	binary		55.9	2	0	89.3
claims (days since) last	numeric		55.9	1,039	1.000	0.340
claims (days since) first	numeric		55.9	1,032	885.0	0.340

Table A.2. Leakage screen: name-based, missingness-based and single-feature AUC checks.

Feature	Name keyword	Missingness auc	Single feature auc	Statistical evidence	Flagged
cancellation reason	yes	1.000	1.000	yes	yes
premium change at last renewal (%)	no	0.501	0.613	no	no
tenure (years)	no		0.601	no	no
policy start date (days since)	no		0.601	no	no

Feature	Name keyword	Missingness auc	Single feature auc	Statistical evidence	Flagged
tenure (months)	no		0.601	no	no
policy start date year	no		0.595	no	no
claims per year of tenure	no		0.590	no	no
tenure band	no		0.588	no	no
claims claim amount sum	no		0.572	no	no
claims in the last 3 years	no		0.570	no	no
claims count	no		0.570	no	no
complaints in the last 12 months	no		0.570	no	no
auto-renewal	no		0.570	no	no
claims status nunique	no	0.567	0.567	no	no
payment method	no		0.566	no	no

Table A.3. Summary statistics of numeric variables (percentiles, skewness, missing share) (first 10 columns; full table in report_tables.xlsx).

Feature	Count	Mean	Std	Min	1%	25%	50%	75%	99%
age	7,992	42.2	12.7	18.0	18.0	33.0	42.0	51.0	72.0
dependents	8,000	1.185	1.090	0	0	0	1.000	2.000	4.000
annual income	7,357	108.58M	70.85M	7.80M	22.00M	61.00M	90.30M	137.00M	365.62M
tenure (months)	8,000	29.2	23.6	1.000	1.000	12.0	23.0	39.0	110.0
annual premium	8,000	5.98M	3.10M	872,000	1.87M	3.76M	5.22M	7.46M	16.12M
premium change at last renewal (%)	7,675	4.110	7.139	-21.5	-12.5	-0.600	4.100	9.000	20.6
sum insured	8,000	475.65M	293.64M	47.00M	108.60M	270.57M	401.10M	597.65M	1.52B
number of ANS products	8,000	1.596	0.770	1.000	1.000	1.000	1.000	2.000	4.000
claims in the last 3 years	8,000	0.590	0.777	0	0	0	0	1.000	3.000
rejected claims	8,000	0.145	0.386	0	0	0	0	0	2.000
average claim settlement time (days)	3,531	14.9	9.665	0	2.000	8.000	13.0	20.0	45.0
complaints in the last 12 months	8,000	0.310	0.573	0	0	0	0	1.000	2.000
late payments in the last 12 months	8,000	0.604	0.811	0	0	0	0	1.000	3.000
digital engagement score	7,190	53.9	20.5	0	6.000	40.0	54.0	68.0	100.0
auto-renewal	8,000	0.546	0.498	0	0	0	1.000	1.000	1.000

Feature	Count	Mean	Std	Min	1%	25%	50%	75%	99%
claims count	8,000	0.590	0.777	0	0	0	0	1.000	3.000
claims claim amount sum	8,000	2.70M	5.06M	0	0	0	0	3.58M	23.26M
claims claim amount mean	3,531	4.53M	4.00M	99,000	558,500	2.08M	3.42M	5.59M	20.19M
claims claim amount max	3,531	5.21M	4.95M	99,000	558,500	2.17M	3.76M	6.47M	24.05M
claims claim amount min	3,531	3.91M	3.88M	99,000	430,500	1.64M	2.78M	4.68M	18.52M
claims claim amount std	969.0	3.06M	3.70M	3,536	32,866	894,490	2.00M	3.86M	18.73M
claims status Approved count	8,000	0.445	0.673	0	0	0	0	1.000	3.000
claims status Rejected count	8,000	0.145	0.386	0	0	0	0	0	2.000
claims status nunique	3,531	0.106	0.309	0	0	0	0	0	1.000
claims (days since) last	3,531	491.5	312.3	1.000	6.000	220.0	467.0	750.0	1,080
claims (days since) first	3,531	602.1	312.2	1.000	10.3	346.0	630.0	874.5	1,088
tenure (years)	8,000	2.430	1.965	0.083	0.083	1.000	1.917	3.250	9.168
new customer (<12 months)	8,000	0.233	0.423	0	0	0	0	0	1.000
premium-to-income ratio	7,357	0.079	0.072	0.002	0.010	0.035	0.059	0.097	0.343
sum insured to premium ratio	8,000	79.5	23.0	40.0	40.8	59.7	79.0	99.1	119.0
has made a claim	8,000	0.441	0.497	0	0	0	0	1.000	1.000
claims per year of tenure	8,000	0.503	1.106	0	0	0	0	0.545	4.800
claim rejection rate	8,000	0.108	0.291	0	0	0	0	0	1.000
has complained	8,000	0.261	0.439	0	0	0	0	1.000	1.000
has late payments	8,000	0.439	0.496	0	0	0	0	1.000	1.000
premium increased at renewal	7,675	0.719	0.450	0	0	0	1.000	1.000	1.000
premium increase above 10%	7,675	0.203	0.402	0	0	0	0	0	1.000

Feature	Count	Mean	Std	Min	1%	25%	50%	75%	99%
holds multiple products	8,000	0.453	0.498	0	0	0	0	1.000	1.000
policy start date (days since)	8,000	886.8	716.7	31.0	31.0	365.0	700.0	1,186	3,344
policy start date year	8,000	2,023	1.976	2,010	2,016	2,022	2,023	2,024	2,025

Automated cleaning log: gender: 2 variasi ejaan/case disatukan (4 $\rightarrow$ 2 level); annual_income: string $\rightarrow$ numeric (100.0% berhasil); payment_method: 4 variasi ejaan/case disatukan (8 $\rightarrow$ 4 level); policy_start_date: parsed sebagai tanggal (100.0%); auto_renew: boolean-like $\rightarrow$ 1/0; claims_status_nunique: binary numeric [np.float64(1.0), np.float64(2.0)] $\rightarrow$ 0/1

Appendix B. Methodology and Validation Design

Architecture of the analytics pipeline, the cross-validation design and the predictive model, plus tuning results.

Statistical screening. For categorical variables we use Pearson's χ^2 test with Cramér's V; for numeric variables the Mann–Whitney U test with the rank-biserial correlation, so that every p-value is paired with an effect size. Because dozens of variables are tested, p-values are converted to Benjamini–Hochberg q-values. Weight of evidence and Information Value, standard in credit and insurance scoring, summarise each binned variable's predictive strength (Siddiqi, 2006); proportions are reported with Wilson 95% intervals.

Survival layer. Tenure is the time axis and active customers are right-censored. Kaplan–Meier curves and log-rank tests describe retention; discrete hazards per tenure interval locate the critical window; the Cox model gives adjusted hazard ratios, its proportional-hazards assumption is checked with Schoenfeld residuals (Schoenfeld, 1982), and a Weibull accelerated failure time model provides time ratios that remain interpretable when the assumption fails (Wei, 1992). Restricted mean survival time expresses differences in months or days retained (Royston & Parmar, 2013).

Predictive layer. Logistic regression, random forest, extra trees, histogram gradient boosting, LightGBM, XGBoost and CatBoost (Ke et al., 2017; Chen & Guestrin, 2016; Prokhorenkova et al., 2018), a glass-box Explainable Boosting Machine (Lou et al., 2013) and a neural network are trained on identical folds. Hyperparameters are tuned with the TPE sampler and median pruning (Akiba et al., 2019) on inner folds; the best families are re-trained with several seeds; features with non-positive permutation importance are tested for removal; and six ensembling strategies, including greedy ensemble selection (Caruana et al., 2004) and stacking, are compared, the simplest

within 0.0005 of the best being preferred. Probabilities are calibrated by isotonic or Platt scaling when this lowers cross-validated log-loss, and calibration is summarised by the Brier score and expected calibration error (Guo et al., 2017).

Model comparison. Correlated AUCs are compared with the DeLong test (DeLong et al., 1988) using the fast algorithm, with Holm correction across comparisons; bootstrap intervals complement the analytic ones. Because AUC is not aligned with campaign profit (Verbeke et al., 2012; Höppner et al., 2020), models are also ranked by the expected maximum profit measure for churn, which integrates campaign profit over a Beta-distributed acceptance rate (Verbraken et al., 2013). The risk score is further validated on time-to-churn with Harrell's C-index (Harrell et al., 1982).

Why not only an interpretable model? Interpretable models should be preferred when they perform comparably (Rudin, 2019); we therefore report the logistic regression and the EBM next to the ensemble and test whether the accuracy gain is significant.

Further methodological support: Uplift models can yield more profitable retention campaigns than churn-risk models (Devriendt et al., 2021).

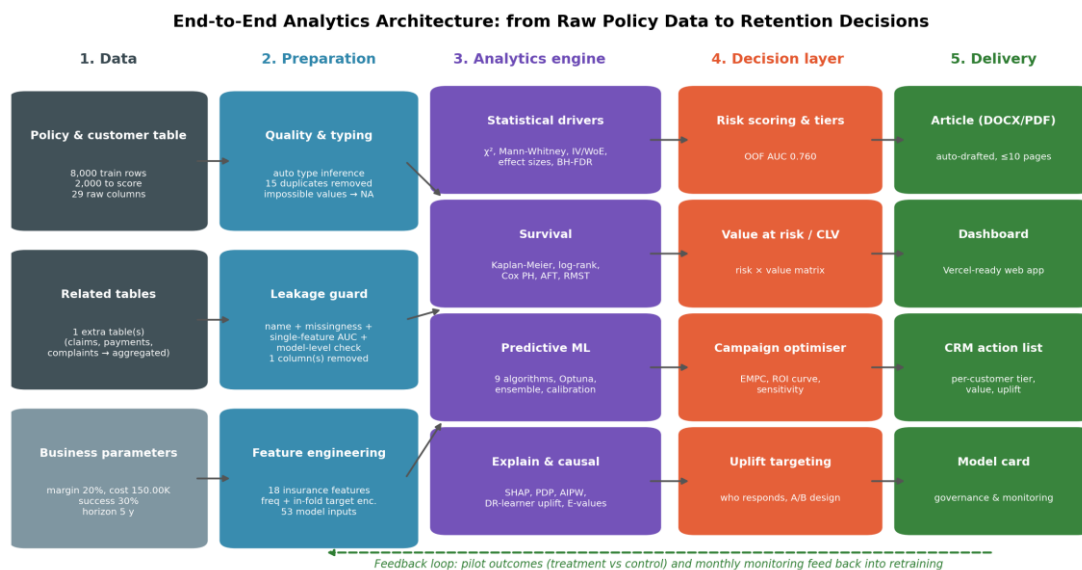


Figure B.1. End-to-end analytics architecture: from raw policy data to retention decisions, with a feedback loop from pilots.

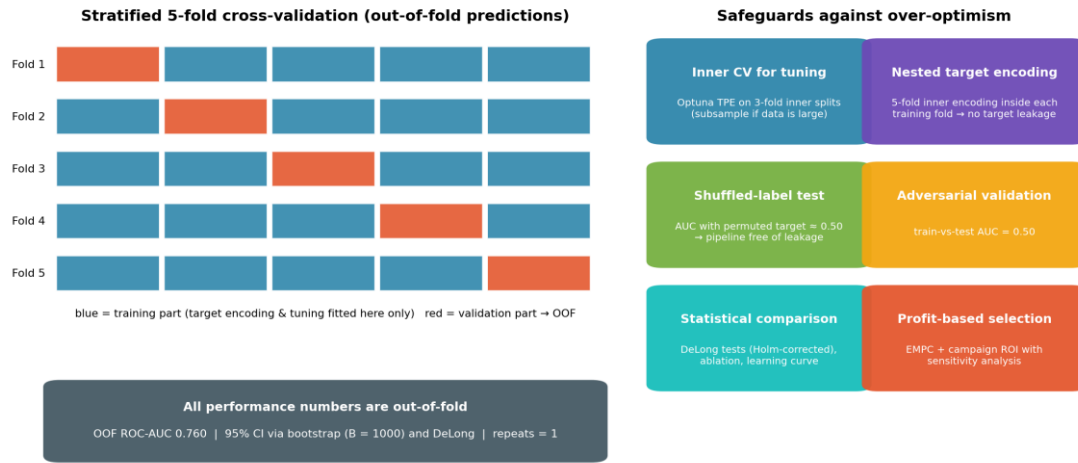


Figure B.2. Validation design: stratified k-fold cross-validation producing out-of-fold predictions for model comparison and evaluation.

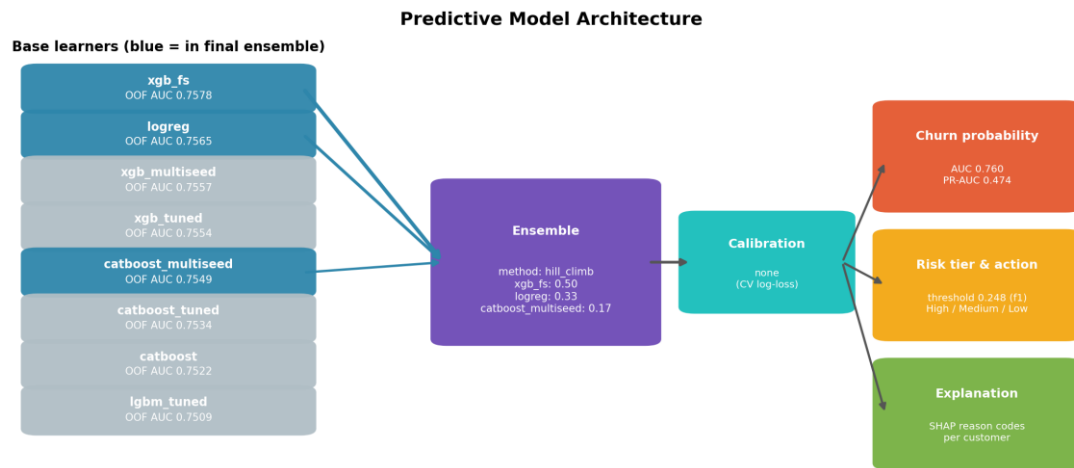


Figure B.3. Predictive Model Architecture

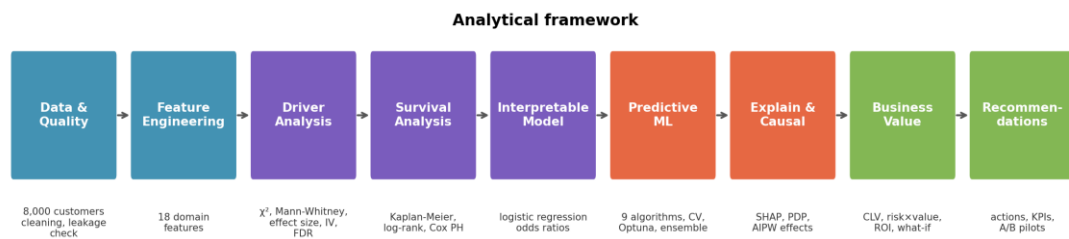


Figure B.4. Analytical framework

Table B.1. Key analysis settings (reproducibility).

Setting	Value
RUN_MODE	full
SEED	42

Setting	Value
N_FOLDS	5
N_REPEATS	1
METRIC	roc_auc
MODELS	['logreg', 'rf', 'et', 'hgb', 'lgbm', 'xgb', 'catboost', 'ebm', 'mlp']
USE_OPTUNA	True
OPTUNA_TRIALS	40
FINAL_SEEDS	[42, 2024, 7]
CALIBRATION	auto
PROFIT_MARGIN	0.2
RETENTION_COST	150000
RETENTION_SUCCESS_RATE	0.3
DISCOUNT_RATE	0.1
CLV_HORIZON_YEARS	5
RISK_TIER_CUTS	(0.5, 0.8)
WHAT_IF_ADOPTION	0.3
PILOT_POWER	0.8
PILOT_ALPHA	0.05

Table B.2. Bayesian hyperparameter optimisation (Optuna) results.

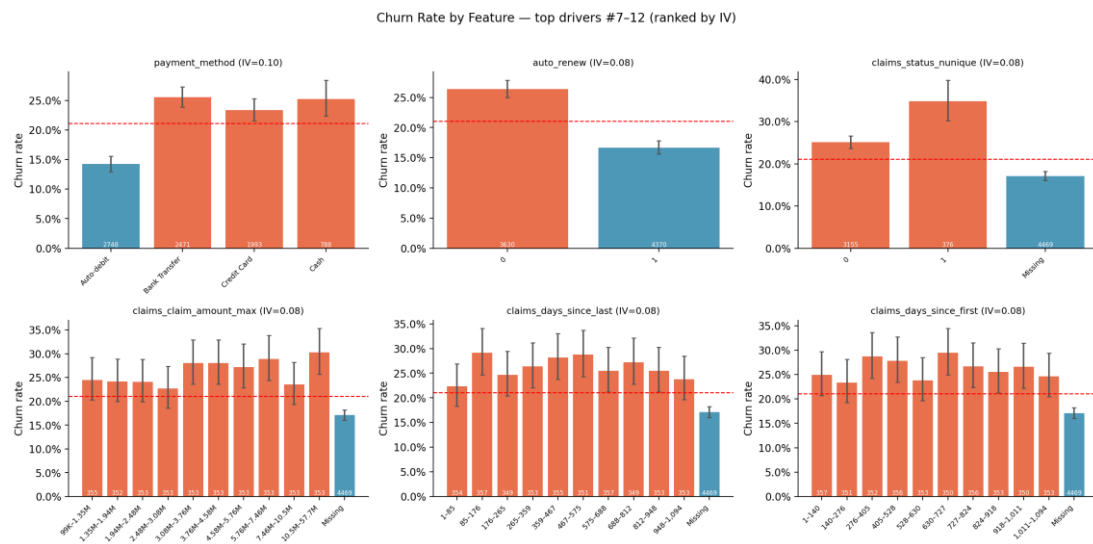
Model	Trials completed	Best tuning score	Default cv auc	Best params	Tuned cv auc	Gain auc
lgbm	39	0.753	0.745	{ "learning_rate": 0.01422387663504771, "num_leaves": 10, "min_child_samples": 38, "subsample": 0.7191667789934336, "colsample_bytree": 0.3516920077513463, "reg_alpha": 0.0007345564742179169, "reg_lambda": 4.2576268745423745e-08, "min_split_gain": 0.5496283858967218, "cat_smooth": 35.775710006609955 }	0.751	0.006
catboost	9	0.754	0.752	{ "learning_rate": 0.0680250643581967, "depth": 4, "l2_leaf_reg": 1.2476394272569447, "random_strength": 2.84665661176, "bagging_temperature": 0.9656320330745594 }	0.753	0.001
xgb	39	0.757	0.747	{ "learning_rate": 0.027776379640539747, "max_depth": 2, "min_child_weight": 1.6444231186105078, "subsample": 0.8192065342459486, "colsample_bytree": 0.30889555098626775, "reg_alpha": 5.7625793724134645e-06, "reg_lambda": 0.7368257037626049, "gamma": 1.3513512844187323 }	0.755	0.008

Appendix C. Driver Analysis (Detailed)

Churn rates for every analysed variable with 95% confidence intervals, predictive strength (Information Value), correlations and interactions.

Key observations. 35 of 52 variables remain significant after Benjamini–Hochberg correction ($q < 0.05$). By Information Value the strongest are premium change at last renewal (%) (0.16), tenure (months) (0.15) and policy start date (days since) (0.15); values above 0.1 indicate medium and above 0.3 strong predictive power (Siddiqi, 2006).

The highest variance inflation factor is 1000799917193443.5 (tenure (months)); the regression in Appendix D therefore reports this variable with caution.



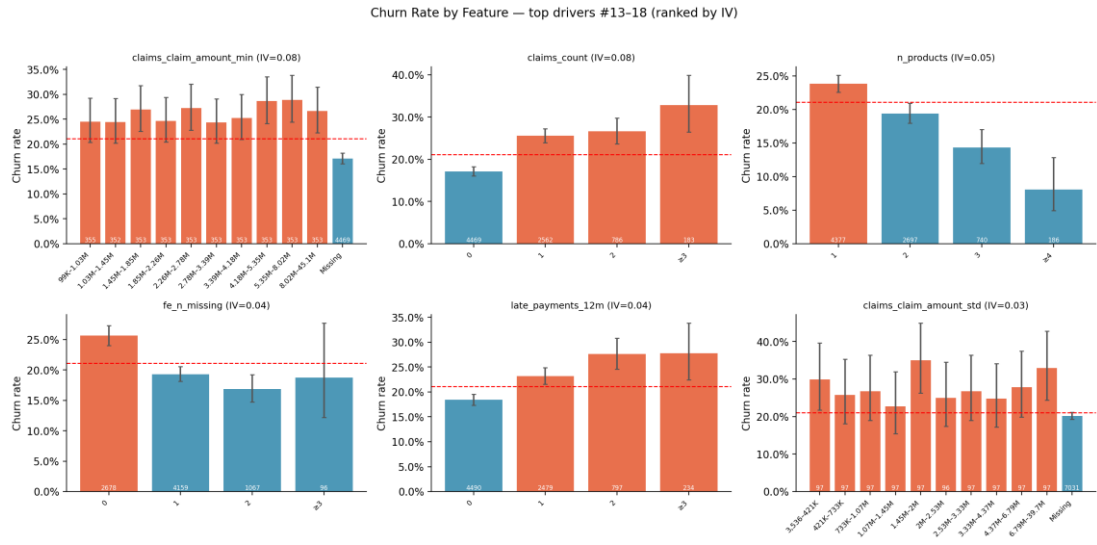


Figure C.2. Churn Rate by Feature — top drivers #13–18 (ranked by IV)

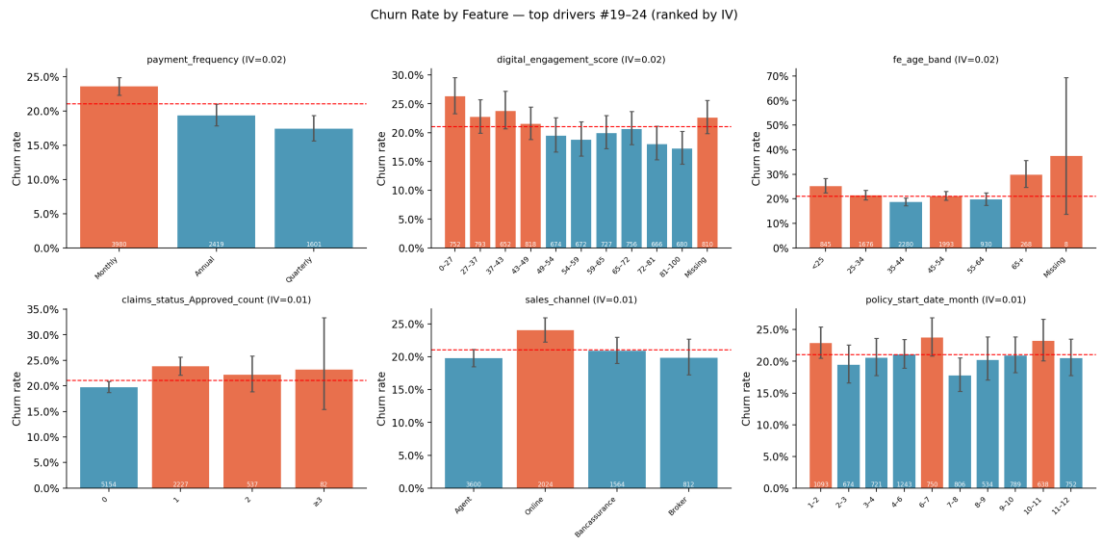


Figure C.3. Churn Rate by Feature — top drivers #19–24 (ranked by IV)

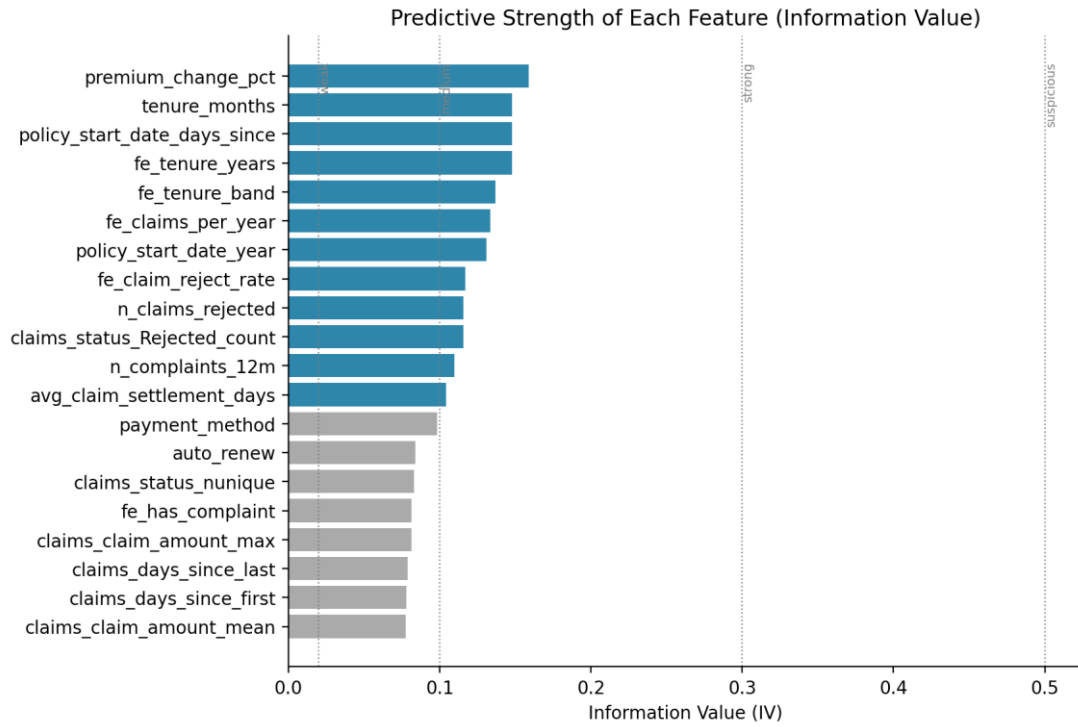


Figure C.4. Predictive Strength of Each Feature (Information Value)

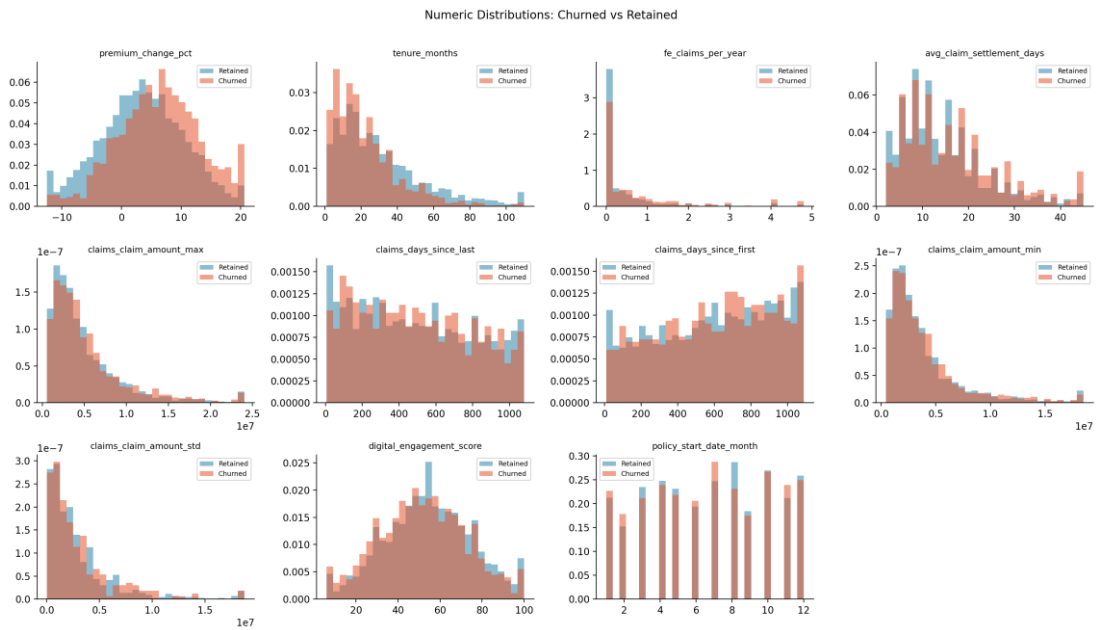


Figure C.5. Numeric Distributions: Churned vs Retained

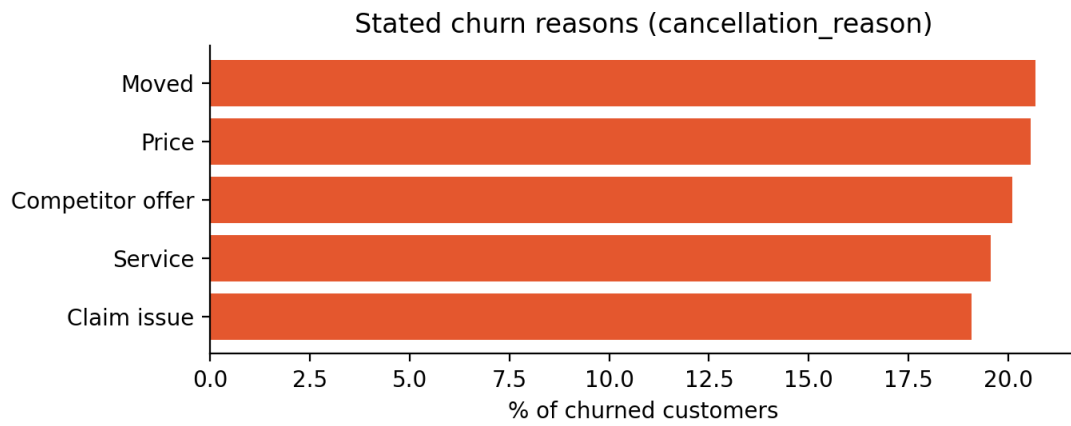


Figure C.6. Stated churn reasons (cancellation reason)

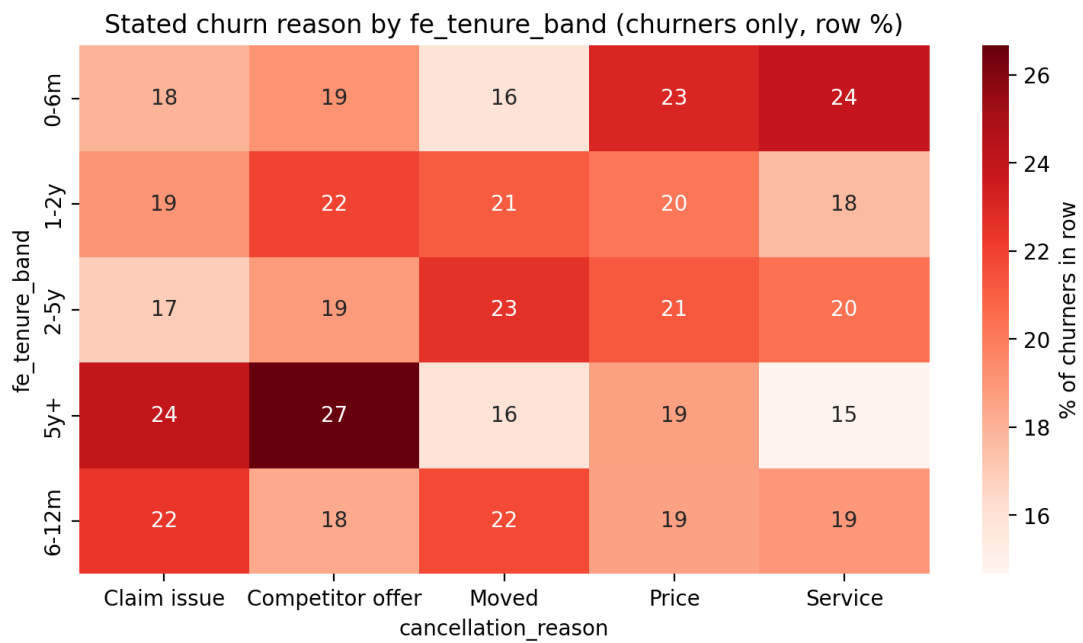


Figure C.7. Stated churn reason by tenure band (churners only, row %)

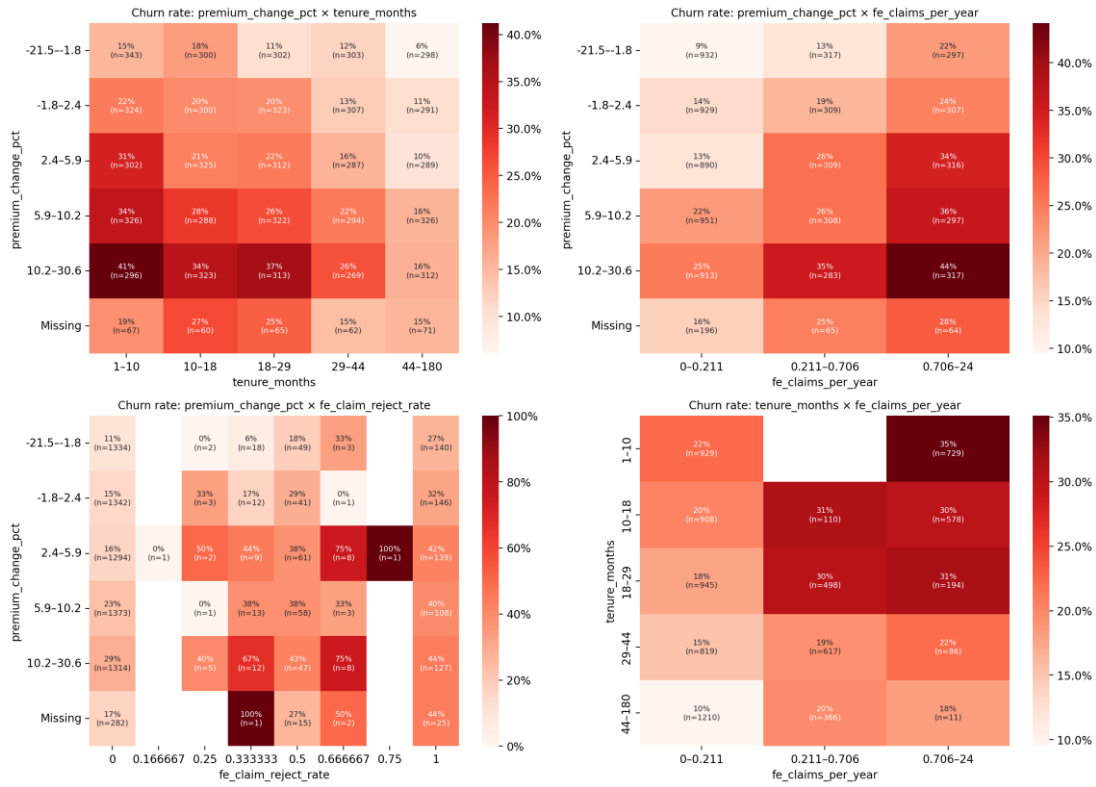


Figure C.8. Churn rate: premium change at last renewal (%) × tenure (months)

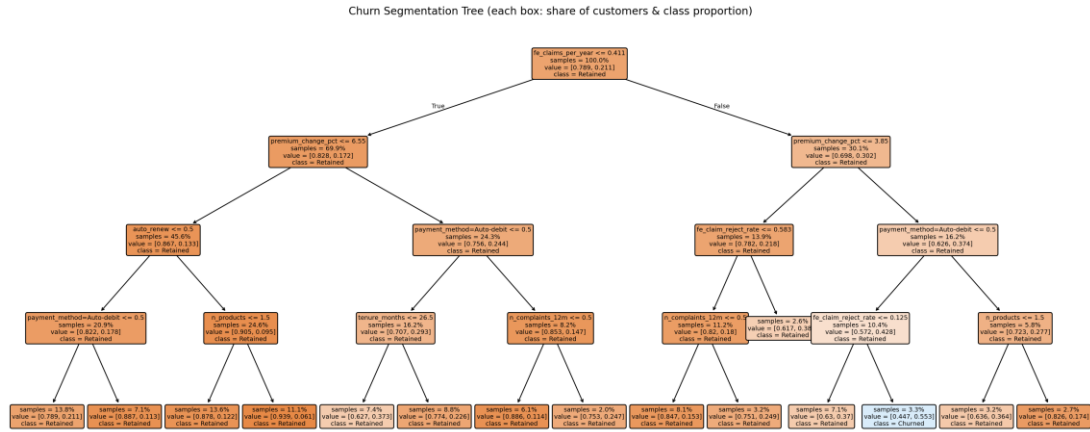


Figure C.9. Churn Segmentation Tree (each box: share of customers & class proportion)

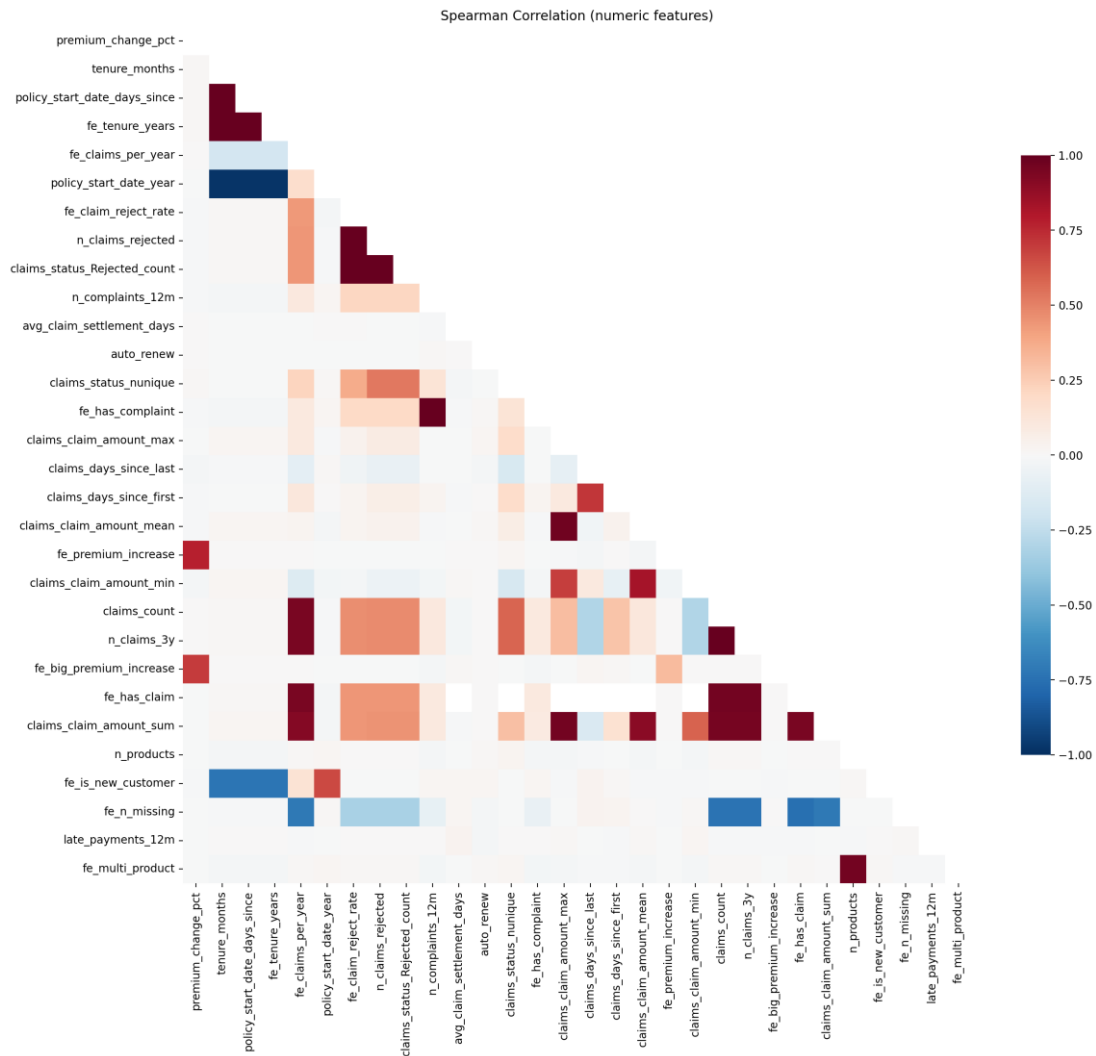


Figure C.10. Spearman Correlation (numeric features)

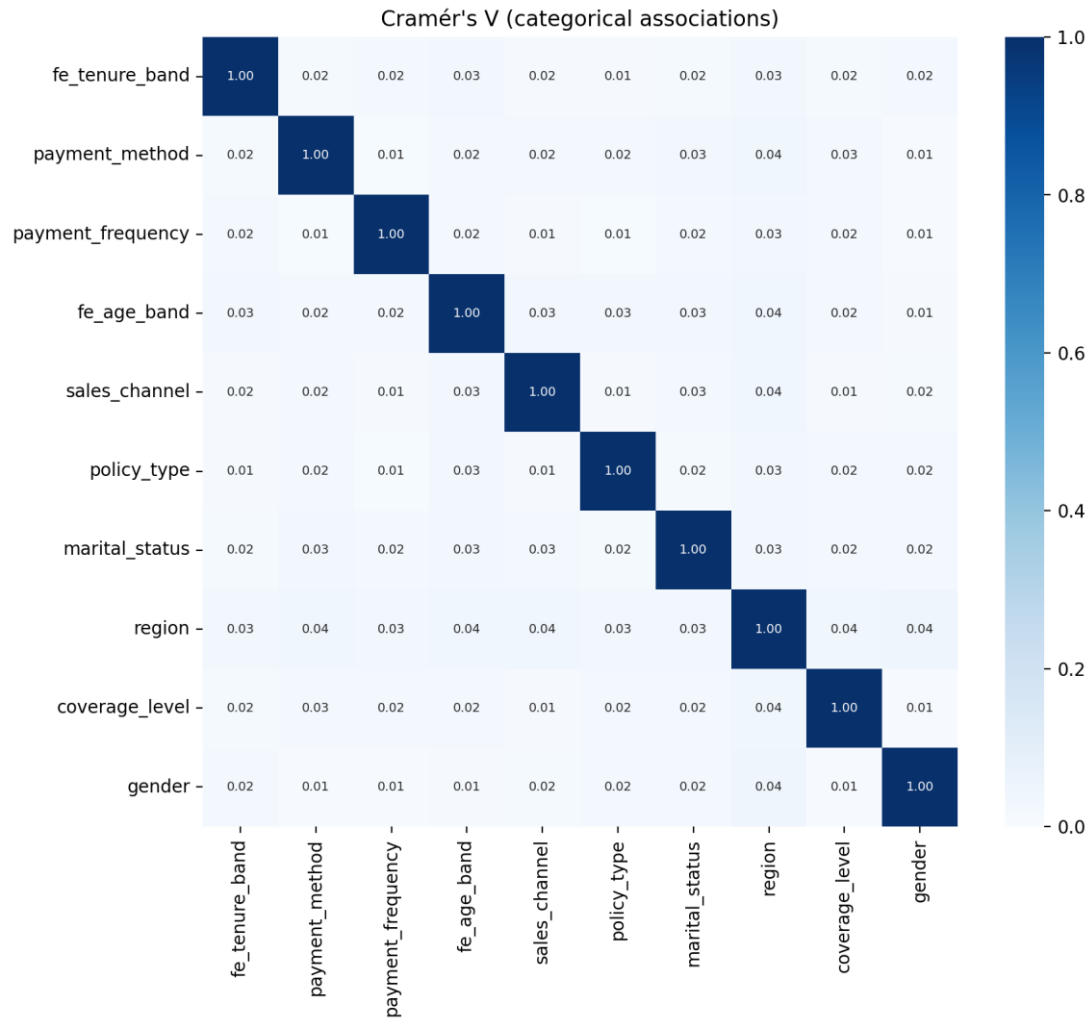


Figure C.11. Cramér's V (categorical associations)

Table C.1. Univariate driver statistics (top 25 by Information Value; q-values Benjamini–Hochberg).

Feature	Type	Iv	Iv strength	Effect metric	Effect size	Q value BH	Univariate auc	Highest risk level	Highest risk rate
premium change at last renewal (%)	numeric	0.159	medium	rank-biserial r	0.231	1.3e-44	0.615	13.2–30.6	0.335
tenure (months)	numeric	0.148	medium	rank-biserial r	-0.201	2.9e-36	0.601	1–6	0.286
policy start date (days since)	numeric	0.148	medium	rank-biserial r	-0.201	2.9e-36	0.601	31–183	0.286
tenure (years)	numeric	0.148	medium	rank-biserial r	-0.201	2.9e-36	0.601	0.0833–0.5	0.286
tenure band	categorical	0.137	medium	Cramér's V	0.142	3.7e-33	0.588	0–6m	0.286
claims per year of tenure	numeric	0.134	medium	rank-biserial r	0.180	3.6e-35	0.590	1.5–24	0.349
policy start	numeric	0.131	medium	rank-	0.191	3.4e-34	0.595	2,024–	0.291

Feature	Type	Iv	Iv strength	Effect metric	Effect size	Q value BH	Univariate auc	Highest risk level	Highest risk rate
date year				biserial r				2,025	
claim rejection rate	numeric	0.117	medium	rank-biserial r	0.122	5.1e-38	0.561	1	0.371
rejected claims	numeric	0.116	medium	rank-biserial r	0.122	2.5e-38	0.561	≥ 2	0.505
claims status Rejected count	numeric	0.116	medium	rank-biserial r	0.122	2.5e-38	0.561	≥ 2	0.505
complaints in the last 12 months	numeric	0.110	medium	rank-biserial r	0.140	3.1e-30	0.570	≥ 2	0.443
average claim settlement time (days)	numeric	0.105	medium	rank-biserial r	0.118	1.8e-07	0.559	21–28	0.346
payment method	categorical	0.098	weak	Cramér's V	0.123	1.3e-25	0.566	Bank Transfer	0.255
auto-renewal	categorical	0.084	weak	Cramér's V	0.118	1.2e-25	0.570	0	0.264
claims status nunique	categorical	0.083	weak	Cramér's V	0.120	2.1e-25	0.567	1	0.348
has complained	categorical	0.081	weak	Cramér's V	0.120	1.9e-26	0.559	1	0.294
claims claim amount max	numeric	0.081	weak	rank-biserial r	0.043	0.078	0.521	10.5M–57.7M	0.303
claims (days since) last	numeric	0.079	weak	rank-biserial r	-1.4e-04	0.995	0.500	85–176	0.291
claims (days since) first	numeric	0.078	weak	rank-biserial r	0.005	0.865	0.503	630–727	0.294
claims claim amount mean	numeric	0.078	weak	rank-biserial r	0.042	0.078	0.521	4.99M–6.3M	0.286
premium increased at renewal	categorical	0.077	weak	Cramér's V	0.108	1.3e-20	0.557	1	0.239
claims claim amount min	numeric	0.077	weak	rank-biserial r	0.029	0.242	0.515	5.35M–8.02M	0.289
claims count	numeric	0.076	weak	rank-biserial r	0.140	7.3e-23	0.570	≥ 3	0.328
claims in the last 3 years	numeric	0.076	weak	rank-biserial r	0.140	7.3e-23	0.570	≥ 3	0.328
premium increase above 10%	categorical	0.074	weak	Cramér's V	0.117	7.9e-24	0.550	1	0.307

Table C.2. All decision-tree segments.

Segment rule	N customers	Share %	Churn rate	Lift	Churners captured %
claims per year of tenure > 0.41 AND premium change at last renewal (%) >	266	3.325	0.553	2.621	8.714

Segment rule	N customers	Share %	Churn rate	Lift	Churners captured %
3.85 AND payment method $\neq$ Auto-debit AND claim rejection rate > 0.12					
claims per year of tenure > 0.41 AND premium change at last renewal (%) ≤ 3.85 AND claim rejection rate > 0.58	209	2.612	0.383	1.815	4.742
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) > 6.55 AND payment method $\neq$ Auto-debit AND tenure (months) ≤ 26.50	590	7.375	0.373	1.768	13.0
claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method $\neq$ Auto-debit AND claim rejection rate ≤ 0.12	568	7.100	0.370	1.753	12.4
claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method = Auto-debit AND number of ANS products ≤ 1.50	253	3.163	0.364	1.724	5.453
claims per year of tenure > 0.41 AND premium change at last renewal (%) ≤ 3.85 AND claim rejection rate ≤ 0.58 AND complaints in the last 12 months > 0.50	253	3.163	0.249	1.181	3.734
claims per year of tenure ≤ 0.41 AND premium	162	2.025	0.247	1.171	2.371

Segment rule	N customers	Share %	Churn rate	Lift	Churners captured %
change at last renewal (%) > 6.55 AND payment method = Auto-debit AND complaints in the last 12 months > 0.50					
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) > 6.55 AND payment method $\neq$ Auto-debit AND tenure (months) > 26.50	705	8.812	0.226	1.070	9.425
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) ≤ 6.55 AND auto-renewal = 0 AND payment method $\neq$ Auto-debit	1,105	13.8	0.211	1.000	13.8
claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method = Auto-debit AND number of ANS products > 1.50	213	2.663	0.174	0.824	2.193
claims per year of tenure > 0.41 AND premium change at last renewal (%) ≤ 3.85 AND claim rejection rate ≤ 0.58 AND complaints in the last 12 months ≤ 0.50	647	8.088	0.153	0.726	5.868
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) ≤ 6.55 AND auto-renewal = 1 AND number of ANS products ≤ 1.50	1,087	13.6	0.122	0.580	7.884

Segment rule	N customers	Share %	Churn rate	Lift	Churners captured %
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) > 6.55 AND payment method = Auto-debit AND complaints in the last 12 months ≤ 0.50	490	6.125	0.114	0.542	3.320
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) ≤ 6.55 AND auto-renewal = 0 AND payment method = Auto-debit	567	7.087	0.113	0.535	3.794
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) ≤ 6.55 AND auto-renewal = 1 AND number of ANS products > 1.50	885	11.1	0.061	0.289	3.201

Table C.3. Variance inflation factors (numeric features).

Feature	VIF
tenure (months)	1,000,799.92B
tenure (years)	1,000,799.92B
claims status Rejected count	1,000,799.92B
rejected claims	1,000,799.92B
claims in the last 3 years	1,000,799.92B
claims count	1,000,799.92B
policy start date (days since)	6.57M
claims claim amount mean	262.4
claims claim amount max	100.1
claims claim amount min	73.5
policy start date year	47.1
claims claim amount sum	34.4
has made a claim	14.1
claim rejection rate	12.8
complaints in the last 12 months	5.996
has complained	5.874
claims (days since) last	5.579
claims (days since) first	5.540
premium change at last renewal (%)	5.055
holds multiple products	3.635
number of ANS products	3.633
premium increased at renewal	2.869
claims status nunique	2.593

Feature	VIF
premium increase above 10%	2.463
claims per year of tenure	2.119
number of missing fields	2.053
new customer (<12 months)	1.630
average claim settlement time (days)	1.026
auto-renewal	1.005
late payments in the last 12 months	1.002

Table C.4. Stated churn reasons (share of churners; descriptive only, the variable is excluded from modelling).

Reason	N churners	Share %
Moved	349	20.7
Price	347	20.6
Competitor offer	339	20.1
Service	330	19.6
Claim issue	322	19.1

Appendix D. Interpretable Logistic Regression

Adjusted odds ratios with 95% confidence intervals; numeric variables are standardised (per 1 SD), categories are compared with the most common level.

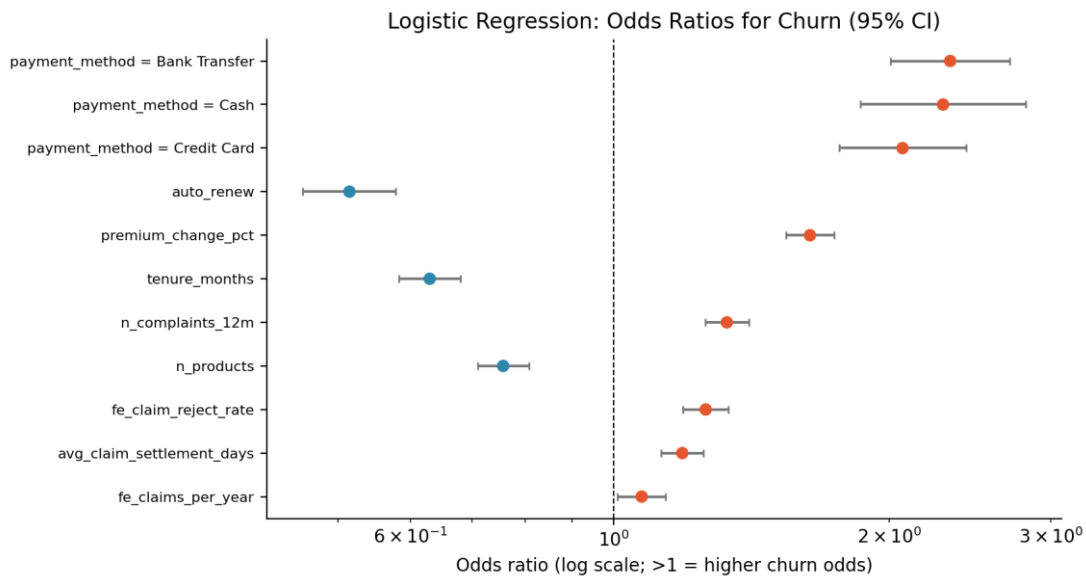


Figure D.1. Logistic Regression: Odds Ratios for Churn (95% CI)

Table D.1. Logistic regression: odds ratios, 95% CI, p-values and average marginal effects.

Term	Coef	Odds ratio	Or ci low	Or ci high	P value	Scale	Avg marginal effect pp	Interpretation
payment method = Bank Transfer	0.846	2.330	2.005	2.709	3.3e-28	vs 'Auto-debit'	12.1	+133% odds of churn (vs 'Auto-debit')

Term	Coef	Odds ratio	Or ci low	Or ci high	P value	Scale	Avg marginal effect pp	Interpretation
payment method = Cash	0.828	2.289	1.860	2.817	5.5e-15	vs 'Auto-debit'	11.8	+129% odds of churn (vs 'Auto-debit')
payment method = Credit Card	0.726	2.067	1.762	2.425	4.7e-19	vs 'Auto-debit'	10.4	+107% odds of churn (vs 'Auto-debit')
auto-renewal	-0.665	0.514	0.458	0.578	7.6e-29	binary (0→1)	-9.486	-49% odds of churn (binary (0→1))
premium change at last renewal (%)	0.493	1.638	1.541	1.741	9.5e-57	per 1 SD (= 6.99)	7.040	+64% odds of churn (per 1 SD (= 6.99))
tenure (months)	-0.463	0.629	0.582	0.680	1.1e-31	per 1 SD (= 23.6)	-6.606	-37% odds of churn (per 1 SD (= 23.6))
complaints in the last 12 months	0.284	1.329	1.258	1.403	2.2e-24	per 1 SD (= 0.573)	4.056	+33% odds of churn (per 1 SD (= 0.573))
number of ANS products	-0.278	0.757	0.710	0.808	3.1e-17	per 1 SD (= 0.77)	-3.972	-24% odds of churn (per 1 SD (= 0.77))
claims status nunique	0.236	1.266	0.937	1.710	0.125	binary (0→1)	3.363	+27% odds of churn (binary (0→1))
claim rejection rate	0.231	1.260	1.189	1.335	3.9e-15	per 1 SD (= 0.291)	3.297	+26% odds of churn (per 1 SD (= 0.291))
average claim settlement time (days)	0.172	1.188	1.126	1.253	2.3e-10	per 1 SD (= 6.49)	2.456	+19% odds of churn (per 1 SD (= 6.49))
claims per year of tenure	0.070	1.073	1.009	1.140	0.024	per 1 SD (= 1.11)	1.000	+7% odds of churn (per 1 SD (= 1.11))
number of missing fields	-0.043	0.958	0.896	1.024	0.211	per 1 SD (= 0.698)	-0.608	-4% odds of churn (per 1 SD (= 0.698))
claims claim amount min	0.037	1.038	0.951	1.133	0.403	per 1 SD (= 2.64e+06)	0.534	+4% odds of churn (per 1 SD (= 2.64e+06))
claims (days since) last	0.026	1.026	0.930	1.133	0.606	per 1 SD (= 208)	0.371	+3% odds of churn (per 1 SD (= 208))
claims (days since) first	-0.015	0.985	0.892	1.088	0.770	per 1 SD (= 208)	-0.211	-1% odds of churn (per 1 SD (= 208))
claims claim amount max	0.012	1.012	0.924	1.107	0.802	per 1 SD (= 3.37e+06)	0.165	+1% odds of churn (per 1 SD (= 3.37e+06))

Appendix E. Survival Analysis

Kaplan–Meier estimates, hazards by tenure, Cox and Weibull AFT models, proportional-hazards diagnostics and validation of risk tiers over time.

Key observations. Persistency ratios, the standard insurance KPI for the share of policies still in force: 13-month 91.5% and 25-month 82.0%.

Schoenfeld tests reject proportional hazards for 0 of 14 terms; the Cox hazard ratios can be read as constant over tenure.

9 Cox terms are significant at 5%; the largest hazard ratio is 1.83 (95% CI 1.61–2.07) for payment method = Bank Transfer.

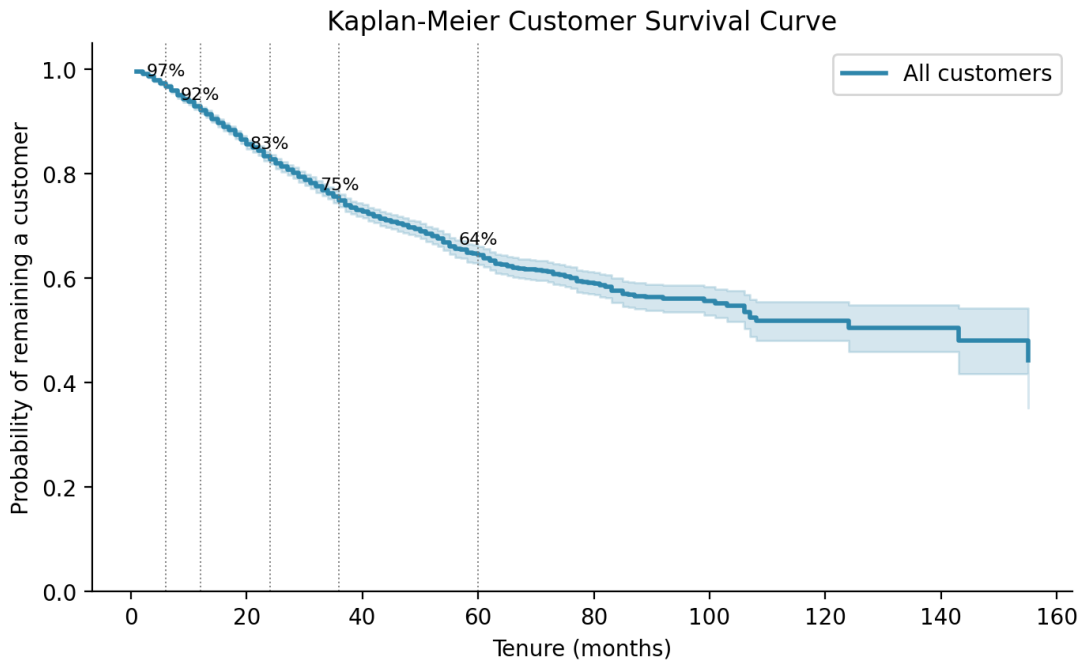


Figure E.1. Kaplan-Meier Customer Survival Curve

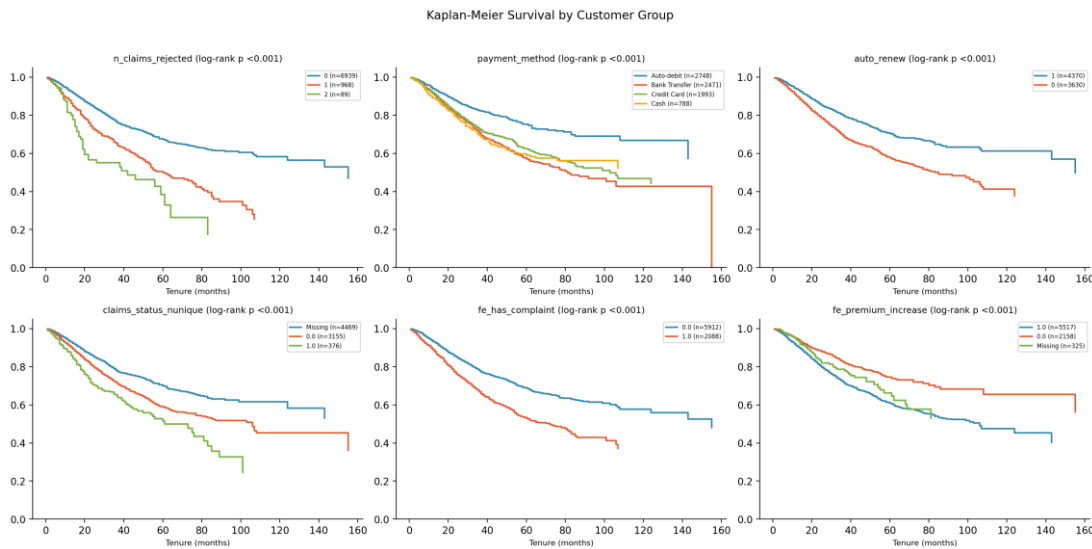


Figure E.2. Kaplan-Meier Survival by Customer Group

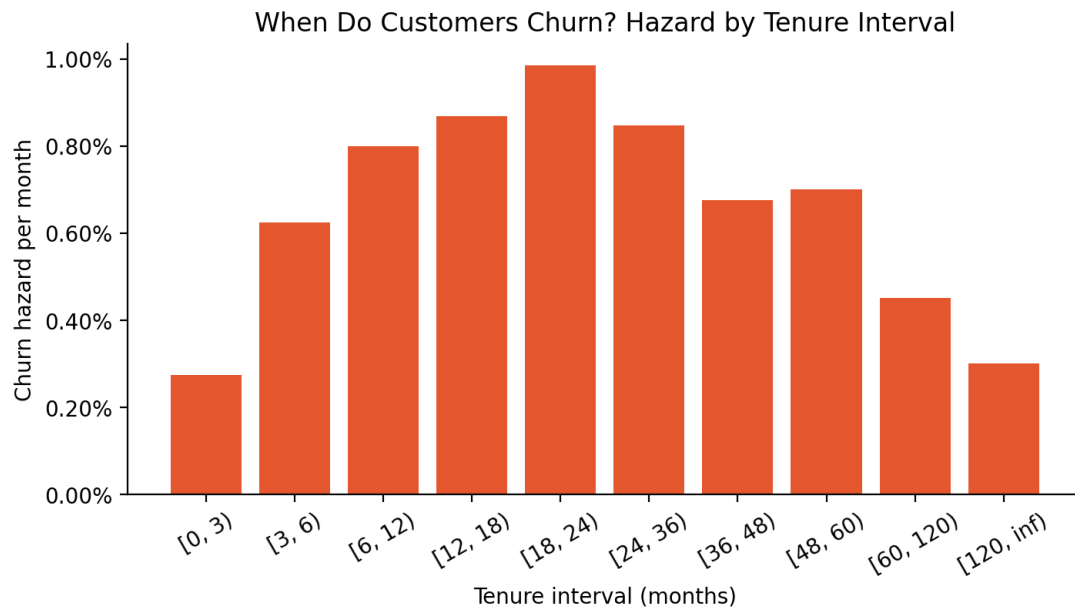


Figure E.3. When Do Customers Churn? Hazard by Tenure Interval

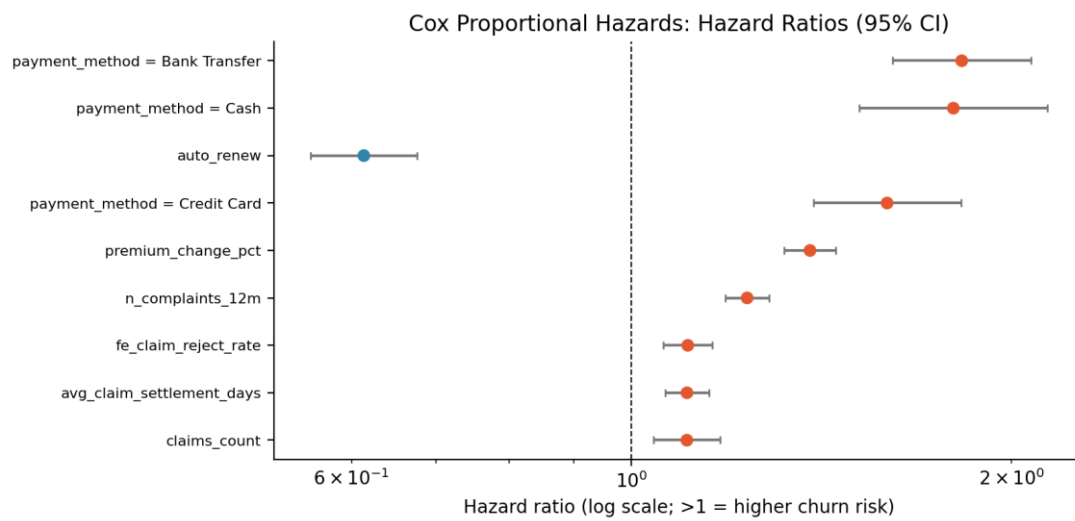


Figure E.4. Cox Proportional Hazards: Hazard Ratios (95% CI)

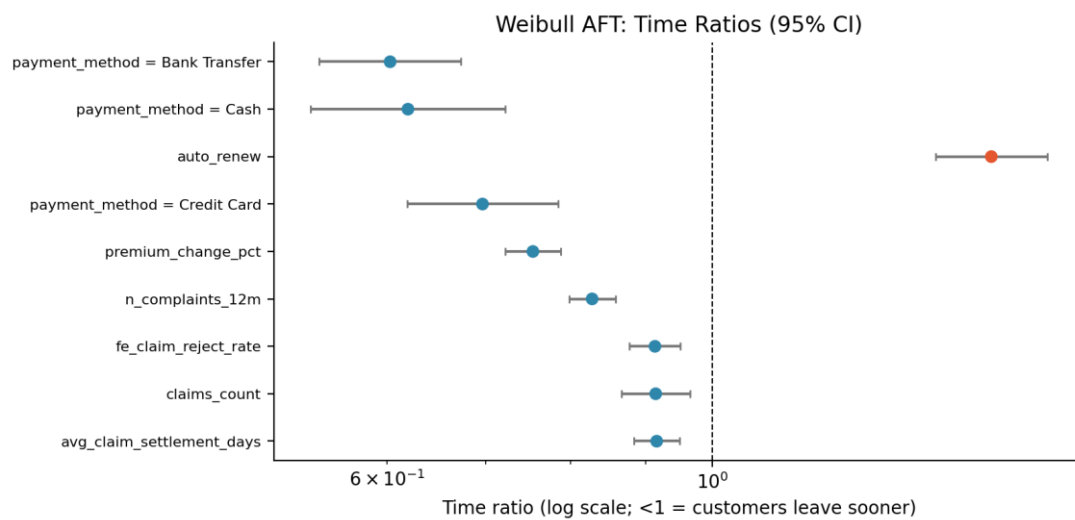


Figure E.5. Weibull AFT: Time Ratios (95% CI)

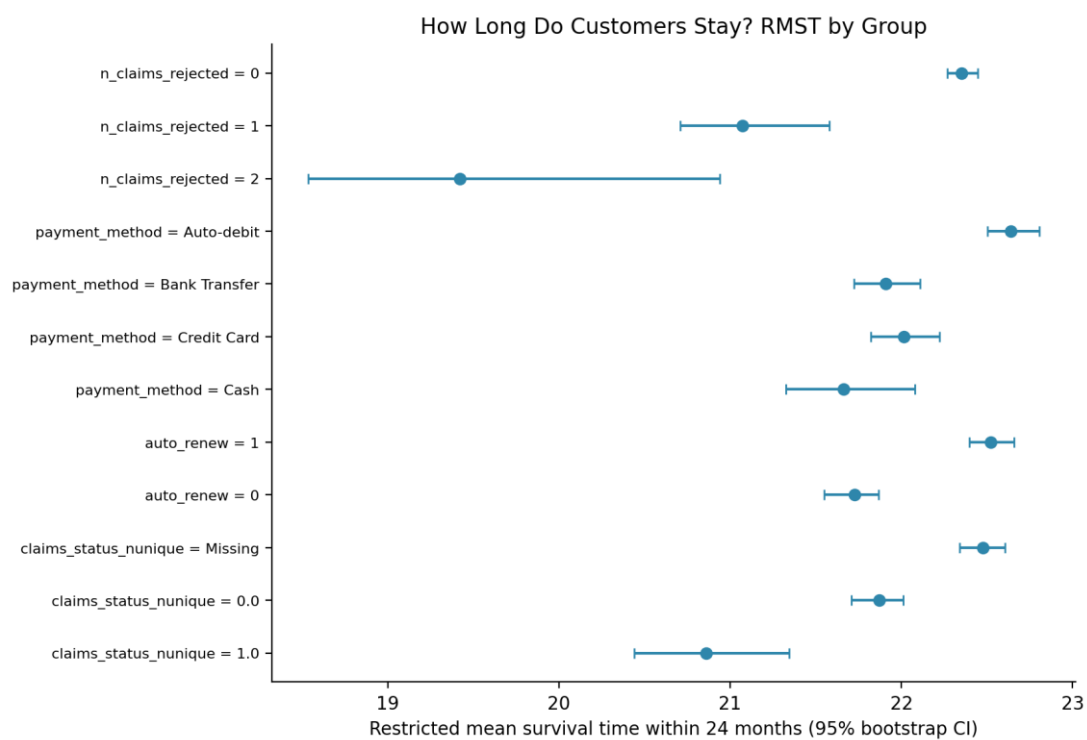


Figure E.6. How Long Do Customers Stay? RMST by Group

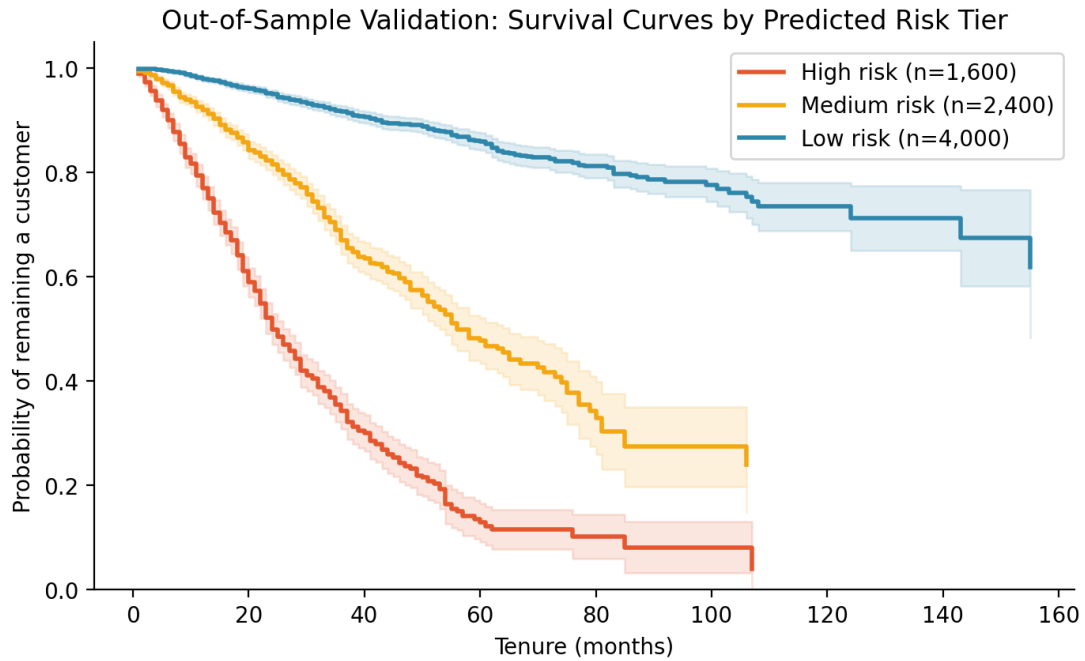


Figure E.7. Out-of-Sample Validation: Survival Curves by Predicted Risk Tier

Table E.1. Hazard by tenure interval.

Interval	At risk start	Churn events	Person time	Hazard
[0, 3)	8,000	65	23,681	0.003
[3, 6)	7,781	140	22,408	0.006
[6, 12)	7,296	318	39,736	0.008
[12, 18)	6,134	283	32,571	0.009
[18, 24)	4,940	256	25,987	0.010
[24, 36)	3,938	310	36,575	0.008
[36, 48)	2,365	145	21,463	0.007
[48, 60)	1,394	91	12,989	0.007
[60, 120)	848	76	16,858	0.005
[120, inf)	47	3	997.0	0.003

Table E.2. Log-rank tests by group.

Feature	Logrank chi2	P value
rejected claims	130.1	0
payment method	94.8	0
auto-renewal	90.7	0
claims status nunique	83.5	0
has complained	107.4	0
premium increased at renewal	72.2	2.2e-16

Table E.3. Cox proportional hazards model.

Term	Coef	Hazard ratio	Hr ci low	Hr ci high	P value	Scale	Interpretation
payment method = Bank Transfer	0.603	1.827	1.610	2.073	9.6e-21	vs 'Auto-debit'	+83% churn hazard (vs 'Auto-debit')

Term	Coef	Hazard ratio	Hr ci low	Hr ci high	P value	Scale	Interpretation
payment method = Cash	0.587	1.799	1.516	2.135	1.8e-11	vs 'Auto-debit'	+80% churn hazard (vs 'Auto-debit')
auto-renewal	-0.488	0.614	0.557	0.676	4.0e-23	binary (0→1)	-39% churn hazard (binary (0→1))
payment method = Credit Card	0.466	1.594	1.393	1.824	1.1e-11	vs 'Auto-debit'	+59% churn hazard (vs 'Auto-debit')
premium change at last renewal (%)	0.326	1.385	1.321	1.452	3.6e-41	per 1 SD (= 6.99)	+39% churn hazard (per 1 SD (= 6.99))
complaints in the last 12 months	0.211	1.235	1.187	1.285	1.9e-25	per 1 SD (= 0.573)	+23% churn hazard (per 1 SD (= 0.573))
claim rejection rate	0.103	1.108	1.060	1.159	5.7e-06	per 1 SD (= 0.291)	+11% churn hazard (per 1 SD (= 0.291))
average claim settlement time (days)	0.101	1.107	1.063	1.151	5.9e-07	per 1 SD (= 6.49)	+11% churn hazard (per 1 SD (= 6.49))
claims count	0.101	1.106	1.041	1.176	0.001	per 1 SD (= 0.777)	+11% churn hazard (per 1 SD (= 0.777))
claims (days since) last	0.042	1.043	0.966	1.126	0.278	per 1 SD (= 208)	+4% churn hazard (per 1 SD (= 208))
claims (days since) first	-0.041	0.960	0.889	1.036	0.296	per 1 SD (= 208)	-4% churn hazard (per 1 SD (= 208))
claims status nunique	0.034	1.035	0.817	1.311	0.778	binary (0→1)	+3% churn hazard (binary (0→1))
claims claim amount min	0.016	1.016	0.953	1.083	0.628	per 1 SD (= 2.64e+06)	+2% churn hazard (per 1 SD (= 2.64e+06))
claims claim amount max	1.7e-04	1.000	0.938	1.066	0.996	per 1 SD (= 3.37e+06)	+0% churn hazard (per 1 SD (= 3.37e+06))

Table E.4. Schoenfeld test of the proportional-hazards assumption.

Term	Test statistic	P	Ph violated
auto-renewal	0.370	0.543	no
average claim settlement time (days)	0.061	0.804	no
claims claim amount max	1.976	0.160	no
claims claim amount min	1.375	0.241	no
claims count	0.026	0.871	no
claims (days since) first	1.184	0.277	no
claims (days since) last	1.272	0.259	no
claims status nunique	0.498	0.480	no
claim rejection rate	0.038	0.846	no
complaints in the last 12 months	9.8e-04	0.975	no
payment method = Bank Transfer	5.992	0.014	no
payment method = Cash	0.141	0.707	no
payment method = Credit Card	0.730	0.393	no
premium change at last renewal (%)	0.436	0.509	no

Table E.5. Weibull accelerated failure time model (time ratios).

Term	Time ratio	Tr ci low	Tr ci high	P value	Scale
payment method = Bank Transfer	0.603	0.539	0.674	4.5e-19	vs 'Auto-debit'
payment method = Cash	0.620	0.532	0.722	7.9e-10	vs 'Auto-debit'
auto-renewal	1.549	1.419	1.691	1.3e-22	binary (0→1)
payment method = Credit Card	0.697	0.619	0.784	2.2e-09	vs 'Auto-debit'
premium change at last renewal (%)	0.754	0.722	0.788	2.1e-36	per 1 SD (= 6.99)
complaints in the last 12 months	0.828	0.798	0.859	2.8e-24	per 1 SD (= 0.573)
claim rejection rate	0.914	0.878	0.951	9.4e-06	per 1 SD (= 0.291)
claims count	0.915	0.867	0.965	0.001	per 1 SD (= 0.777)
average claim settlement time (days)	0.916	0.883	0.950	2.2e-06	per 1 SD (= 6.49)
claims status nunique	0.961	0.779	1.185	0.709	binary (0→1)
claims (days since) first	1.031	0.966	1.099	0.360	per 1 SD (= 208)
claims (days since) last	0.972	0.912	1.037	0.387	per 1 SD (= 208)
claims claim amount min	0.983	0.931	1.038	0.539	per 1 SD (= 2.64e+06)
claims claim amount max	1.003	0.951	1.058	0.920	per 1 SD (= 3.37e+06)

Table E.6. Restricted mean survival time by group.

Feature	Level	N	Rmst	Rmst ci low	Rmst ci high	Diff vs best	Reference level	Tau
rejected claims	0	6,939	22.4	22.3	22.4	0	0	24.0
rejected claims	1	968	21.1	20.7	21.6	-1.281	0	24.0
rejected claims	2	89	19.4	18.5	20.9	-2.930	0	24.0
payment method	Auto-debit	2,748	22.6	22.5	22.8	0	Auto-debit	24.0
payment method	Bank Transfer	2,471	21.9	21.7	22.1	-0.730	Auto-debit	24.0
payment method	Credit Card	1,993	22.0	21.8	22.2	-0.624	Auto-debit	24.0
payment method	Cash	788	21.7	21.3	22.1	-0.976	Auto-debit	24.0
auto-renewal	1	4,370	22.5	22.4	22.7	0	1	24.0
auto-renewal	0	3,630	21.7	21.5	21.9	-0.795	1	24.0
claims status nunique	Missing	4,469	22.5	22.3	22.6	0	Missing	24.0
claims status nunique	0.0	3,155	21.9	21.7	22.0	-0.604	Missing	24.0

Feature	Level	N	Rmst	Rmst ci low	Rmst ci high	Diff vs best	Reference level	Tau
claims status nunique	1.0	376	20.9	20.4	21.3	-1.614	Missing	24.0

Table E.7. Survival probability by predicted risk tier.

Tier	N	S(6)	S(12)	S(24)	S(36)	S(60)
High	1,600	0.900	0.770	0.500	0.342	0.129
Medium	2,400	0.968	0.920	0.815	0.671	0.478
Low	4,000	0.994	0.980	0.950	0.917	0.859

Appendix F. Model Development, Comparison and Validation

All logged experiments (trial-and-error), statistical comparison of models, ablation study and diagnostic checks.

Key observations. The development journey logged 32 experiments. The best default-parameter model reached ROC-AUC 0.757; tuning, multi-seed averaging, feature selection and ensembling raised the final out-of-fold AUC to 0.760 (+0.003).

In the ablation study, removing premium_change costs the most accuracy (Δ AUC 0.033, Holm-adjusted $p < 0.001$).

The largest train-test shift is PSI 0.275 (agent id); this feature should be monitored after deployment.

Across segments with at least 100 customers, AUC ranges from 0.734 to 0.841; the weakest segment is tenure band = 6-12m ($n = 1,184$).

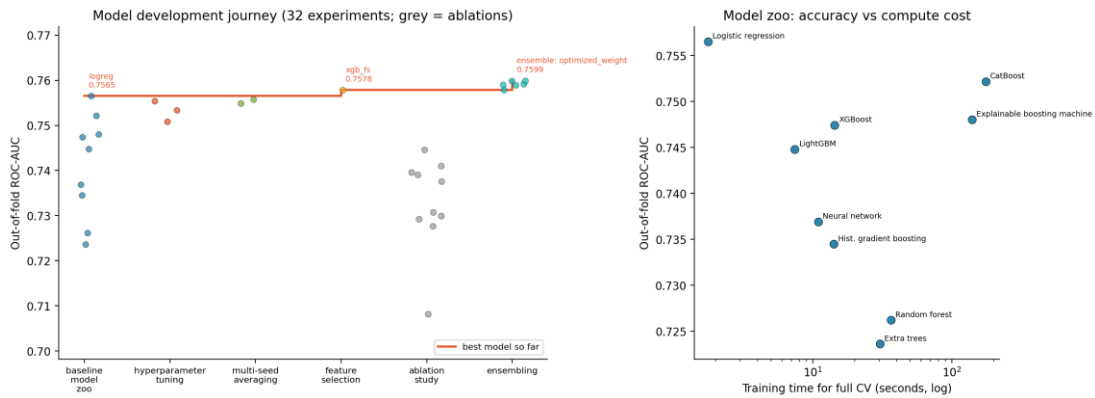


Figure F.1. Model development journey (32 experiments; grey = ablations)

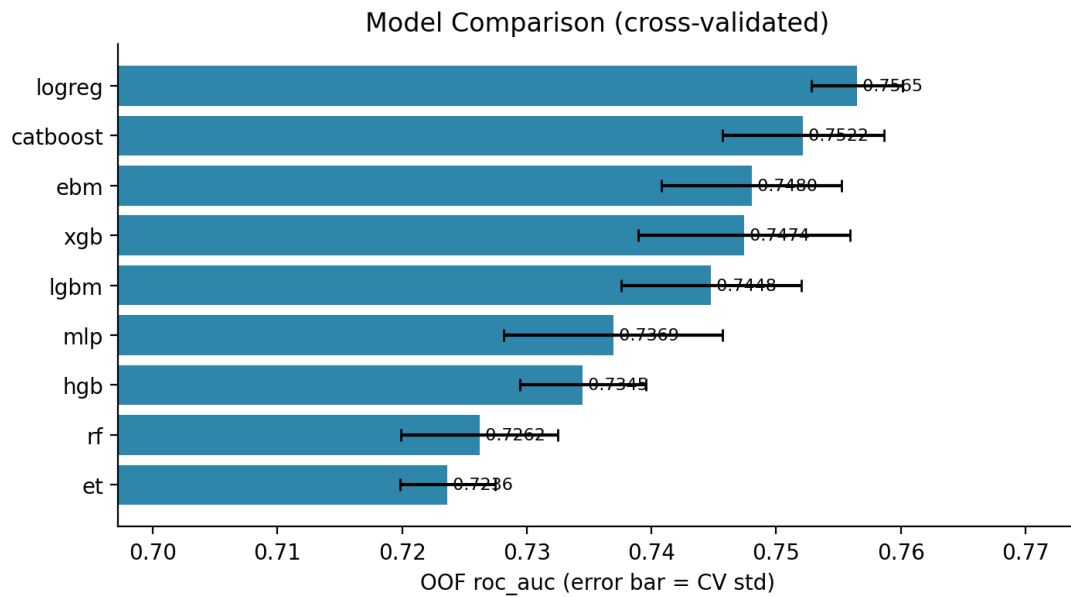


Figure F.2. Model Comparison (cross-validated)

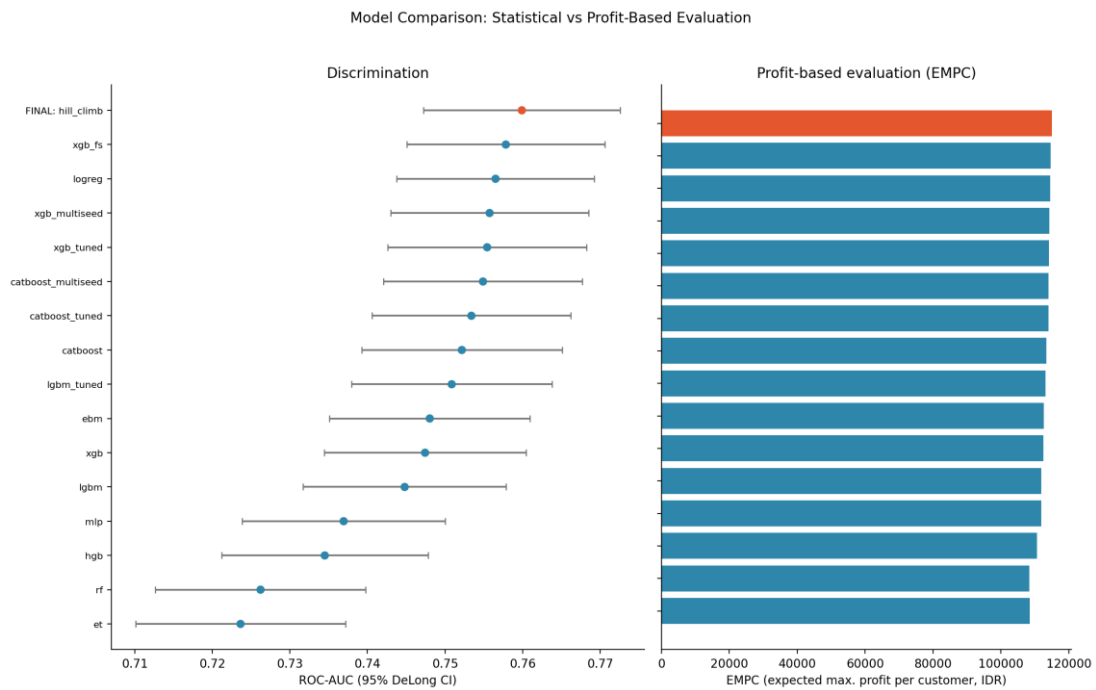


Figure F.3. Model Comparison: Statistical vs Profit-Based Evaluation

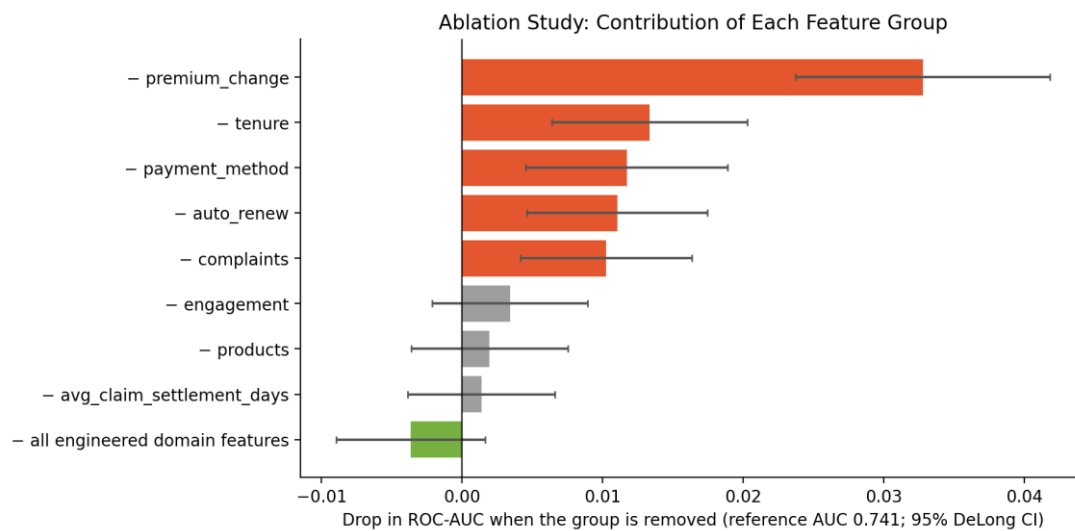


Figure F.4. Ablation Study: Contribution of Each Feature Group

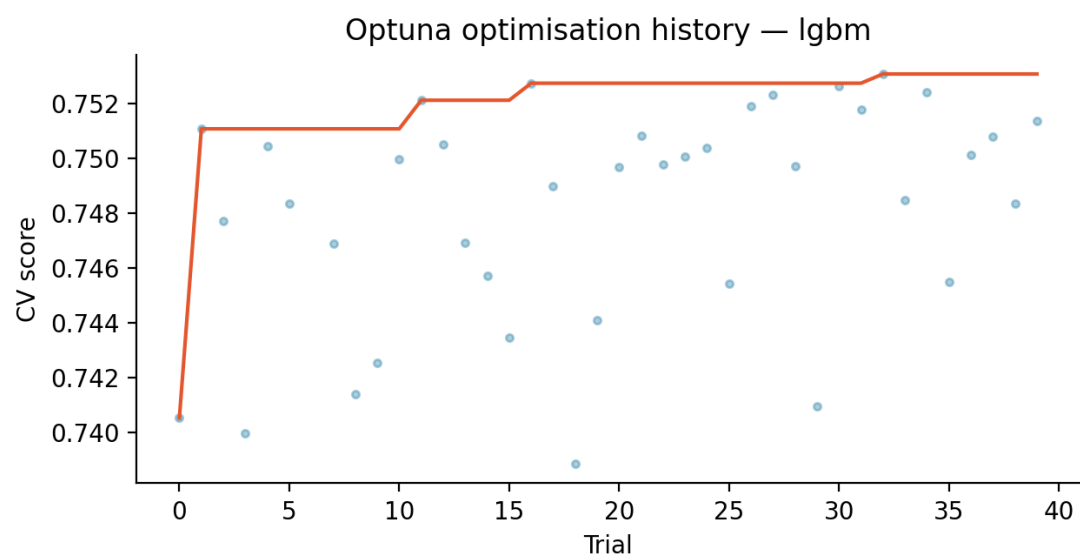


Figure F.5. Optuna optimisation history — lgbm

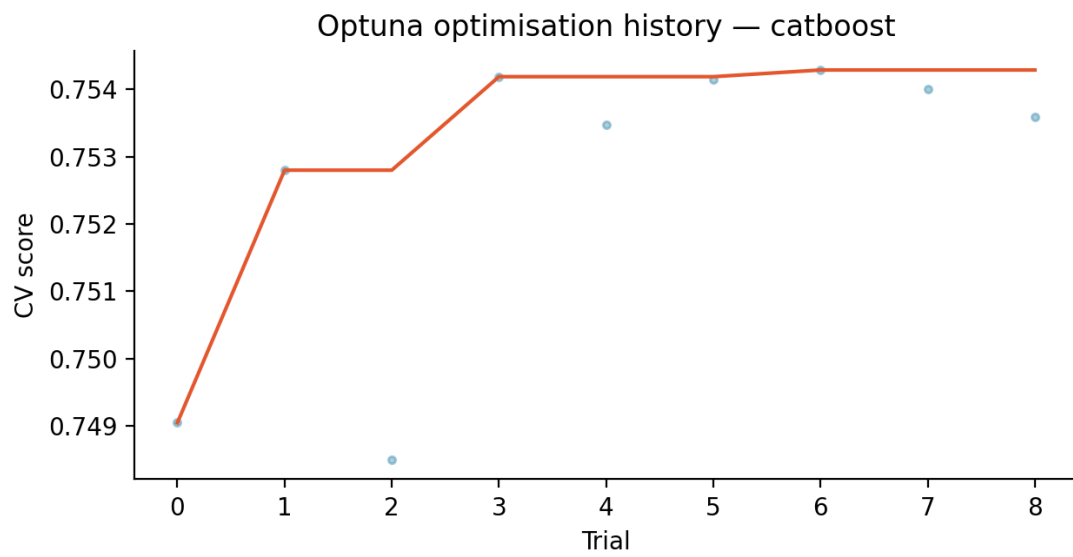


Figure F.6. Optuna optimisation history — catboost

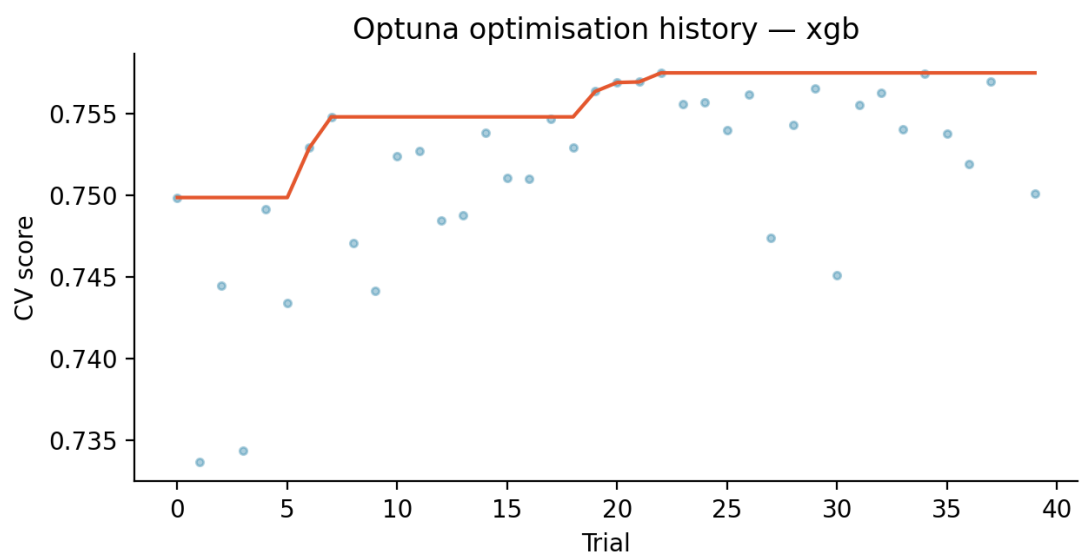


Figure F.7. Optuna optimisation history — xgb

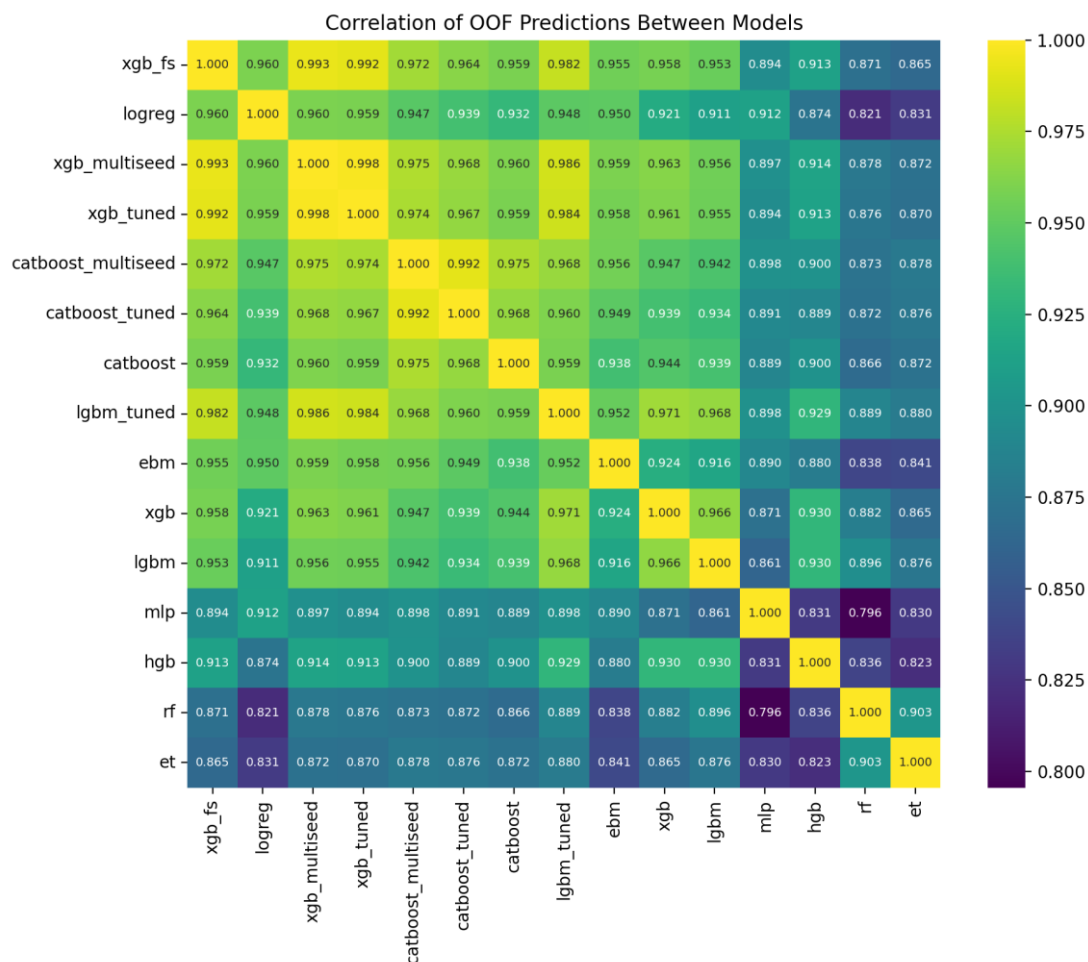


Figure F.8. Correlation of OOF Predictions Between Models

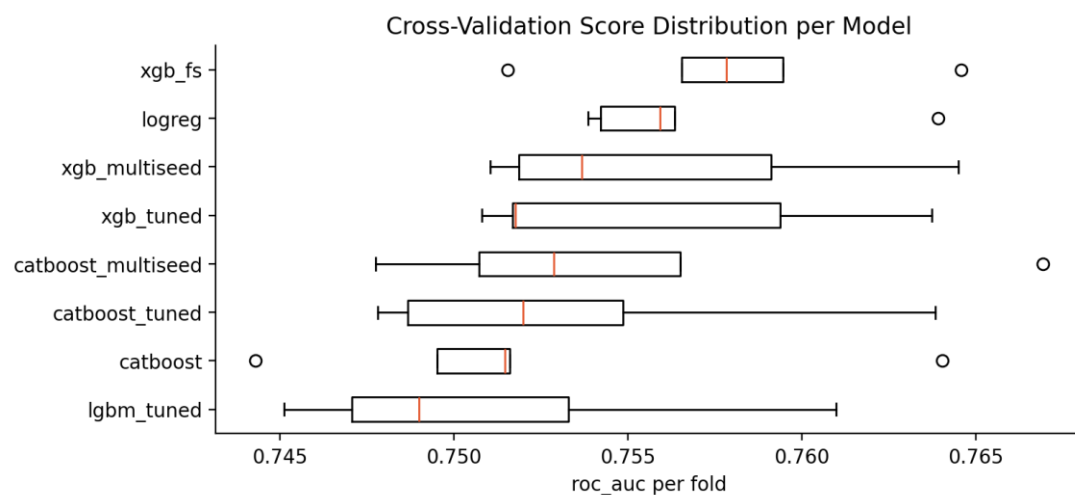


Figure F.9. Cross-Validation Score Distribution per Model

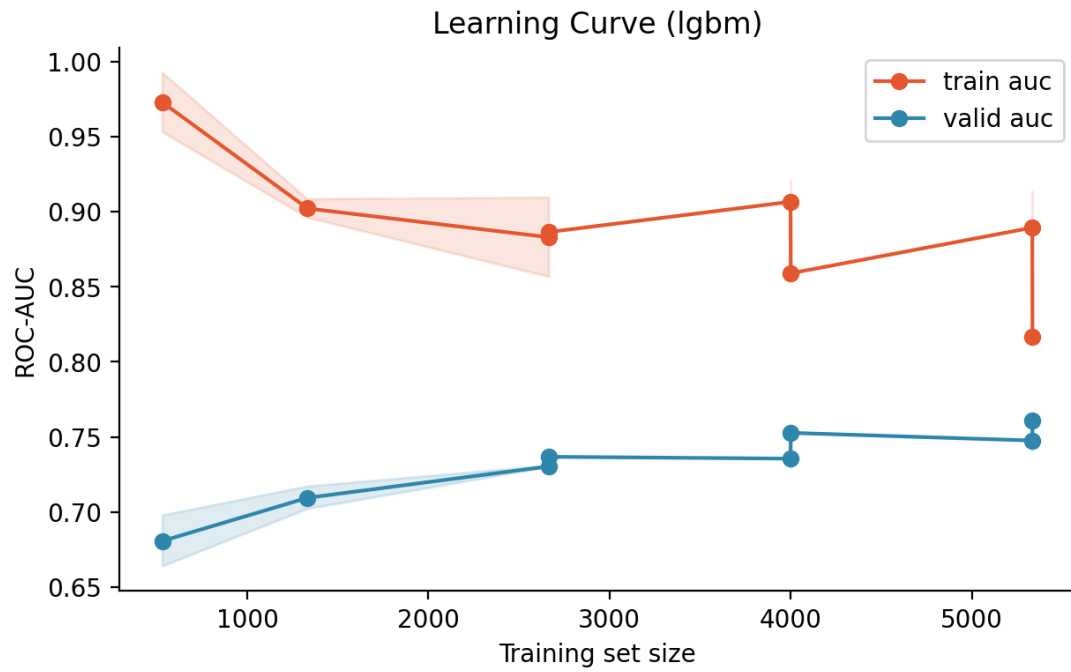


Figure F.10. Learning Curve (lgbm)

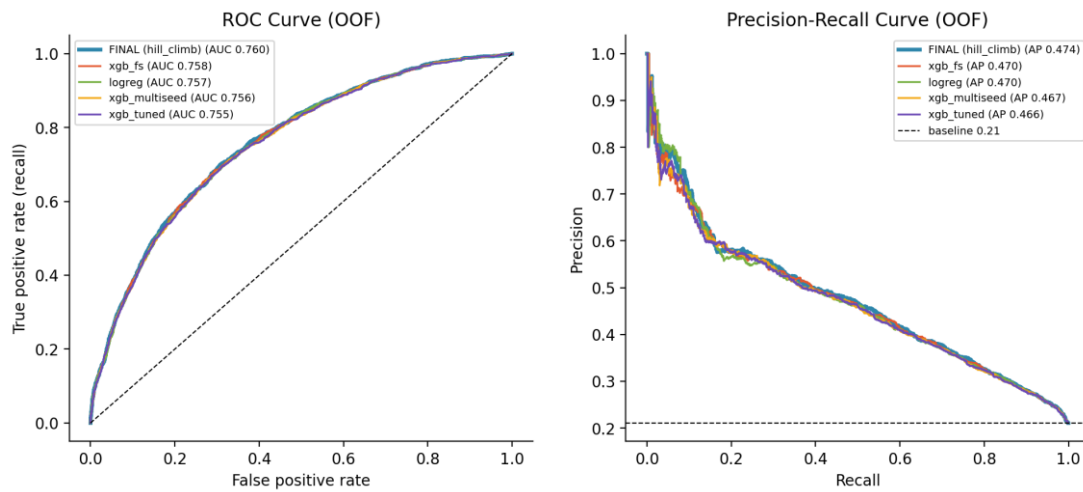


Figure F.11. ROC Curve (OOF)

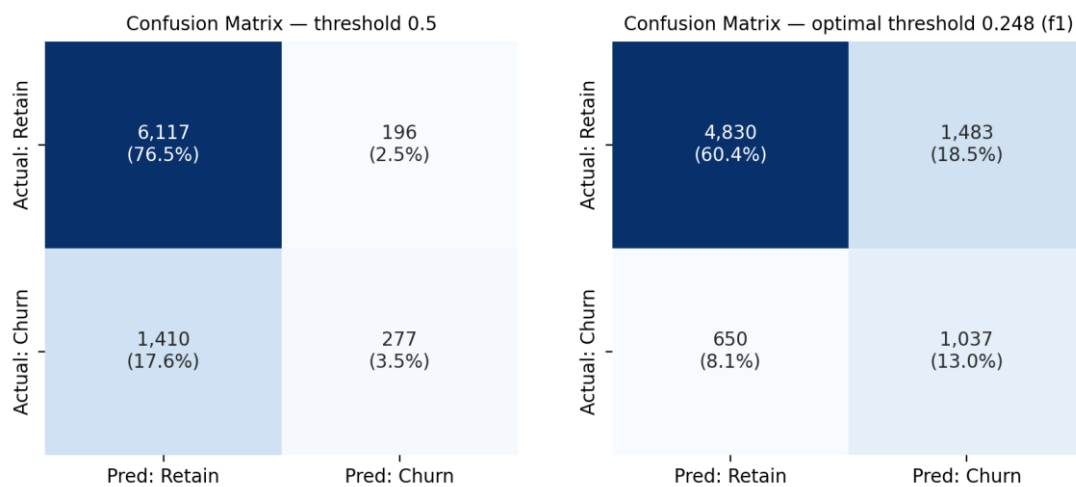


Figure F.12. Confusion Matrix — threshold 0.5

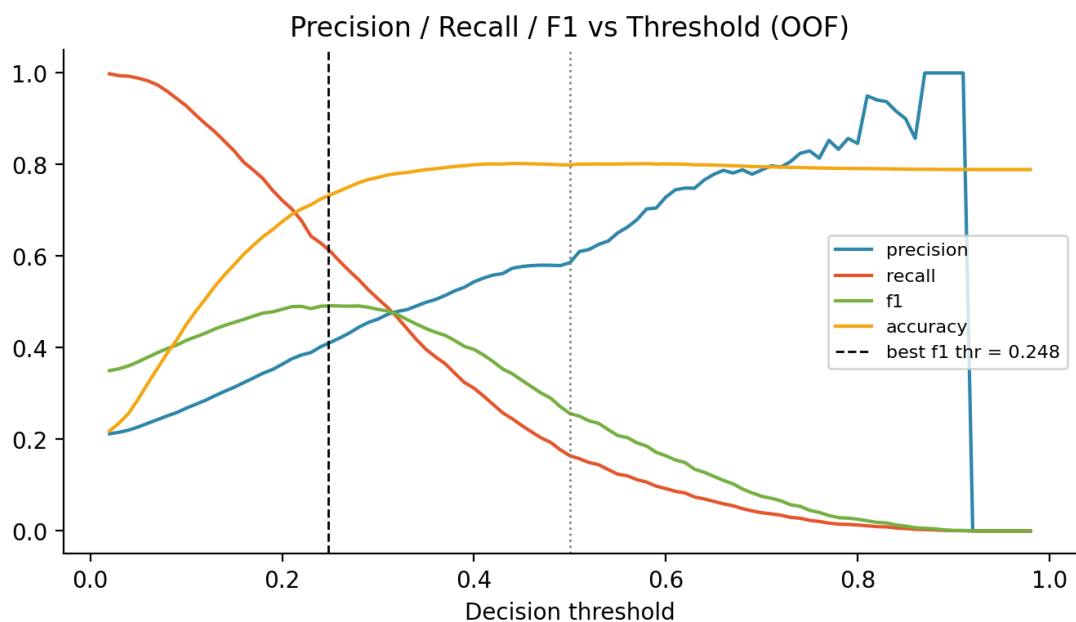


Figure F.13. Precision / Recall / F1 vs Threshold (OOF)

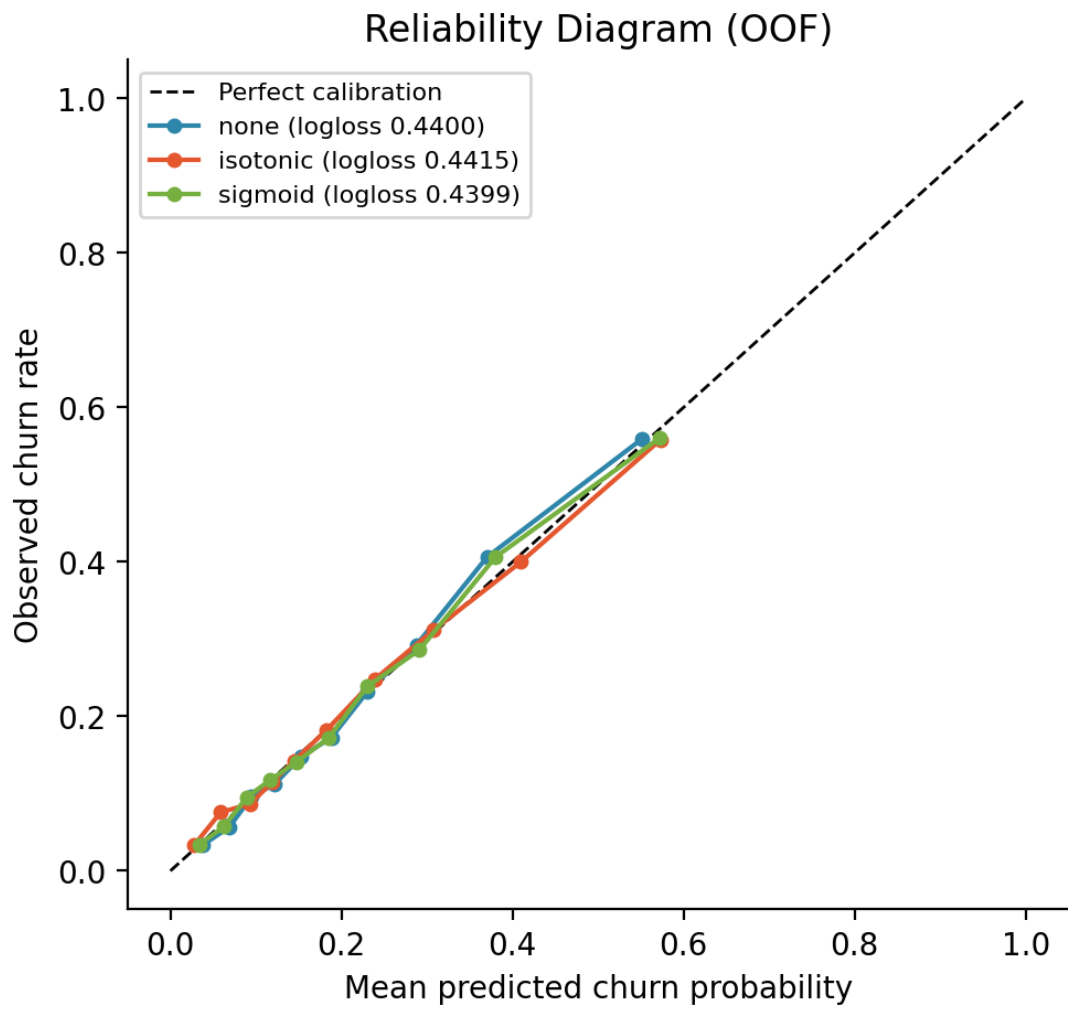


Figure F.14. Reliability Diagram (OOF)

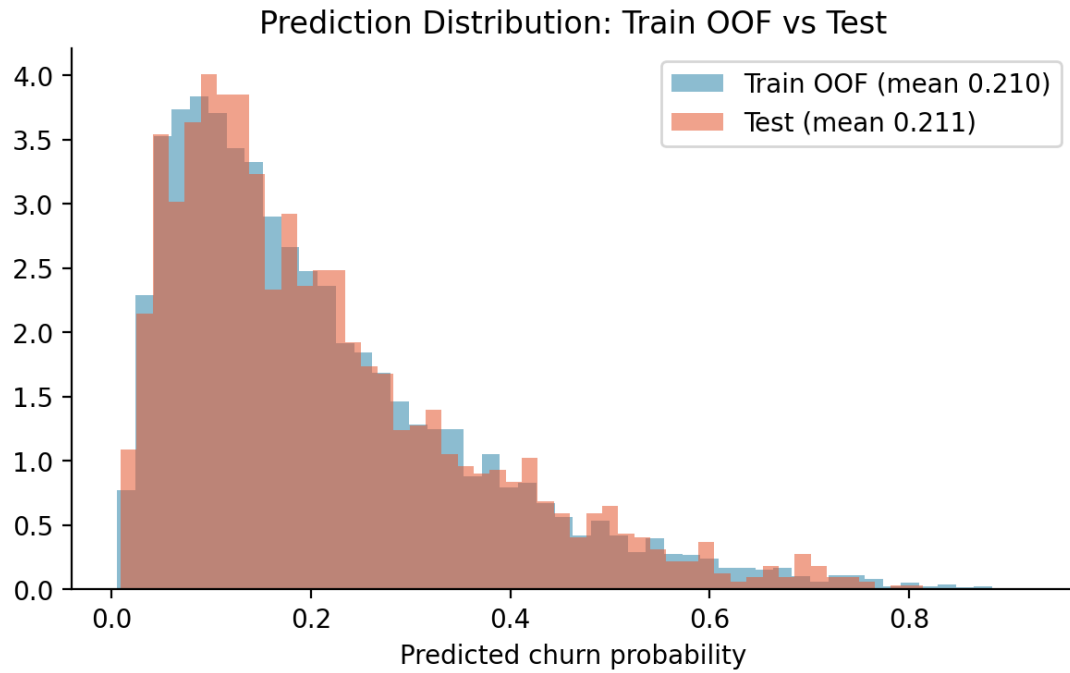


Figure F.15. Prediction Distribution: Train OOF vs Test

Table F.1. Highest-risk segments and the most loyal segment.

Segment (decision-tree rule)	Customers	Churn rate	Lift
claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method ≠ Auto-debit AND claim rejection rate > 0.12	266	55.3%	2.6×
claims per year of tenure > 0.41 AND premium change at last renewal (%) ≤ 3.85 AND claim rejection rate > 0.58	209	38.3%	1.8×
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) > 6.55 AND payment method ≠ Auto-debit AND tenure (months) ≤ 26	590	37.3%	1.8×
claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) ≤ 6.55 AND auto-renewal = Yes AND number of ANS products ≥ 2	885	6.1%	0.3×

Table F.2. Full model comparison: AUC with DeLong CI, PR-AUC, Brier, ECE, KS and EMPC.

Model	Roc auc	Auc ci low	Auc ci high	Pr auc	Brier	Ece	Ks	Empe per customer	Delong p holm
FINAL: hill_climb	0.760	0.747	0.773	0.474	0.140	0.010	0.392	114,994	
xgb fs	0.758	0.745	0.771	0.470	0.141	0.013	0.390	114,608	0.022
logreg	0.757	0.744	0.769	0.470	0.141	0.012	0.382	114,505	0.016
xgb multiseed	0.756	0.743	0.769	0.467	0.142	0.015	0.384	114,240	2.6e-04
xgb tuned	0.755	0.743	0.768	0.466	0.142	0.015	0.383	114,163	3.0e-04
catboost multiseed	0.755	0.742	0.768	0.468	0.141	0.014	0.380	114,026	0.002
catboost tuned	0.753	0.741	0.766	0.463	0.142	0.014	0.371	113,994	6.0e-04
catboost	0.752	0.739	0.765	0.466	0.142	0.012	0.380	113,417	2.6e-04
lgbm tuned	0.751	0.738	0.764	0.462	0.142	0.012	0.379	113,073	5.0e-09
ebm	0.748	0.735	0.761	0.461	0.142	0.004	0.365	112,655	8.6e-08
xgb	0.747	0.734	0.760	0.453	0.143	0.011	0.374	112,499	6.4e-08
lgbm	0.745	0.732	0.758	0.450	0.144	0.014	0.374	111,854	9.4e-10
mlp	0.737	0.724	0.750	0.436	0.146	0.025	0.355	111,843	6.5e-12
hgb	0.734	0.721	0.748	0.435	0.146	0.022	0.352	110,558	5.3e-15
rf	0.726	0.713	0.740	0.426	0.147	0.023	0.338	108,382	6.2e-19
et	0.724	0.710	0.737	0.418	0.148	0.019	0.338	108,529	5.9e-20

Table F.3. Ablation study by feature group (DeLong tests, Holm-corrected) (first 10 columns; full table in report_tables.xlsx).

Ablation	N features removed	Features	Auc full	Auc without	Auc drop	Ci low	Ci high	Delong p	Holm p
– premium_change	3	premium change at last renewal (%), premium increased at renewal, premium increase above 10%	0.741	0.708	0.033	0.024	0.042	1.2e-12	1.1e-11
– tenure	6	tenure (months), tenure (years), new customer (<12 months), policy start date (days since), policy start date year, tenure band	0.741	0.728	0.013	0.006	0.020	1.7e-04	0.001
– payment method	1	payment method	0.741	0.729	0.012	0.005	0.019	0.001	0.007
– auto-renewal	1	auto-renewal	0.741	0.730	0.011	0.005	0.017	7.6e-04	0.005
– complaints	2	complaints in the last 12 months, has complained	0.741	0.731	0.010	0.004	0.016	0.001	0.006
– engagement	1	digital engagement score	0.741	0.738	0.003	-0.002	0.009	0.226	0.714
– products	2	number of ANS	0.741	0.739	0.002	-0.004	0.008	0.489	0.978

Ablation	N features removed	Features	Auc full	Auc without	Auc drop	Ci low	Ci high	Delong p	Holm p
		products, holds multiple products							
– average claim settlement time (days)	1	average claim settlement time (days)	0.741	0.740	0.001	-0.004	0.007	0.607	0.978
– all engineered domain features	18	tenure (years), new customer (<12 months), premium-to-income ratio, sum insured to premium ratio, has made a claim, claims per year of tenure, claim rejection rate, has complained	0.741	0.745	-0.004	-0.009	0.002	0.179	0.714

Table F.4. Ensemble methods compared.

Method	Oof roc auc	Oof auc	Weights
best_single	0.758	0.758	{'xgb_fs': 1.0}
mean_top3	0.759	0.759	{'xgb_fs': 0.333, 'logreg': 0.333, 'xgb_multiseed': 0.333}
rank_mean_top3	0.759	0.759	{'xgb_fs': 0.333, 'logreg': 0.333, 'xgb_multiseed': 0.333}
hill_climb	0.760	0.760	{'xgb_fs': 0.5, 'logreg': 0.333, 'catboost_multiseed': 0.167}
optimized_weights	0.760	0.760	{'xgb_fs': 0.446, 'logreg': 0.283, 'catboost_tuned': 0.193, 'catboost': 0.078}
stacking_logreg	0.759	0.759	{'xgb_fs': 0.71, 'logreg': 0.429, 'xgb_multiseed': -0.041, 'xgb_tuned': -0.224, 'catboost_multiseed': 0.294, 'catboost_tuned': 0.023, 'catboost': 0.178, 'lgbm_tuned': -0.35, 'ebm': -0.06, 'xgb': 0.078}

Table F.5. Bootstrap 95% confidence intervals for the final model.

Metric	Estimate	Ci low	Ci high
roc_auc	0.760	0.748	0.772
pr_auc	0.474	0.450	0.500

Metric	Estimate	Ci low	Ci high
logloss	0.440	0.428	0.452

Table F.6. All models: out-of-fold and training AUC, overfitting gap and cross-validation variability (first 10 columns; full table in report_tables.xlsx).

Model	Algorithm	Oof roc auc	Cv mean	Cv std	Oof auc	Train auc	Overfit gap auc	Pr auc	Logloss
logreg	Logistic regression	0.757	0.757	0.004	0.757	0.776	0.019	0.470	0.442
xgb multiseed	XGBoost	0.756	0.756	0.005	0.756	0.807	0.051	0.467	0.443
xgb tuned	XGBoost	0.755	0.755	0.005	0.755	0.807	0.051	0.466	0.443
catboost multiseed	CatBoost	0.755	0.755	0.007	0.755	0.835	0.080	0.468	0.443
catboost tuned	CatBoost	0.753	0.753	0.006	0.753	0.824	0.070	0.463	0.444
catboost	CatBoost	0.752	0.752	0.006	0.752	0.894	0.142	0.466	0.444
lgbm tuned	LightGBM	0.751	0.751	0.006	0.751	0.862	0.111	0.462	0.445
ebm	Explainable boosting machine (EBM)	0.748	0.748	0.007	0.748	0.808	0.060	0.461	0.446
xgb	XGBoost	0.747	0.748	0.008	0.747	0.912	0.165	0.453	0.448
lgbm	LightGBM	0.745	0.745	0.007	0.745	0.922	0.177	0.450	0.449
mlp	Neural network (MLP)	0.737	0.739	0.009	0.737	0.827	0.090	0.436	0.458
hgb	Hist. gradient boosting	0.734	0.735	0.005	0.734	0.961	0.226	0.435	0.457
rf	Random forest	0.726	0.726	0.006	0.726	1.000	0.274	0.426	0.461
et	Extra trees	0.724	0.724	0.004	0.724	1.000	0.276	0.418	0.462
dummy	Baseline (prior)	0.500	0.500	0	0.500	0.500	4.5e-04	0.211	0.515

Table F.7. Baseline models with default hyperparameters (starting point of the development journey) (first 10 columns; full table in report_tables.xlsx).

Model	Algorithm	Oof roc auc	Cv mean	Cv std	Oof auc	Train auc	Overfit gap auc	Pr auc	Logloss
logreg	Logistic regression	0.757	0.757	0.004	0.757	0.776	0.019	0.470	0.442
catboost	CatBoost	0.752	0.752	0.006	0.752	0.894	0.142	0.466	0.444
ebm	Explainable boosting machine (EBM)	0.748	0.748	0.007	0.748	0.808	0.060	0.461	0.446
xgb	XGBoost	0.747	0.748	0.008	0.747	0.912	0.165	0.453	0.448
lgbm	LightGBM	0.745	0.745	0.007	0.745	0.922	0.177	0.450	0.449
mlp	Neural network (MLP)	0.737	0.739	0.009	0.737	0.827	0.090	0.436	0.458
hgb	Hist. gradient boosting	0.734	0.735	0.005	0.734	0.961	0.226	0.435	0.457
rf	Random forest	0.726	0.726	0.006	0.726	1.000	0.274	0.426	0.461
et	Extra trees	0.724	0.724	0.004	0.724	1.000	0.276	0.418	0.462
dummy	Baseline (prior)	0.500	0.500	0	0.500	0.500	4.5e-04	0.211	0.515

Table F.8. Paired bootstrap comparison of each candidate model with the final model.

Final vs	Diff roc auc	Ci low	Ci high	P value one sided	Significant
xgb fs	0.002	3.7e-04	0.004	0.008	yes
logreg	0.003	0.001	0.006	0.002	yes
xgb multiseed	0.004	0.002	0.006	0	yes
xgb tuned	0.004	0.002	0.007	0	yes

Table F.9. Out-of-fold permutation importance used for feature selection.

Feature model	Importance mean	Importance std	Feature
premium change at last renewal (%)	0.053	0.008	premium change at last renewal (%)
auto-renewal	0.028	0.006	auto-renewal
complaints in the last 12 months	0.014	0.002	complaints in the last 12 months
tenure (months)	0.009	0.004	tenure (months)
number of ANS products	0.008	0.003	number of ANS products
digital engagement score	0.007	0.002	digital engagement score
average claim settlement time (days)	0.007	0.002	average claim settlement time (days)
payment method freq	0.005	0.004	payment method freq
tenure (years)	0.003	0.002	tenure (years)
payment frequency freq	0.003	0.002	payment frequency freq
has complained	0.003	0.001	has complained
late payments in the last 12 months	0.003	0.002	late payments in the last 12 months
holds multiple products	0.002	0.002	holds multiple products
policy start date (days since)	0.002	9.8e-04	policy start date (days since)
claim rejection rate	0.002	0.002	claim rejection rate
age	0.001	6.5e-04	age
annual income	0.001	5.4e-04	annual income
payment frequency	0.001	7.8e-04	payment frequency
rejected claims	9.9e-04	5.6e-04	rejected claims
has late payments	9.9e-04	0.001	has late payments
payment method	9.4e-04	0.003	payment method
premium increased at renewal	6.2e-04	4.6e-04	premium increased at renewal
claims per year of tenure	4.6e-04	0.002	claims per year of tenure
sales channel	4.2e-04	0.001	sales channel
premium-to-income ratio	3.1e-04	3.1e-04	premium-to-income ratio
tenure band	2.8e-04	5.6e-04	tenure band
claims claim amount min	2.5e-04	3.5e-04	claims claim amount min
claims status Rejected count	2.4e-04	0.001	claims status Rejected count
sum insured	2.2e-04	6.3e-04	sum insured
policy start date year	2.2e-04	6.0e-04	policy start date year
sum insured to premium ratio	2.2e-04	3.0e-04	sum insured to premium ratio
annual premium	2.0e-04	7.4e-05	annual premium
marital status	1.4e-04	2.2e-04	marital status
coverage level	1.4e-04	1.6e-04	coverage level
marital status freq	1.4e-04	3.9e-04	marital status freq
claims (days since) first	9.6e-05	2.6e-04	claims (days since) first
new customer (<12 months)	8.2e-05	1.1e-04	new customer (<12 months)
policy type	6.4e-05	2.8e-04	policy type

Feature model	Importance mean	Importance std	Feature
number of missing fields	4.5e-05	5.3e-05	number of missing fields
policy start date month	3.7e-05	6.5e-05	policy start date month

Table F.10. Model performance by customer segment (AUC and churn rate per level).

Feature	Level	N	Churn rate	Auc
tenure band	2-5y	2,960	0.175	0.742
tenure band	1-2y	2,154	0.241	0.749
tenure band	6-12m	1,184	0.272	0.734
tenure band	0-6m	879	0.286	0.757
tenure band	5y+	823	0.091	0.743
payment method	Auto-debit	2,748	0.142	0.748
payment method	Bank Transfer	2,471	0.255	0.737
payment method	Credit Card	1,993	0.234	0.745
payment method	Cash	788	0.253	0.793
payment frequency	Monthly	3,980	0.236	0.754
payment frequency	Annual	2,419	0.194	0.761
payment frequency	Quarterly	1,601	0.174	0.760
age band	35-44	2,280	0.187	0.754
age band	45-54	1,993	0.211	0.773
age band	25-34	1,676	0.214	0.745
age band	55-64	930	0.198	0.739
age band	<25	845	0.252	0.744
age band	65+	268	0.299	0.841

Table F.11. Error analysis: share of true/false positives and negatives by feature.

Feature	TP	FN	FP	TN
premium change at last renewal (%)	8.169	3.570	7.145	2.362
tenure (months)	18.7	29.2	20.0	34.2
claims per year of tenure	1.070	0.341	0.857	0.294
claim rejection rate	0.259	0.075	0.198	0.052
complaints in the last 12 months	0.618	0.248	0.482	0.199
average claim settlement time (days)	17.5	14.0	16.2	13.4
payment method	Bank Transfer	Auto-debit	Bank Transfer	Auto-debit
auto-renewal	0.351	0.562	0.384	0.636
claims status nunique	0.177	0.051	0.125	0.078
claims claim amount max	5.54M	4.96M	5.32M	5.06M
claims (days since) last	485.5	501.7	497.1	489.3
claims (days since) first	608.2	598.1	603.5	599.6

Table F.12. Experiment log (all logged runs).

Exp id	Stage	Experiment	Note	Roc auc	Pr auc	Logloss	Time s
1	1 baseline model zoo	dummy	Baseline (prior), default hyperparameters	0.500	0.211	0.515	0.255
2	1 baseline model zoo	logreg	Logistic regression, default	0.757	0.470	0.442	1.763

Exp id	Stage	Experiment	Note	Roc auc	Pr auc	Logloss	Time s
			hyperparameters				
3	1 baseline model zoo	rf	Random forest, default hyperparameters	0.726	0.426	0.461	36.6
4	1 baseline model zoo	et	Extra trees, default hyperparameters	0.724	0.418	0.462	30.4
5	1 baseline model zoo	hgb	Hist. gradient boosting, default hyperparameters	0.734	0.435	0.457	14.1
6	1 baseline model zoo	lgbm	LightGBM, default hyperparameters	0.745	0.450	0.449	7.415
7	1 baseline model zoo	xgb	XGBoost, default hyperparameters	0.747	0.453	0.448	14.3
8	1 baseline model zoo	catboost	CatBoost, default hyperparameters	0.752	0.466	0.444	175.6
9	1 baseline model zoo	ebm	Explainable boosting machine (EBM), default hyperparameters	0.748	0.461	0.446	139.9
10	1 baseline model zoo	mlp	Neural network (MLP), default hyperparameters	0.737	0.436	0.458	10.9
11	2 hyperparameter tuning	lgbm_tuned	Optuna TPE, 39 trials, 3-fold inner CV	0.751	0.462	0.445	10.7
12	2 hyperparameter tuning	catboost_tuned	Optuna TPE, 9 trials, 3-fold inner CV	0.753	0.463	0.444	110.0
13	2 hyperparameter tuning	xgb_tuned	Optuna TPE, 39 trials, 3-fold inner CV	0.755	0.466	0.443	16.9
14	3 multi-seed averaging	xgb_multiseed	average of 3 seeds (base: xgb_tuned)	0.756	0.467	0.443	48.7
15	3 multi-seed averaging	catboost_multiseed	average of 3 seeds (base: catboost_tuned)	0.755	0.468	0.443	297.5
16	4 feature selection	xgb_fs	drop 20 features with permutation importance ≤ 0	0.758	0.470	0.442	39.6
17	5 ablation study	ablation reference (all features)	reference for ablation (same folds & params)	0.741	0.444	0.452	4.579
18	5 ablation study	ablation – premium_change	drop 3 feature(s): premium change at last renewal (%), premium increased at renewal, premium increase above 10%	0.708	0.402	0.468	4.033
19	5 ablation study	ablation – auto-renewal	drop 1 feature(s): auto-renewal	0.730	0.425	0.458	4.453
20	5 ablation study	ablation – complaints	drop 2 feature(s): complaints in the last 12 months, has complained	0.731	0.419	0.458	4.765
21	5 ablation study	ablation – tenure	drop 6 feature(s): tenure (months), tenure (years), new customer (<12 months), policy start date (days since)...	0.728	0.423	0.459	4.104
22	5 ablation study	ablation – products	drop 2 feature(s): number of ANS products, holds multiple products	0.739	0.438	0.453	3.802
23	5 ablation study	ablation – engagement	drop 1 feature(s): digital engagement	0.738	0.439	0.453	3.659

Exp id	Stage	Experiment	Note	Roc auc	Pr auc	Logloss	Time s
			score				
24	5 ablation study	ablation – average claim settlement time (days)	drop 1 feature(s): average claim settlement time (days)	0.740	0.443	0.452	4.298
25	5 ablation study	ablation – payment method	drop 1 feature(s): payment method	0.729	0.424	0.458	4.334
26	5 ablation study	ablation – all engineered domain features	drop 18 feature(s): tenure (years), new customer (<12 months), premium-to-income ratio, sum insured to premium ratio...	0.745	0.447	0.450	3.941
27	6 ensembling	ensemble: best_single	xgb_fs 1.00	0.758	0.470	0.442	
28	6 ensembling	ensemble: mean_top3	xgb_fs 0.33, logreg 0.33, xgb_multiseed 0.33	0.759	0.473	0.441	
29	6 ensembling	ensemble: rank_mean_top3	xgb_fs 0.33, logreg 0.33, xgb_multiseed 0.33	0.759	0.472	0.723	
30	6 ensembling	ensemble: hill_climb	xgb_fs 0.50, logreg 0.33, catboost_multiseed 0.17	0.760	0.474	0.440	
31	6 ensembling	ensemble: optimized_weights	xgb_fs 0.45, logreg 0.28, catboost_tuned 0.19, catboost 0.08	0.760	0.474	0.440	
32	6 ensembling	ensemble: stacking_logreg	xgb_fs 0.71, logreg 0.43, xgb_multiseed - 0.04, xgb_tuned - 0.22, catboost_multiseed 0.29, catboost_tuned 0.02, catboost 0.18, lgbm_tuned - 0.35, ebm -0.06, xgb 0.	0.759	0.472	0.441	

Table F.13. Train–test drift (PSI, KS; top 15).

Feature	Psi	Ks stat	Ks p	N unseen levels	Unseen rows %	Psi flag
agent id	0.275			0	0	MAJOR shift
claims claim amount std	0.062	0.059	0.503			stable
annual income	0.012	0.026	0.285			stable
claims (days since) last	0.010	0.030	0.524			stable
average claim settlement time (days)	0.009	0.042	0.159			stable
tenure (years)	0.009	0.022	0.410			stable
policy start date (days since)	0.009	0.022	0.410			stable
tenure (months)	0.009	0.022	0.410			stable
claims claim amount mean	0.009	0.035	0.345			stable

Feature	Psi	Ks stat	Ks p	N unseen levels	Unseen rows %	Psi flag
claims claim amount max	0.009	0.023	0.820			stable
claims (days since) first	0.008	0.032	0.437			stable
complaints in the last 12 months	0.007	0.028	0.161			stable
annual premium	0.007	0.019	0.573			stable
policy start date month	0.007	0.017	0.732			stable
policy start date year	0.006	0.020	0.565			stable

Appendix G. Explainability

Global and local explanations: gain, permutation and SHAP importance, dependence and partial dependence, individual explanations and segment heterogeneity.

Key observations. The top five SHAP features account for 37% of total attribution; premium change at last renewal (%), auto-renewal and payment method (frequency) lead the ranking, in line with the Information Value ranking in Appendix C.

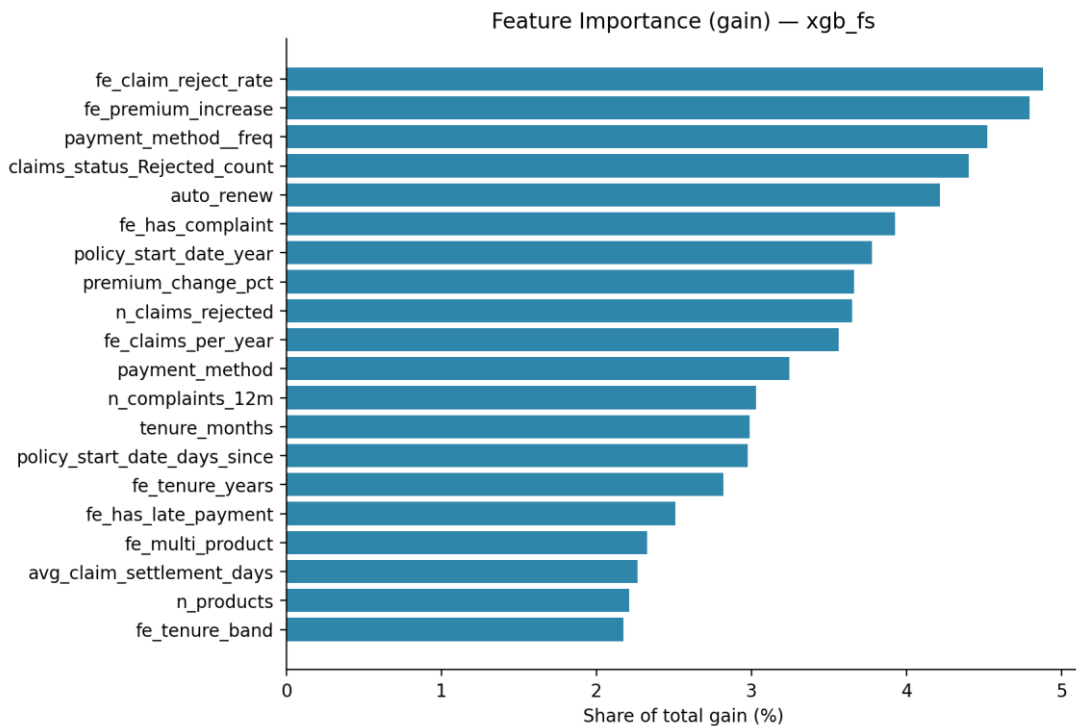


Figure G.1. Feature Importance (gain) — xgb_fs

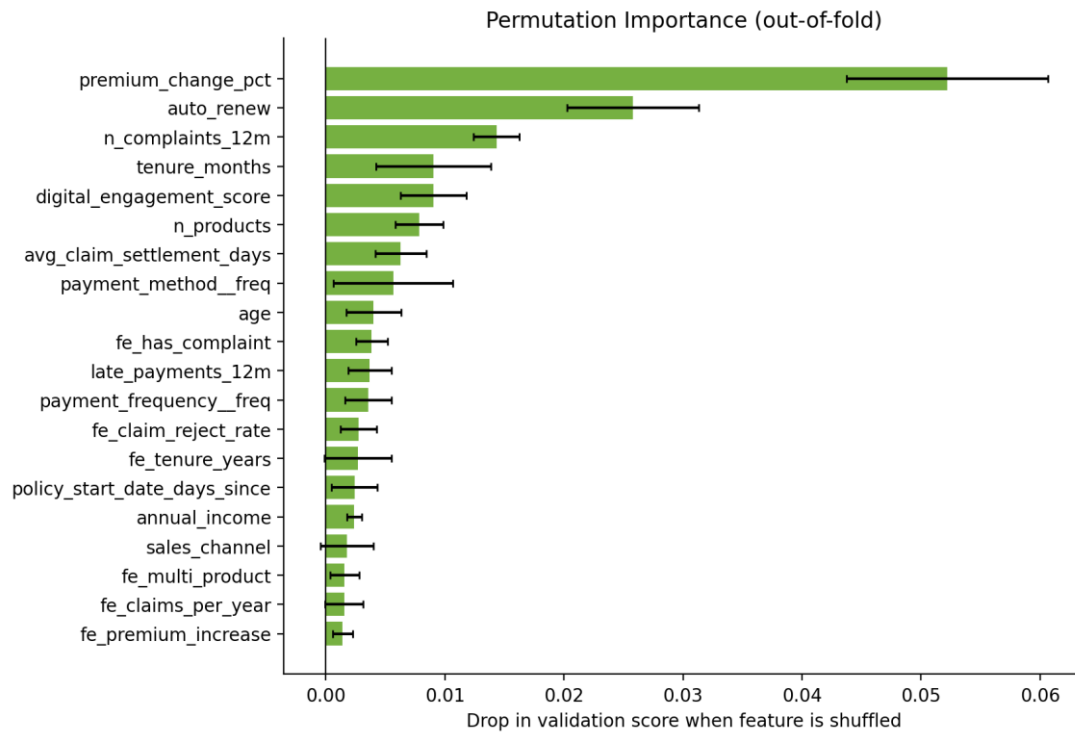


Figure G.2. Permutation Importance (out-of-fold)

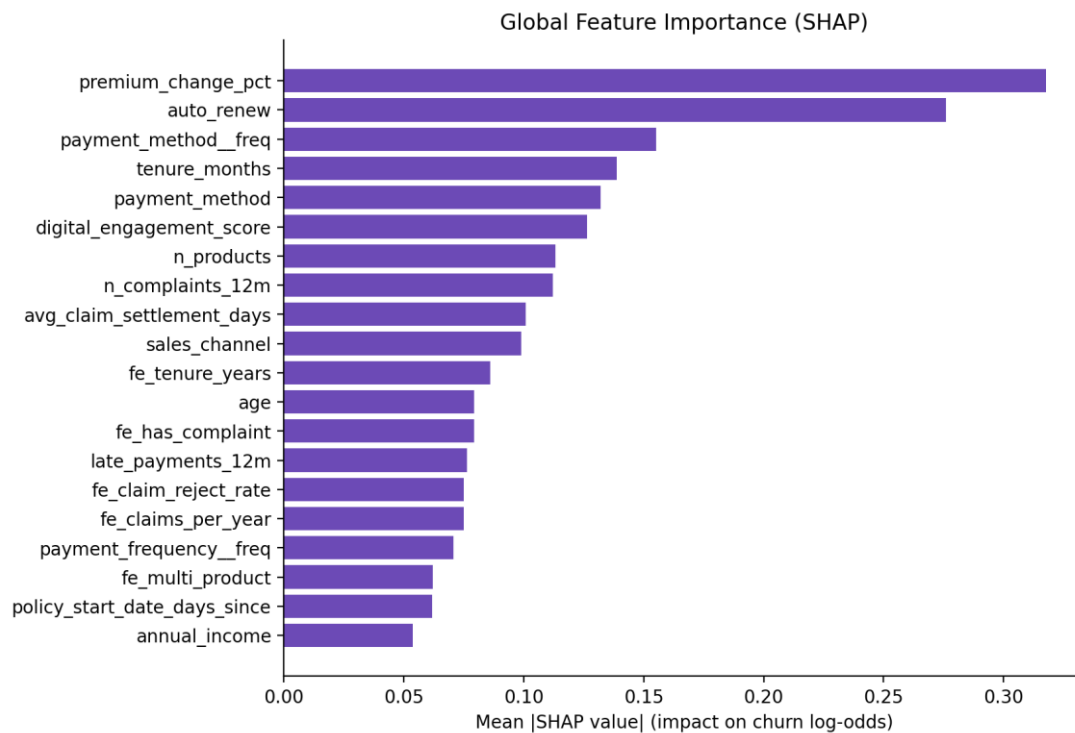


Figure G.3. Global Feature Importance (SHAP)

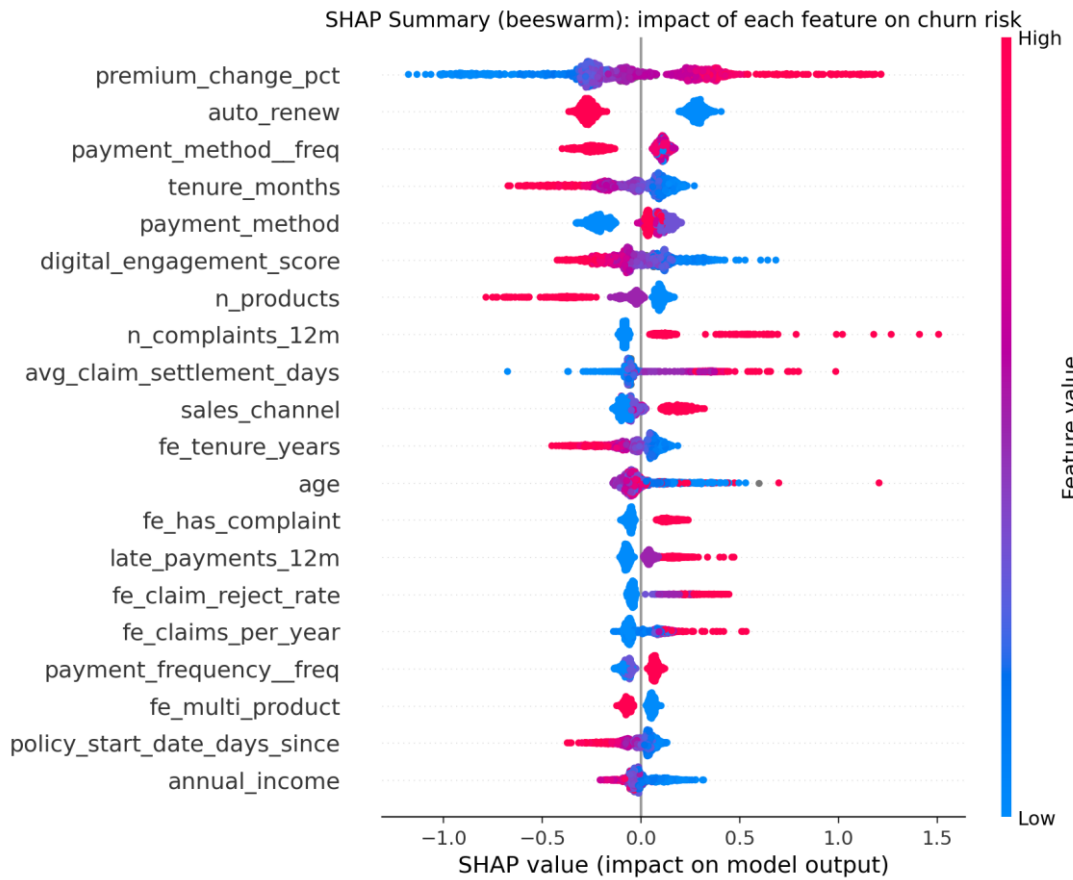


Figure G.4. SHAP summary: each dot is a customer; positive values push churn risk up (colour = feature value).

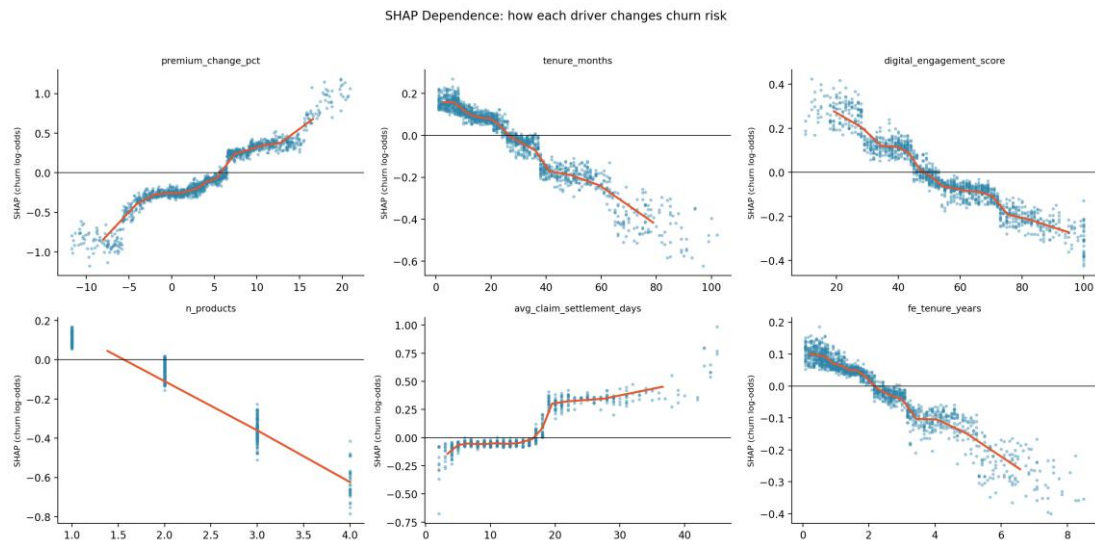


Figure G.5. SHAP Dependence: how each driver changes churn risk

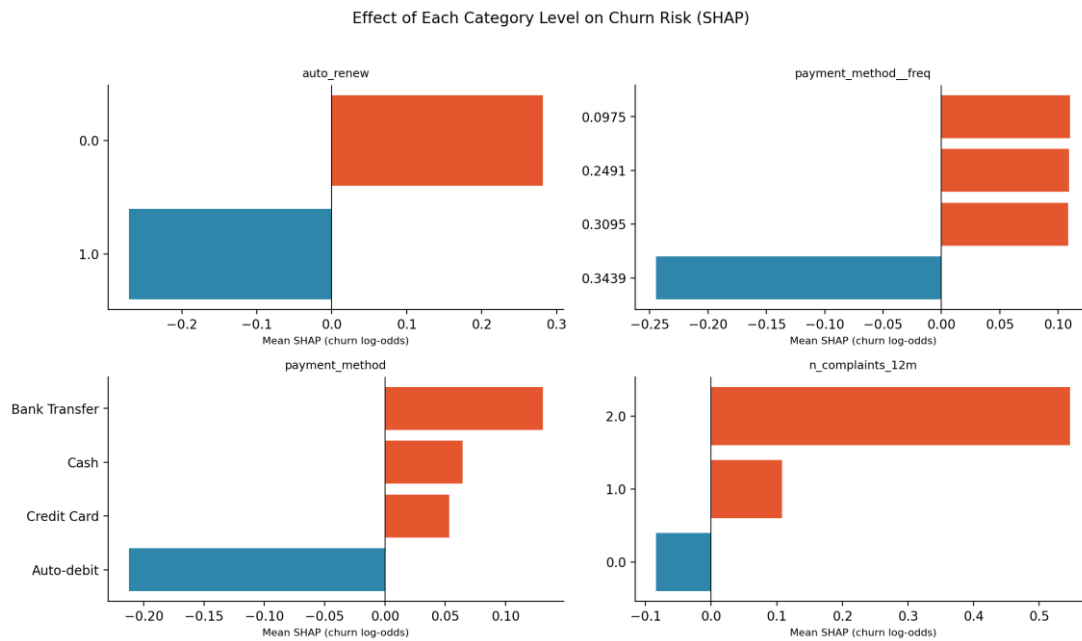


Figure G.6. Effect of Each Category Level on Churn Risk (SHAP)

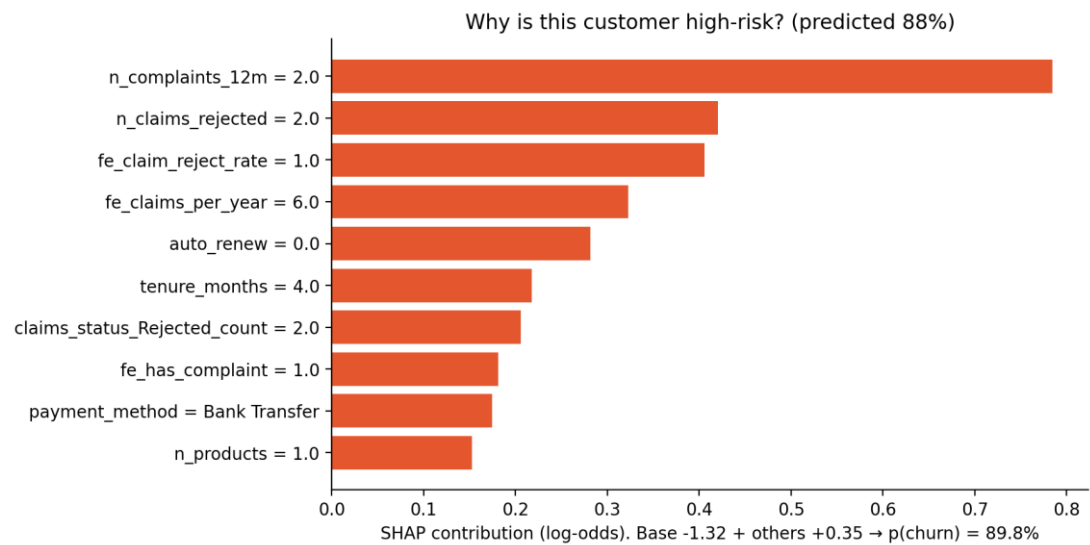


Figure G.7. Why is this customer high-risk? (predicted 88%)

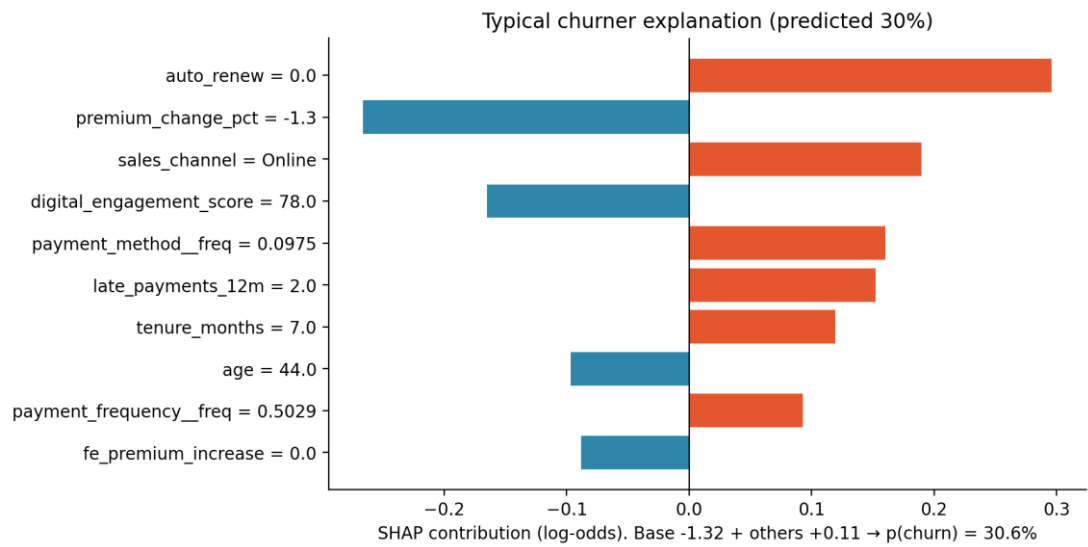


Figure G.8. Typical churning explanation (predicted 30%)

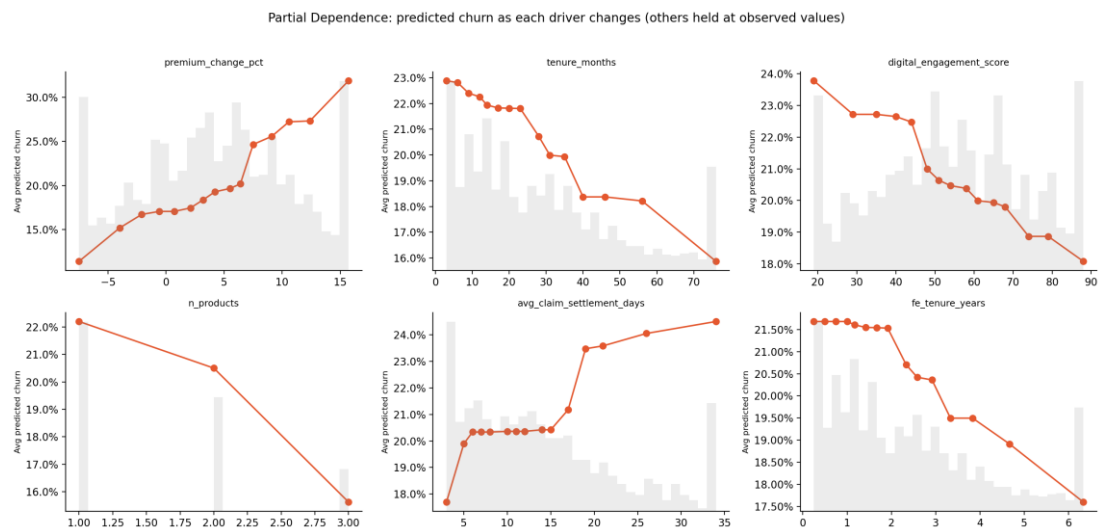


Figure G.9. Partial Dependence: predicted churn as each driver changes (others held at observed values)

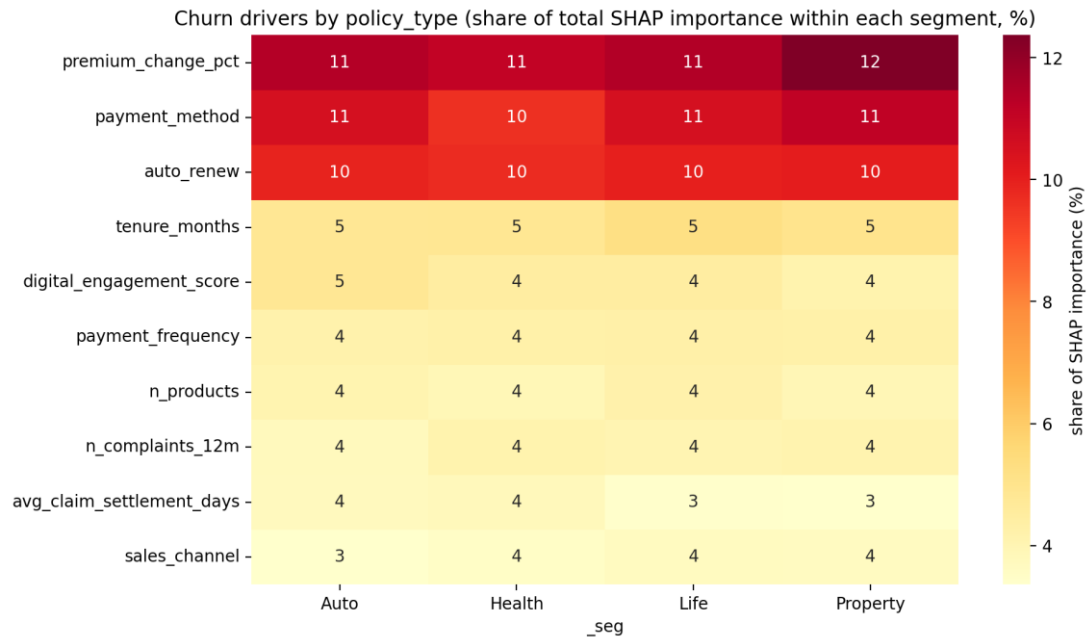


Figure G.10. Churn drivers by policy type (share of total SHAP importance within each segment, %)

Table G.1. Share of SHAP importance by customer segment (%), top drivers.

Feature	Auto	Health	Life	Property
premium change at last renewal (%)	11.4	11.1	11.5	12.4
payment method	10.6	9.652	10.6	11.1
auto-renewal	9.923	9.755	9.985	10.1
tenure (months)	4.819	4.869	5.225	4.990
digital engagement score	4.808	4.481	4.461	4.163
payment frequency	4.202	4.254	4.281	4.243
number of ANS products	4.078	3.915	4.233	3.948
complaints in the last 12 months	3.775	4.158	4.032	4.117
average claim settlement time (days)	3.724	3.789	3.397	3.363
sales channel	3.357	3.533	3.680	3.655

Table G.2. SHAP importance and direction of effect (top 20).

Feature	Mean abs shap	Share %	Direction
premium change at last renewal (%)	0.318	11.4	higher value → higher churn risk
auto-renewal	0.276	9.898	riskiest: '0.0' (+0.28); safest: '1.0' (-0.27)
payment_method_freq	0.155	5.564	higher value → lower churn risk
tenure (months)	0.139	4.974	higher value → lower churn risk
payment method	0.132	4.736	riskiest: 'Bank Transfer' (+0.13); safest: 'Auto-debit'

Feature	Mean abs shap	Share %	Direction
			(-0.21)
digital engagement score	0.126	4.530	higher value → lower churn risk
number of ANS products	0.113	4.055	higher value → lower churn risk
complaints in the last 12 months	0.112	4.018	higher value → higher churn risk
average claim settlement time (days)	0.101	3.615	higher value → higher churn risk
sales channel	0.099	3.543	riskiest: 'Online' (+0.19); safest: 'Agent' (-0.09)
tenure (years)	0.086	3.081	higher value → lower churn risk
age	0.079	2.847	higher value → lower churn risk
has complained	0.079	2.838	riskiest: '1.0' (+0.14); safest: '0.0' (-0.06)
late payments in the last 12 months	0.076	2.737	higher value → higher churn risk
claim rejection rate	0.075	2.686	higher value → higher churn risk
claims per year of tenure	0.075	2.685	higher value → higher churn risk
payment_frequency_freq	0.071	2.537	higher value → higher churn risk
holds multiple products	0.062	2.228	riskiest: '0.0' (+0.05); safest: '1.0' (-0.07)
policy start date (days since)	0.062	2.216	higher value → lower churn risk
annual income	0.054	1.921	higher value → lower churn risk

Table G.3. Consolidated driver evidence matrix (first 10 columns; full table in report_tables.xlsx).

Feature	Iv	Effect size	Q value BH	Highest risk level	Highest risk rate	Odds ratio	Hazard ratio	Shap rank	Evidence score
premium change at last renewal (%)	0.159	0.231	1.3e-44	13.2–30.6	0.335	1.638	1.385	1.000	4
payment method	0.098	0.123	1.3e-25	Bank Transfer	0.255	2.330	1.827	2.000	4
auto-renewal	0.084	0.118	1.2e-25	0	0.264	0.514	0.614	3.000	4
complaints in the last 12 months	0.110	0.140	3.1e-30	≥2	0.443	1.329	1.235	8.000	4
average claim settlement time (days)	0.105	0.118	1.8e-07	21–28	0.346	1.188	1.107	9.000	4
tenure (months)	0.148	-0.201	2.9e-36	1–6	0.286	0.629		4.000	3
number of ANS products	0.055	-0.106	1.6e-13	1	0.238	0.757		7.000	3
claim rejection rate	0.117	0.122	5.1e-38	1	0.371	1.260	1.108	15.0	3
digital engagement score	0.023	-0.080	3.0e-06	0–27	0.263			5.000	2

Feature	Iv	Effect size	Q value BH	Highest risk level	Highest risk rate	Odds ratio	Hazard ratio	Shap rank	Evidence score
payment frequency	0.024	0.063	1.9e-07	Monthly	0.236			6.000	2
sales channel	0.011	0.044	0.003	Online	0.241			10.0	2
claims per year of tenure	0.134	0.180	3.6e-35	1.5–24	0.349	1.073		16.0	2
claims count	0.076	0.140	7.3e-23	≥3	0.328		1.106		2
tenure (years)	0.148	-0.201	2.9e-36	0.0833–0.5	0.286			11.0	1
has complained	0.081	0.120	1.9e-26	1	0.294			13.0	1
late payments in the last 12 months	0.038	0.101	1.7e-12	≥3	0.278			14.0	1
holds multiple products	0.033	0.074	7.5e-11	0	0.238			17.0	1
policy start date (days since)	0.148	-0.201	2.9e-36	31–183	0.286			18.0	1
annual income	0.009	-0.040	0.024	7.8M–42.1M	0.241			19.0	1
has late payments	0.033	0.074	7.3e-11	1	0.245			20.0	1
premium increased at renewal	0.077	0.108	1.3e-20	1	0.239			22.0	1
rejected claims	0.116	0.122	2.5e-38	≥2	0.505			24.0	1
claims status Rejected count	0.116	0.122	2.5e-38	≥2	0.505			25.0	1
policy type	0.009	0.039	0.011	Health	0.232			27.0	1
policy start date year	0.131	0.191	3.4e-34	2,024–2,025	0.291			28.0	1

Appendix H. Causal Inference and Uplift

Doubly robust effect estimates, identification assumptions (DAG), overlap, sensitivity (E-values), placebo tests and heterogeneous effects.

Key observations. 5 of 5 estimated effects pass every robustness check (placebo effect close to zero, overlap problems below the threshold and E-value above 1.25). The smallest E-value among them is 1.51, so an unmeasured confounder would need at least this risk ratio with both treatment and churn to explain the effect away.

Identification. For each controllable lever a binary 'favourable condition' T is defined. Under no unmeasured confounding and overlap, the augmented inverse probability weighting estimator combines a propensity model $e(X)$ and outcome models $\mu_1(X)$, $\mu_0(X)$ and stays consistent if either is correct (Robins et al., 1994); all nuisance models are cross-fitted with five folds to avoid own-observation bias (Chernozhukov et al.,

2018). Propensity scores are clipped at 0.02/0.98 and the share of customers outside [0.05, 0.95] is reported.

Robustness. A placebo test re-estimates the effect with a randomly permuted treatment (should be ≈ 0). E-values quantify how strong an unmeasured confounder would have to be, on the risk-ratio scale, to explain away the estimate (VanderWeele & Ding, 2017). Only effects that pass all checks are labelled 'robust'.

Heterogeneity and uplift. The doubly robust pseudo-outcome is regressed on customer features with cross-fitting (DR-learner; Kennedy, 2023), a meta-learner in the family studied by Künzel et al. (2019). Customers are sorted into quintiles of predicted uplift and the average pseudo-outcome per quintile (with standard errors) tests whether effects differ. Uplift-based targeting is compared with risk-based targeting, following the uplift literature for insurance retention (Guelman et al., 2015; Devriendt et al., 2021).

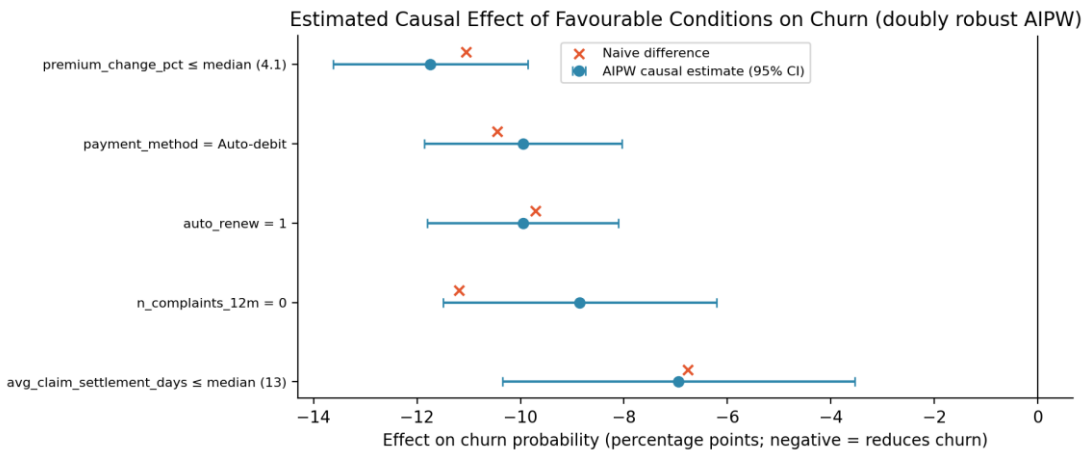


Figure H.1. Estimated Causal Effect of Favourable Conditions on Churn (doubly robust AIPW)

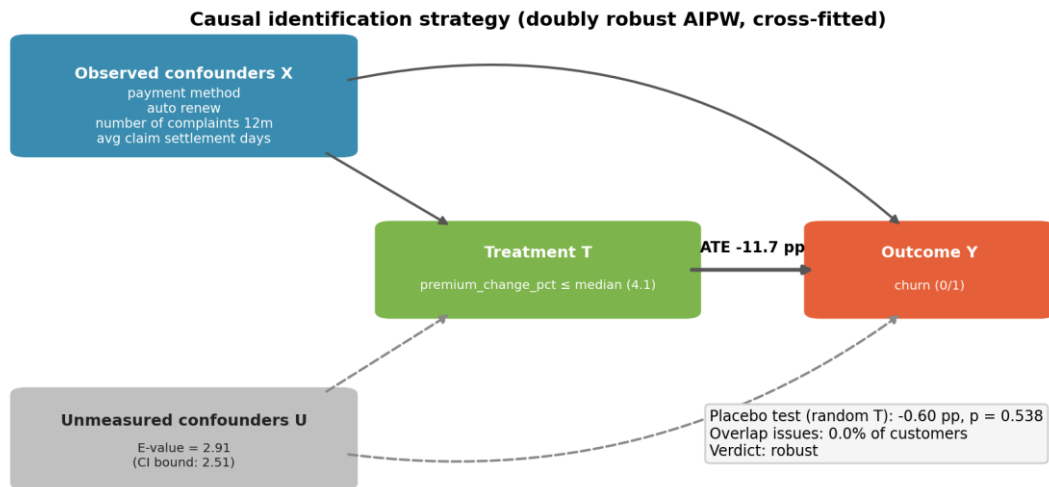


Figure H.2. Assumed causal diagram (DAG) behind the doubly robust effect estimates: lever, confounders and churn.

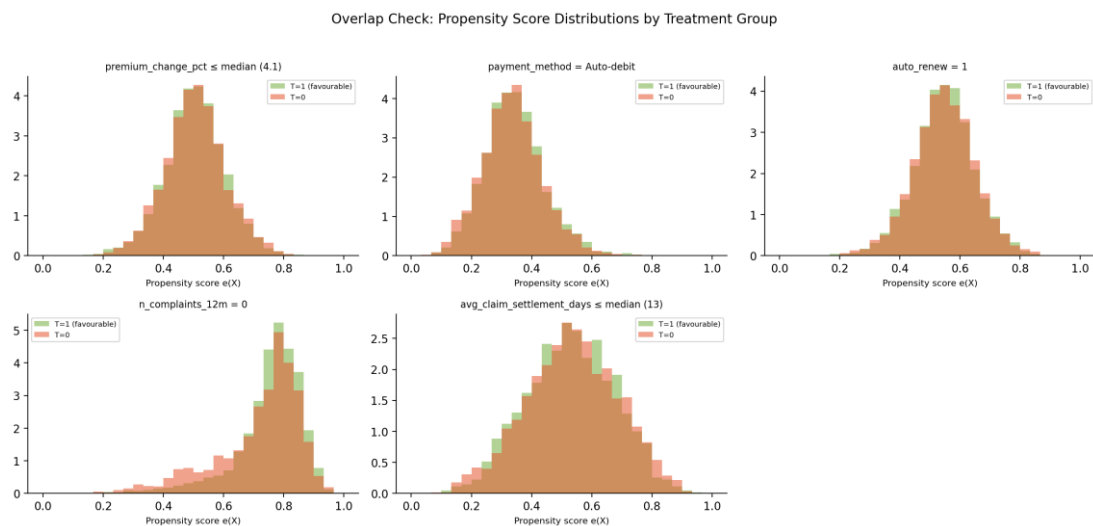


Figure H.3. Overlap Check: Propensity Score Distributions by Treatment Group

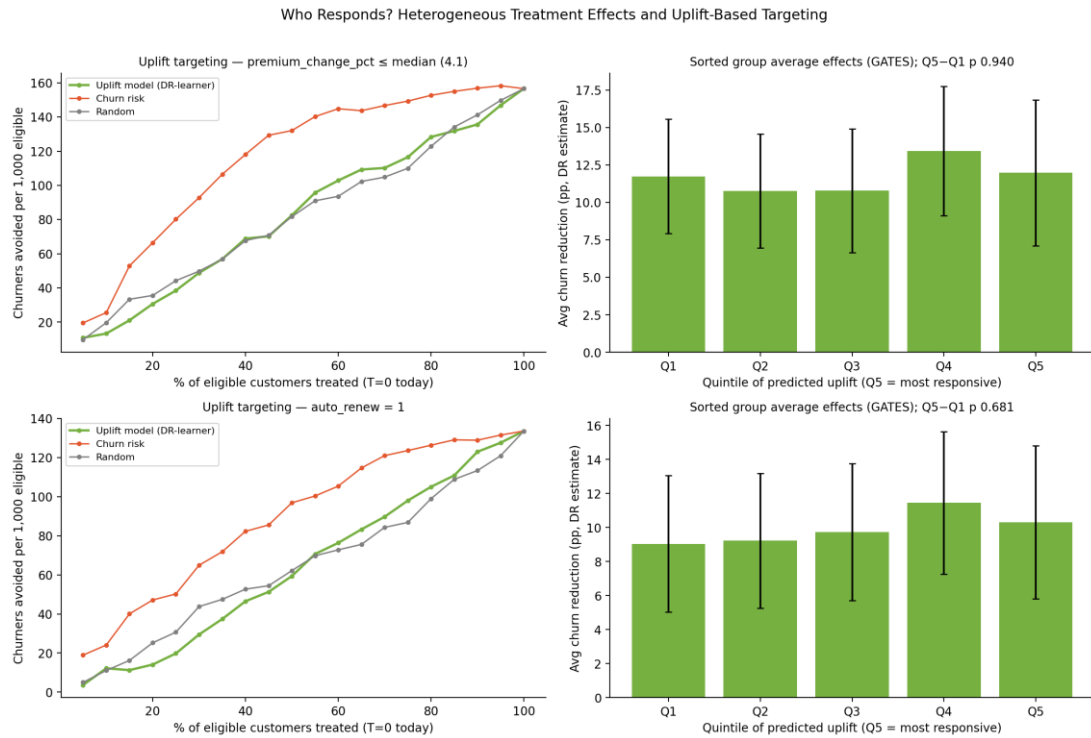


Figure H.4. Who Responds? Heterogeneous Treatment Effects and Uplift-Based Targeting

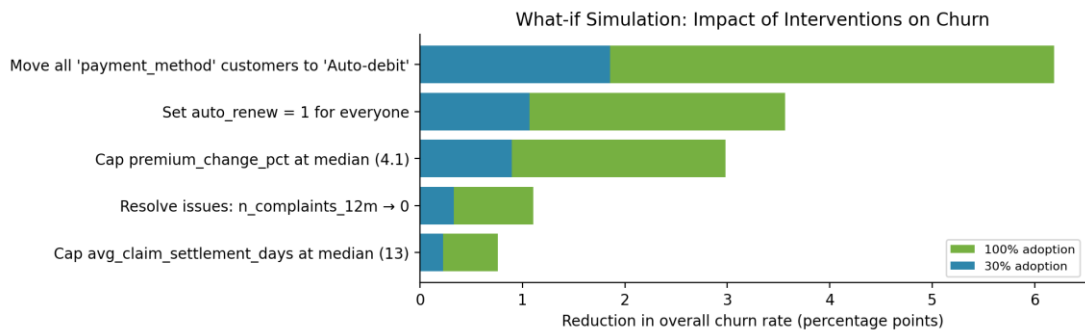


Figure H.5. What-if Simulation: Impact of Interventions on Churn

Table H.1. AIPW average treatment effects.

Feature	Favorable condition (T=1)	Treated %	Naive diff pp	Ate pp	Ci low pp	Ci high pp	P value	Overlap issue %
premium change at last renewal (%)	premium change at last renewal (%) ≤ median (4.1)	50.4	-11.1	-11.7	-13.6	-9.863	0	0
payment method	payment method = Auto-debit	34.4	-10.4	-9.949	-11.9	-8.036	0	0.025
auto-renewal	auto-renewal = 1	54.6	-9.709	-9.954	-11.8	-8.102	0	0
complaints	complaints	73.9	-11.2	-8.852	-11.5	-6.207	5.4e-11	0.137

Feature	Favorable condition (T=1)	Treated %	Naive diff pp	Ate pp	Ci low pp	Ci high pp	P value	Overlap issue %
in the last 12 months	in the last 12 months = 0							
average claim settlement time (days)	average claim settlement time (days) ≤ median (13)	52.8	-6.763	-6.943	-10.3	-3.537	6.4e-05	0

Table H.2. Robustness of causal estimates (first 10 columns; full table in report_tables.xlsx).

Treatment (T=1)	Feature	Ate pp	P value	Mu1 churn if T1	Mu0 churn if T0	Risk ratio	E value point	E value ci	Placebo ate pp
premium change at last renewal (%) ≤ median (4.1)	premium change at last renewal (%)	-11.7	0	0.155	0.272	0.569	2.911	2.511	-0.595
payment method = Auto-debit	payment method	-9.949	0	0.145	0.245	0.594	2.758	2.341	0.699
auto-renewal = 1	auto-renewal	-9.954	0	0.166	0.266	0.626	2.576	2.232	1.178
complaints in the last 12 months = 0	complaints in the last 12 months	-8.852	5.4e-11	0.188	0.276	0.679	2.306	1.902	-2.134
average claim settlement time (days) ≤ median (13)	average claim settlement time (days)	-6.943	6.4e-05	0.237	0.307	0.774	1.908	1.514	-1.700

Table H.3. Uplift (DR-learner) summary.

Treatment	Feature	Eligible customers	Ate pp	Gates q5 pp	Gates q1 pp	Heterogeneity p	Avoided per1000 top20 uplift	Avoided per1000 top20 risk	Avoided per1000 top20 random
premium change at last renewal (%) ≤ median (4.1)	premium change at last renewal (%)	3,809	-11.7	12.0	11.7	0.940	30.6	66.5	35.6
auto-renewal = 1	auto-renewal	3,630	-9.954	10.3	9.041	0.681	14.1	47.1	25.2

Table H.4. Sorted group average treatment effects (GATES).

Treatment	Uplift quintile	Avg churn reduction pp	Ci low pp	Ci high pp	N
premium change	Q1	11.7	7.918	15.6	1,535

Treatment	Uplift quintile	Avg churn reduction pp	Ci low pp	Ci high pp	N
at last renewal (%) $\leq$ median (4.1)					
premium change at last renewal (%) $\leq$ median (4.1)	Q2	10.8	6.973	14.6	1,535
premium change at last renewal (%) $\leq$ median (4.1)	Q3	10.8	6.663	14.9	1,535
premium change at last renewal (%) $\leq$ median (4.1)	Q4	13.4	9.130	17.7	1,535
premium change at last renewal (%) $\leq$ median (4.1)	Q5	12.0	7.115	16.8	1,535
auto-renewal = 1	Q1	9.041	5.026	13.1	1,600
auto-renewal = 1	Q2	9.232	5.266	13.2	1,600
auto-renewal = 1	Q3	9.742	5.723	13.8	1,600
auto-renewal = 1	Q4	11.5	7.263	15.6	1,600
auto-renewal = 1	Q5	10.3	5.807	14.8	1,600

Table H.5. Model-based what-if simulation (associational).

Scenario	Feature	Affected %	Churn affected before	Churn affected after	Overall reduction pp full	Overall reduction pp 30pct adoption	Churners avoided per 1000
Move all 'payment method' customers to 'Auto-debit'	payment method	67.2	0.237	0.145	6.188	1.857	18.6
Set auto-renewal = 1 for everyone	auto-renewal	45.4	0.246	0.167	3.563	1.069	10.7
Cap premium change at last renewal (%) at median (4.1)	premium change at last renewal (%)	48.4	0.255	0.193	2.984	0.895	8.951
Resolve issues: complaints in the last 12 months $\rightarrow$ 0	complaints in the last 12 months	26.0	0.270	0.228	1.107	0.332	3.320
Cap average claim settlement time (days) at median (13)	average claim settlement time (days)	19.2	0.277	0.237	0.758	0.227	2.275

Appendix I. Business Analytics

Lift and gains, risk tiers, risk $\times$ value prioritisation, campaign simulation and sensitivity, churn personas.

Key observations. The top risk decile churns at 55.9% (2.6× the average) and the bottom decile at 3.4%, a 17-fold spread that makes targeted retention economical.



Figure I.1. Customer Churn — Executive Summary

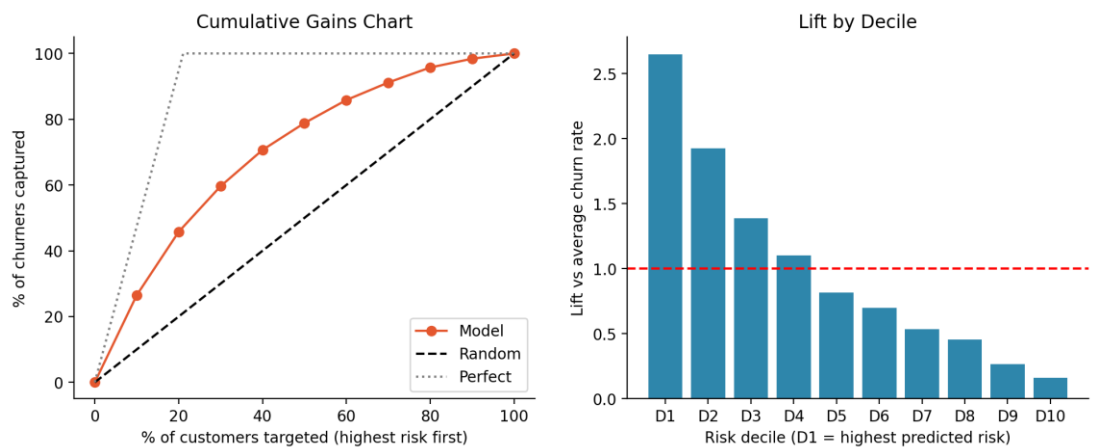


Figure I.2. Cumulative Gains Chart

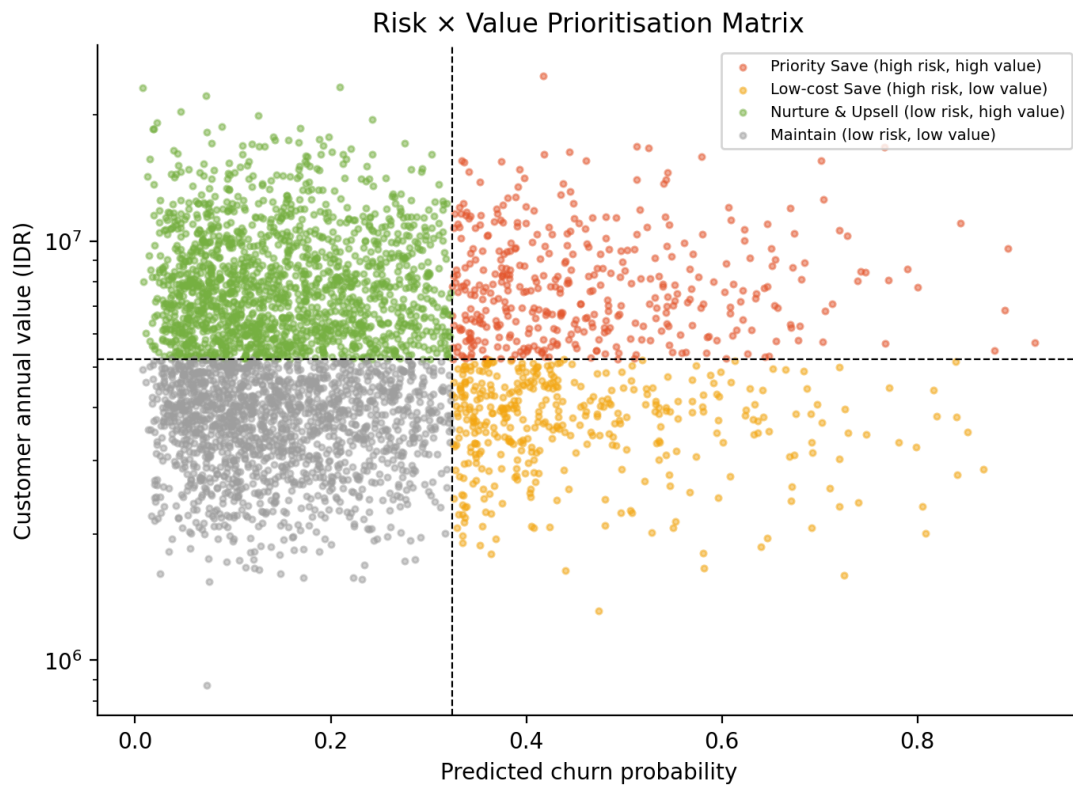


Figure I.3. Risk x Value Prioritisation Matrix

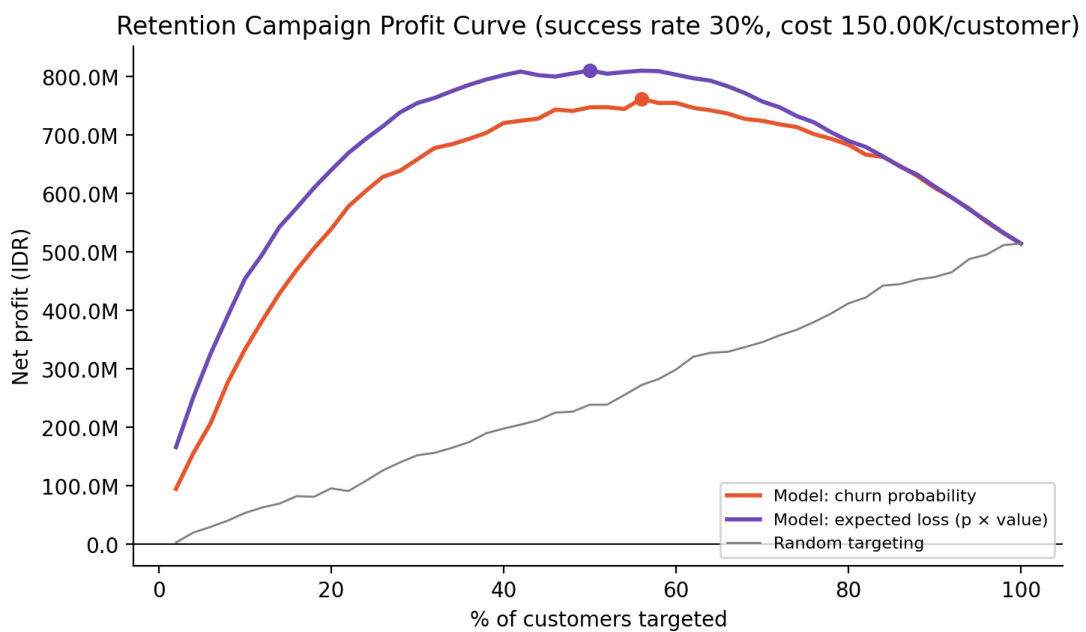


Figure I.4. Retention Campaign Profit Curve (success rate 30%, cost 150.00K/customer)

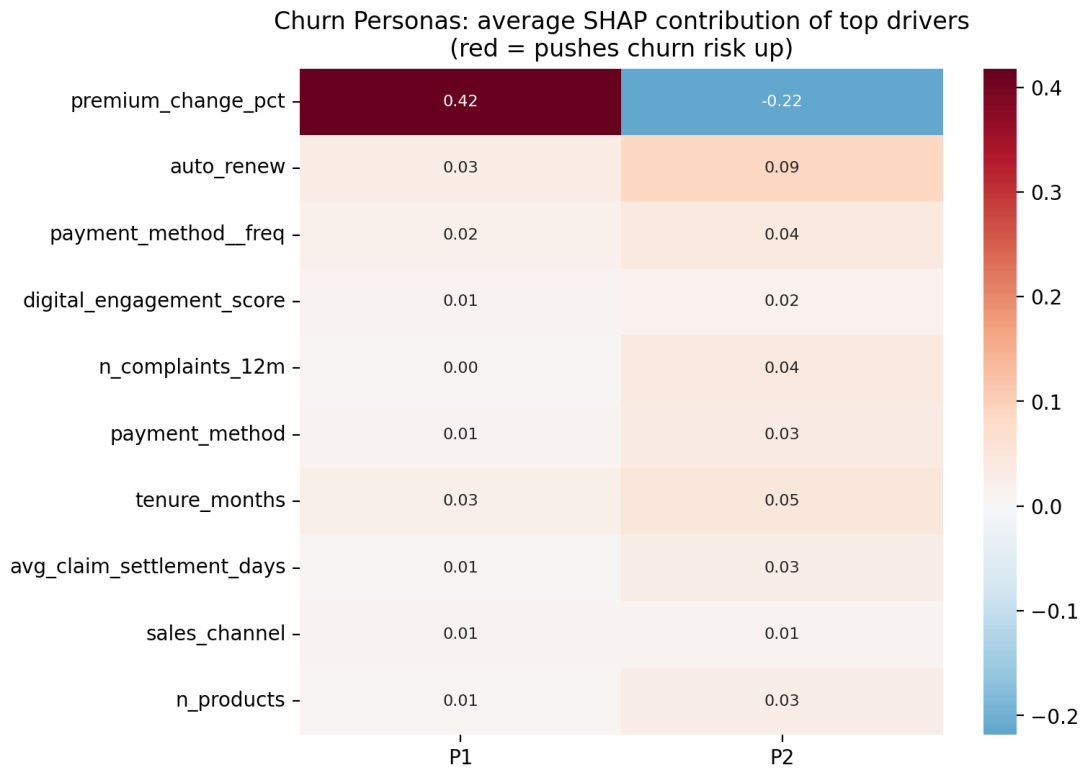


Figure I.5. Churn Personas: average SHAP contribution of top drivers
(red = pushes churn risk up)

Table I.1. Lift by risk decile.

Decile	N	Churners	Avg pred	Min pred	Churn rate	Lift	Cum customers %	Cum churners captured %	Cum lift
D1	800	447	0.551	0.425	0.559	2.650	10.0	26.5	2.650
D2	800	325	0.370	0.324	0.406	1.926	20.0	45.8	2.288
D3	800	234	0.288	0.257	0.292	1.387	30.0	59.6	1.988
D4	800	186	0.231	0.208	0.233	1.103	40.0	70.7	1.766
D5	800	138	0.188	0.169	0.172	0.818	50.0	78.8	1.577
D6	800	118	0.152	0.136	0.147	0.699	60.0	85.8	1.431
D7	800	90	0.122	0.108	0.113	0.533	70.0	91.2	1.302
D8	800	77	0.095	0.082	0.096	0.456	80.0	95.7	1.197
D9	800	45	0.069	0.055	0.056	0.267	90.0	98.4	1.093
D10	800	27	0.038	0.006	0.034	0.160	100.0	100.0	1.000

Table I.2. Risk tiers.

Tier	N train	Actual churn rate	Avg predicted	Share churners %	N test
High	1,600	0.482	0.461	45.8	405
Medium	2,400	0.233	0.236	33.1	615
Low	4,000	0.089	0.095	21.2	980

Table I.3. Risk × value quadrants.

Quadrant	N	Actual churn rate	Avg pred	Total value	Value at risk	Share value at risk %
Low-cost Save (high risk, low value)	818	0.480	0.461	3.00B	1.38B	13.8
Maintain (low risk, low value)	3,182	0.143	0.147	11.79B	1.73B	17.4
Nurture & Upsell (low risk, high value)	3,218	0.143	0.149	26.68B	3.94B	39.5
Priority Save (high risk, high value)	782	0.485	0.460	6.36B	2.92B	29.3

Table I.4. Optimal campaign size by strategy.

Strategy	Target %	N targeted	Churners reached	Benefit	Cost	Net profit	Roi
Model: churn probability	56.0	4,480	1,405	1.43B	672.00M	762.12M	1.134
Model: expected loss (p × value)	50.0	4,000	1,247	1.41B	600.00M	810.79M	1.351
Random targeting	100.0	8,000	1,687	1.71B	1.20B	514.76M	0.429

Table I.5. Campaign sensitivity analysis.

Success rate	Cost per contact	Optimal target %	Max net profit	Roi
0.150	75,000	55.1	404.25M	1.223
0.150	150,000	22.4	203.51M	0.757
0.150	300,000	6.083	56.06M	0.385
0.300	75,000	67.3	1.18B	2.921
0.300	150,000	55.1	808.51M	1.223
0.300	300,000	22.4	407.01M	0.757
0.450	75,000	87.8	2.01B	3.809
0.450	150,000	63.3	1.57B	2.073
0.450	300,000	38.8	962.65M	1.035

Table I.6. Churn personas (first 10 columns; full table in report_tables.xlsx).

Persona	N in sample	Share of at risk %	Avg pred churn	Actual churn rate	Top drivers	Premium change at last renewal (%)	Payment method	Auto-renewal	Complaints in the last 12 months
P2	418	52.9	0.298	0.325	auto-renewal (+0.09); payment method (+0.08)	1.210	Bank Transfer	0.360	0.500
P1	372	47.1	0.358	0.368	premium change at last renewal (%) (+0.42)	11.6	Bank Transfer	0.450	0.400

Appendix J. Governance, Deployment and Implementation

Fairness audit, model card, deployment loop, pilot design and implementation roadmap.

Key observations. The largest discrimination gap across groups is 0.127 AUC (region); calibration ranges from 0.91 to 1.07 across its levels. Selection differences that follow genuine differences in churn are expected; the score is used only to decide who receives retention outreach.

The model card follows Mitchell et al. (2019). The fairness audit compares calibration (mean predicted vs actual churn), discrimination (AUC) and selection into the High-risk tier across customer groups; differences in selection that mirror genuine differences in churn are expected, whereas calibration gaps would indicate bias. The score is restricted to retention outreach.

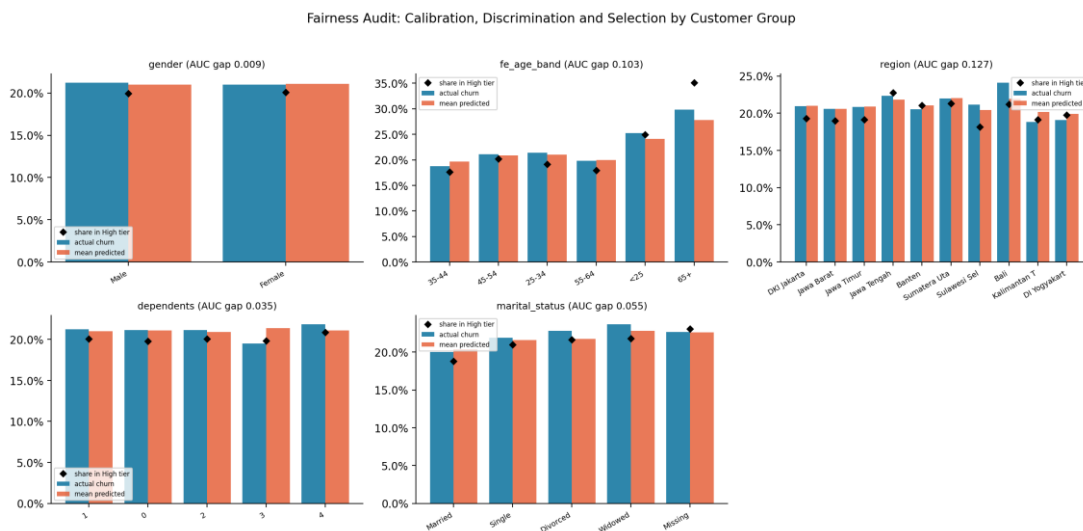


Figure J.1. Fairness Audit: Calibration, Discrimination and Selection by Customer Group

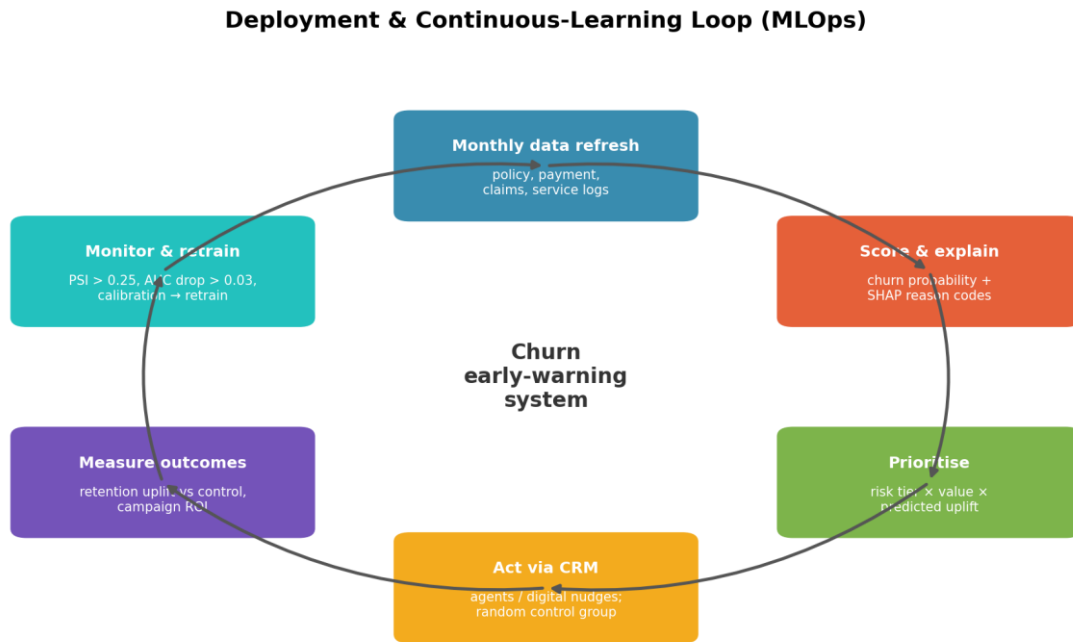


Figure J.2. Deployment & Continuous-Learning Loop (MLOps)

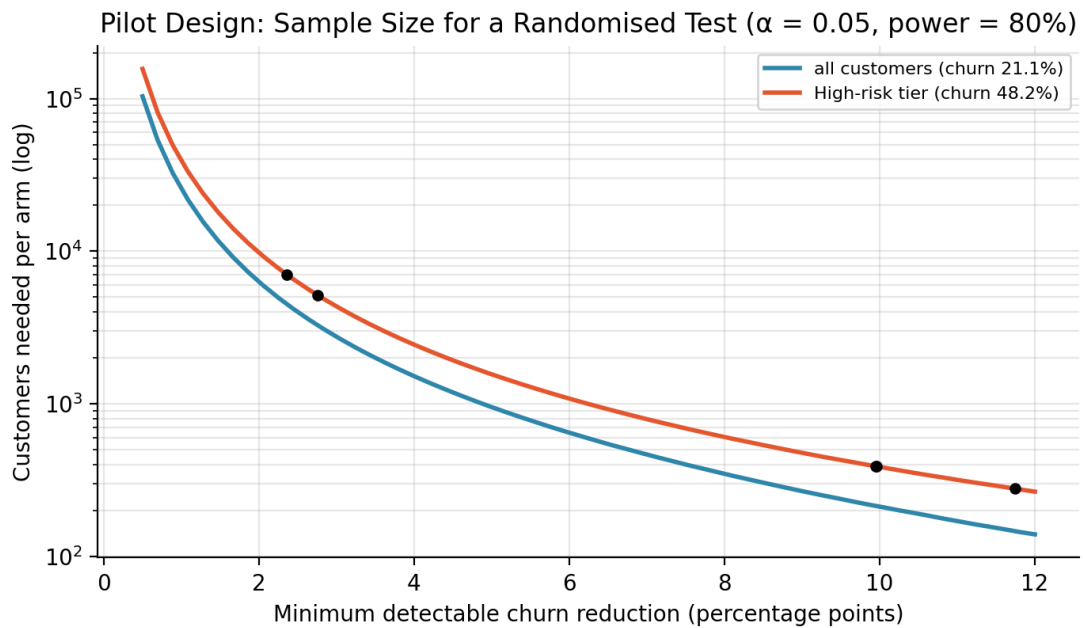


Figure J.3. Pilot Design: Sample Size for a Randomised Test ($\alpha = 0.05$, power = 80%)

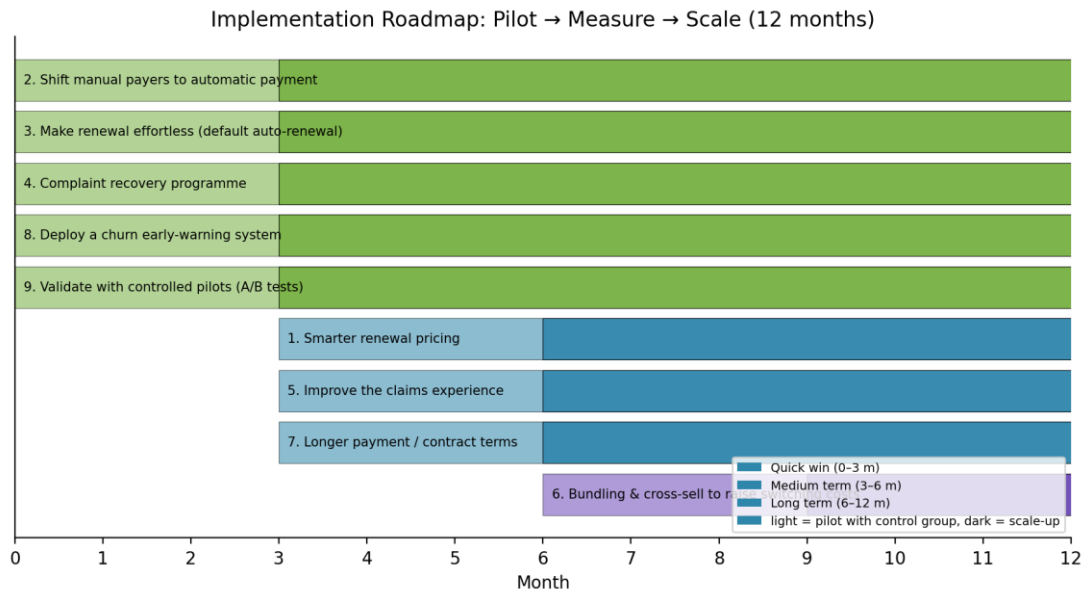


Figure J.4. Twelve-month implementation roadmap: each initiative is piloted against a control group before scale-up.

Table J.1. Model card (Mitchell et al., 2019).

Item	Description
Model	hill_climb (xgb_fs 0.50, logreg 0.33, catboost_multiseed 0.17)
Intended use	Rank active policyholders by 12-month churn risk to prioritise retention outreach and allocate retention budget.
Out-of-scope use	Not for premium pricing, underwriting, claims decisions or denial of service; not validated for other products/markets.
Training data	8,000 customers, 53 input features; target 'churn' (churn = Yes); churn rate 21.1%.
Excluded variables	cancellation reason (post-event leakage)
Validation	Stratified 5-fold CV (out-of-fold), target encoding inside folds, shuffled-label & adversarial checks, bootstrap & DeLong CIs.
Performance	ROC-AUC 0.760 (95% CI 0.748–0.772); PR-AUC 0.474; Brier 0.140; ECE 0.010; EMPC 114.99K IDR/customer.
Calibration	none (selected by cross-validated log-loss).
Decision rule	Risk tiers at the 50%/80% quantiles; campaign size chosen by expected profit (see Section 4).
Explainability	Global: SHAP, permutation importance, PDP, EBM shape functions (if used). Local: SHAP waterfall per customer for agents.
Fairness	Audited on gender, age band, region, dependents, marital status; max AUC gap 0.127.
Monitoring	Monthly: PSI per feature (alert > 0.25), score distribution, realised churn by tier, calibration; retrain quarterly or if AUC drops > 0.03.
Limitations	Observational snapshot data; causal effects assume no unmeasured confounding; ROI depends on stated cost/acceptance assumptions.

Table J.2. Fairness summary by attribute.

Attribute	Min sel	Max sel	Min auc	Max auc	Min cal	Max cal	Selection ratio	Auc gap
dependents	0.198	0.209	0.732	0.767	0.966	1.094	0.947	0.035
age band	0.176	0.351	0.739	0.841	0.932	1.050	0.503	0.103
gender	0.199	0.201	0.756	0.764	0.991	1.005	0.993	0.009
marital status	0.188	0.231	0.722	0.778	0.955	1.018	0.815	0.055
region	0.181	0.228	0.708	0.834	0.911	1.073	0.797	0.127

Table J.3. Pilot design: sample size per arm.

Intervention	Effect source	Baseline churn target segment	Expected reduction pp	N per arm	Total pilot size	Share of high tier %
Shift to: premium change at last renewal (%) $\leq$ median (4.1)	AIPW causal estimate	0.482	11.7	277	554	136.8
Shift to: auto-renewal = 1	AIPW causal estimate	0.482	9.954	388	776	191.6
Shift to: payment method = Auto-debit	AIPW causal estimate	0.482	9.949	389	778	192.1
Move all 'payment method' customers to 'Auto-debit'	what-if $\times$ 30% adoption	0.482	2.761	5,129	10,258	2,533
Set auto-renewal = 1 for everyone	what-if $\times$ 30% adoption	0.482	2.356	7,048	14,096	3,480

Table J.4. Implementation roadmap.

Priority	Initiative	Phase	Start month	Pilot months	Scale months	Kpi
2	Shift manual payers to automatic payment	Quick win (0–3 m)	0	3	3	% of policies on auto-pay; lapse rate of converted vs control
3	Make renewal effortless (default auto-renewal)	Quick win (0–3 m)	0	3	3	Auto-renew adoption; renewal rate
4	Complaint recovery programme	Quick win (0–3 m)	0	3	3	Complaint resolution time; churn rate of complainants
8	Deploy a churn early-warning system	Quick win (0–3 m)	0	3	3	Precision/recall of alerts; retained revenue vs control group
9	Validate with controlled pilots (A/B tests)	Quick win (0–3 m)	0	3	3	Uplift in retention rate (treated – control)
1	Smarter renewal pricing	Medium term (3–6 m)	3	3	3	Churn rate among customers with increases; price elasticity
5	Improve the claims	Medium term (3–6 m)	3	3	3	Claim settlement days;

Priority	Initiative	Phase	Start month	Pilot months	Scale months	Kpi
	experience					post-claim churn rate; claim NPS
7	Longer payment / contract terms	Medium term (3–6 m)	3	3	3	Share of annual payers; churn by payment frequency
6	Bundling & cross-sell to raise switching costs	Long term (6–12 m)	6	3	6	Products per customer; churn of bundled vs single

Table J.5. Full recommendation list with evidence and KPIs.

Priority	Recommendation	Action	Evidence	Estimated impact	Kpi
1	Smarter renewal pricing	Cap or phase in premium increases for at-risk loyal customers, explain the value behind increases, and offer coverage/deductible adjustments instead of lapse.	premium change at last renewal (%) = 13.2–30.6: churn 33.5% (1.6× avg); IV 0.16; OR 1.64; HR 1.39; SHAP rank #1; causal (AIPW) -11.7 pp [-13.6, -9.9]	-2.98 pp churn at full adoption (≈9 churners avoided /1,000 at 30% adoption)	Churn rate among customers with increases; price elasticity
2	Shift manual payers to automatic payment	Offer a small incentive (e.g. premium discount/cashback) and one-click enrolment for auto-debit/recurring card payment; send pre-due reminders to remaining manual payers.	payment method = Bank Transfer: churn 25.5% (1.2× avg); IV 0.10; OR 2.33; HR 1.83; SHAP rank #2; causal (AIPW) -9.9 pp [-11.9, -8.0]	-6.19 pp churn at full adoption (≈19 churners avoided /1,000 at 30% adoption)	% of policies on auto-pay; lapse rate of converted vs control
3	Make renewal effortless (default auto-renewal)	Default new and renewing policies to auto-renewal with clear opt-out; 30/14/7-day renewal reminders with a one-click renew link.	auto-renewal = No: churn 26.4% (1.3× avg); IV 0.08; OR 0.51; HR 0.61; SHAP rank #3; causal (AIPW) -10.0 pp [-11.8, -8.1]	-3.56 pp churn at full adoption (≈11 churners avoided /1,000 at 30% adoption)	Auto-renew adoption; renewal rate
4	Complaint recovery programme	Trigger a proactive call within 48h of any complaint, enforce resolution SLAs, offer service-recovery gestures to high-value complainants, and fix top root causes.	complaints in the last 12 months = ≥2: churn 44.3% (2.1× avg); IV 0.11; OR 1.33; HR 1.23; SHAP rank #8; causal (AIPW) -8.9 pp [-11.5, -6.2]	-1.11 pp churn at full adoption (≈3 churners avoided /1,000 at 30% adoption)	Complaint resolution time; churn rate of complainants
5	Improve the claims experience	Fast-track settlement, explain rejections transparently with an appeal path, and assign a claims	average claim settlement time (days) = 21–28: churn 34.6% (1.6× avg); IV 0.10; OR 1.19;	-0.76 pp churn at full adoption (≈2 churners avoided /1,000 at 30% adoption)	Claim settlement days; post-claim churn rate; claim NPS

Priority	Recommendation	Action	Evidence	Estimated impact	Kpi
		concierge for high-value / high-risk customers after a claim event.	HR 1.11; SHAP rank #9; causal (AIPW) -6.9 pp [-10.3, -3.5]		
6	Bundling & cross-sell to raise switching costs	Offer multi-policy discounts and relevant riders to single-product customers with good risk profiles.	number of ANS products = 1: churn 23.8% (1.1× avg); IV 0.05; OR 0.76; SHAP rank #7		Products per customer; churn of bundled vs single
7	Longer payment / contract terms	Encourage annual/quarterly payment or longer contracts with a discount; monthly payers get extra reminders.	payment frequency = Monthly: churn 23.6% (1.1× avg); IV 0.02; SHAP rank #6		Share of annual payers; churn by payment frequency
8	Deploy a churn early-warning system	Score all active policies monthly with this model; route High-risk/Priority-Save customers to retention teams, lower tiers to automated digital nudges; refresh the model quarterly.	Model ROC-AUC 0.760; top-20% captures 46% of churners	optimal campaign: target top 50% → net 810.79M	Precision/recall of alerts; retained revenue vs control group
9	Validate with controlled pilots (A/B tests)	For each intervention keep a random control group (e.g. 20% of targeted customers) to measure true uplift before scaling.	What-if results are associational		Uplift in retention rate (treated – control)

Appendix K. Methodological Details and Formulas

This appendix states the estimators behind the main results so that every number can be recomputed.

Information Value and Wilson intervals. For each bin j of a variable, the weight of evidence and Information Value are (Siddiqi, 2006):

$$\text{WoE}_j = \ln(\% \text{non-churners}_j / \% \text{churners}_j) \quad (3)$$

$$\text{IV} = \sum_j (\% \text{non-churners}_j - \% \text{churners}_j) \cdot \text{WoE}_j \quad (4)$$

Values above 0.1 indicate medium and above 0.3 strong predictive power. Churn rates carry Wilson score intervals:

$$\text{CI} = [\hat{p} + z^2/(2n) \pm z \cdot \sqrt{ (\hat{p}(1 - \hat{p})/n + z^2/(4n^2)) }] / (1 + z^2/n) \quad (5)$$

Multiple testing. Univariate p-values, ordered $p_{(1)} \leq \dots \leq p_{(m)}$, are converted to Benjamini–Hochberg q-values (Benjamini & Hochberg, 1995); model comparisons use the Holm step-down correction:

$$q_{(i)} = \min\{ m \cdot p_{(k)} / k : k \geq i \} \quad (6)$$

Survival. With d_i churn events and n_i customers at risk at time t_i , the Kaplan–Meier estimator, the Cox model and the restricted mean survival time are (Kaplan & Meier, 1958; Cox, 1972; Royston & Parmar, 2013):

$$\hat{S}(t) = \prod (1 - d_i / n_i), \text{ product over all } t_i \leq t \quad (7)$$

$$h(t | x) = h_0(t) \cdot \exp(\beta'x) \quad (8)$$

$$\text{RMST}(\tau) = \int \hat{S}(t) dt, \text{ integrated from } 0 \text{ to } \tau \quad (9)$$

$\exp(\beta_k)$ is a hazard ratio, and the 13- and 25-month persistency ratios are $\hat{S}(13)$ and $\hat{S}(25)$ with tenure in months.

Discrimination and calibration. ROC-AUC is the probability that a random churning scores higher than a random non-churner; correlated AUCs are compared with the DeLong variance estimator (DeLong et al., 1988). Calibration uses the Brier score and the expected calibration error over probability bins k (Guo et al., 2017):

$$\text{BS} = (1/n) \sum_i (p_i - y_i)^2 \quad (10)$$

$$\text{ECE} = \sum_k (n_k / n) \cdot |\bar{y}_k - \bar{p}_k| \quad (11)$$

Profit-based evaluation. The expected maximum profit measure integrates campaign profit over a Beta-distributed acceptance rate γ , where η is the share of customers targeted (Verbraken et al., 2013); customer value is the discounted margin while the policy stays in force:

$$\text{EMPC} = \int \max\{ P(\eta; \gamma) : \eta \} \cdot h(\gamma) d\gamma \quad (12)$$

$$\text{CLV}_i = m \cdot P_i \cdot (1 - \rho_i^H) / (1 - \rho_i), \quad \rho_i = (1 - p_i) / (1 + d) \quad (13)$$

with margin m , annual premium P_i , churn probability p_i , discount rate d and horizon H .

Causal effects. With treatment T , outcome Y , propensity $\hat{e}(X)$ and outcome models $\hat{\mu}_0(X)$, the augmented inverse probability weighting estimator is consistent if either nuisance model is correct (Robins et al., 1994); nuisance models are cross-fitted (Chernozhukov et al., 2018):

$$\text{ATE} = (1/n) \sum_i [\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + T_i(Y_i - \hat{\mu}_1(X_i))/\hat{e}(X_i) - (1 - T_i)(Y_i - \hat{\mu}_0(X_i))/(1 - \hat{e}(X_i))] \quad (14)$$

The E-value for a risk ratio RR states how strong an unmeasured confounder would have to be to explain the effect away (VanderWeele & Ding, 2017):

$$E = RR + \sqrt{(RR \cdot (RR - 1))} \quad (15)$$

Drift and pilot size. The population stability index compares the share of customers per bin k in scoring data (a_k) and training data (e_k); values below 0.1 indicate no material shift. Pilot arms use the two-proportion sample-size formula:

$$PSI = \sum_k (a_k - e_k) \cdot \ln(a_k / e_k) \quad (16)$$

$$n = (z_{1-\alpha/2} + z_{1-\beta})^2 \cdot [p_1(1 - p_1) + p_2(1 - p_2)] / (p_1 - p_2)^2 \quad (17)$$

Appendix L. Reproducibility

All results were produced by one notebook executed top-to-bottom (Restart & Run All) with random seed 42, 5-fold stratified cross-validation and the settings in Appendix B. Software: Python 3.11.9, pandas 2.3.3, scikit-learn 1.6.1, LightGBM 4.7.0, XGBoost 3.2.0, CatBoost 1.2.10, SHAP 0.51.0, statsmodels 0.15.0, lifelines 0.30.3, Optuna 4.9.0; platform Windows 10. The submission package contains the executed notebook with all outputs, every figure and table (CSV and Excel), the model card, the per-customer action list and an interactive dashboard.

References for the Appendices

- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (pp. 2623–2631). <https://doi.org/10.1145/3292500.3330701>
- Caruana, R., Niculescu-Mizil, A., Crew, G., & Ksikes, A. (2004). Ensemble selection from libraries of models. In *Proceedings of the 21st International Conference on Machine Learning* (p. 18). <https://doi.org/10.1145/1015330.1015432>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>
- Devriendt, F., Berrevoets, J., & Verbeke, W. (2021). Why you should stop predicting customer churn and start using uplift models. *Information Sciences*, 548, 497–515. <https://doi.org/10.1016/j.ins.2019.12.075>
- Guelman, L., Guillén, M., & Pérez-Marín, A. M. (2015). Uplift random forests. *Cybernetics and Systems*, 46(3–4), 230–248. <https://doi.org/10.1080/01969722.2015.1012892>
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning* (PMLR 70, pp. 1321–1330).

- Harrell, F. E., Califf, R. M., Pryor, D. B., Lee, K. L., & Rosati, R. A. (1982). Evaluating the yield of medical tests. *JAMA*, 247(18), 2543–2546.
<https://doi.org/10.1001/jama.1982.03320430047030>
- Höppner, S., Stripling, E., Baesens, B., vanden Broucke, S., & Verdonck, T. (2020). Profit driven decision trees for churn prediction. *European Journal of Operational Research*, 284(3), 920–933. <https://doi.org/10.1016/j.ejor.2018.11.072>
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems* 30.
https://papers.nips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10), 4156–4165.
<https://doi.org/10.1073/pnas.1804597116>
- Lou, Y., Caruana, R., Gehrke, J., & Hooker, G. (2013). Accurate intelligible models with pairwise interactions. In *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 623–631). <https://doi.org/10.1145/2487575.2487579>
- Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 220–229). <https://doi.org/10.1145/3287560.3287596>
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems* 31.
- Robins, J. M., Rotnitzky, A., & Zhao, L. P. (1994). Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427), 846–866.
<https://doi.org/10.1080/01621459.1994.10476818>
- Royston, P., & Parmar, M. K. B. (2013). Restricted mean survival time: An alternative to the hazard ratio for the design and analysis of randomized trials with a time-to-event outcome. *BMC Medical Research Methodology*, 13, 152.
<https://doi.org/10.1186/1471-2288-13-152>
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5), 206–215. <https://doi.org/10.1038/s42256-019-0048-x>

- Schoenfeld, D. (1982). Partial residuals for the proportional hazards regression model. *Biometrika*, 69(1), 239–241. <https://doi.org/10.1093/biomet/69.1.239>
- Siddiqi, N. (2006). *Credit risk scorecards: Developing and implementing intelligent credit scoring*. John Wiley & Sons.
- Verbeke, W., Dejaeger, K., Martens, D., Hur, J., & Baesens, B. (2012). New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. *European Journal of Operational Research*, 218(1), 211–229. <https://doi.org/10.1016/j.ejor.2011.09.031>
- Wei, L. J. (1992). The accelerated failure time model: A useful alternative to the Cox regression model in survival analysis. *Statistics in Medicine*, 11(14–15), 1871–1879. <https://doi.org/10.1002/sim.4780111409>